EXAM PACK

EEE459

Compound Semiconductor and Hetero-Junction Devices



Exam Pack

Session 2017–18 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2021, and Term Test 1*

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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2017–18

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PAST PAPER SOLUTIONS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Paper Solutions

Session 2017–18

4th Year 1st Semester Final Examination, 2021, with Term Test 1

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7
20 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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November 2026

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The paper at a glance

3 h	100	2 of 6	90 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part A				
Q1	(a) Single-Crystal, Polycrystalline and Amorphous Solids	3	5 min	☐
	(b) Effective Mass from the E–k Diagram	4	7 min	
	(c) Band Gap of GaAsP Against Composition and Light Emission	9	16 min	
	(d) Composition of InGaAsSb on InP and Its Critical Thickness	9	16 min	
Q2	(a) Anderson's Model and the n-Ge / p-GaAs Heterojunction	7	13 min	☐
	(b) Band Diagrams of Al0.3Ga0.7As / GaAs Heterojunctions	8	14 min	
	(c) Tungsten on n-Type GaAs Schottky Diode	10	18 min	
Q3	(a) Conditions and Band Diagrams for Rectifying Metal–Semiconductor Contacts	9	16 min	☐
	(b) Critical Thickness, and Optical Transitions in Si and GaAs	9	16 min	
	(c) Tunnelling Junction and Why It Is Ohmic	7	13 min	
Part B				
Q4	(a) Bipolar Transistor Operation from the Ebers–Moll Circuit	8	14 min	☐
	(b) Bandgap Narrowing	3	5 min	
	(c) Open-Base Breakdown Voltage	9	16 min	
	(d) Emitter Minority Holes with Bandgap Narrowing	5	9 min	
Q5	(a) Operation of the Tunnelling-Emitter Bipolar Transistor (TEBT)	4	7 min	☐
	(b) Two-Dimensional Electron Gas (2DEG) and the Operation of the HEMT	5	9 min	
	(c) Kirk Effect, and Abrupt and Graded Heterojunction Bipolar Transistors	11	20 min	
	(d) Two-Dimensional Electron Concentration in an AlGaAs-GaAs Heterojunction	5	9 min	
Q6	(a) Channel Space-Charge Profiles of a Single-Gate n-Channel JFET	8	14 min	☐
	(b) Internal Pinch-Off Voltage of a MESFET	6	11 min	
	(c) Internal Pinch-Off and Pinch-Off Voltage of a Silicon p-Channel JFET	5	9 min	
	(d) Transconductance and the Small-Signal Equivalent Circuit of a JFET	6	11 min	
Term Test 1 · 20 marks · 30 min				
Q7	(a) Composition of InGaAs on GaAs from a Critical Thickness of 50 Å	8	12 min	☐
	(b) Effective Mass of Electrons in GaAs and Si, Mobility and Its Field Dependence	12	18 min	

Formulas and constants follow Neamen, Semiconductor Physics and Devices (4th ed.); compound-semiconductor data follow Sze and Ng, Physics of Semiconductor Devices.

Shahjalal University of Science & Technology

Department of Electrical and Electronic Engineering
4th year 1st Semester Final Examination, 2021
Course Code: EEE 459
Course Title: Compound Semiconductor and Heterojunction Devices

Total Marks: 100 **Credit: 3.0** **Time: 3 Hours**

Answer any two questions from each part.

Part A

1. (a) Define mono crystalline, polycrystalline and amorphous materials. 3

   (b) What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram. 4

   (c) How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap. 9

   (d) For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system. 7+2

Semiconductor	Lattice Constant (Å)	Bandgap (eV)
InAs	6.06	0.38
GaAs	5.65	1.43
InSb	6.48	0.18
GaSb	6.09	0.7

2. (a) What are the assumptions made in Anderson's model? Determine ΔE_C, ΔE_V and V_{bi} for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, n_i = [unclear: 2.5×10^{13}] cm^{-3} (for Ge), $N_d = 1.2\times10^{16}$ cm^{-3}, and $N_a = 1\times10^{16}$ cm^{-3}. 2+5

   (b) Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for:

   (i) [unclear: N^+]-AlGaAs, intrinsic GaAs.
   (ii) P^+-AlGaAs, n-GaAs.

   Assume E_g of AlAs = 2.1 eV and $\Delta E_C = \Delta E_V$. 4+4

   (c) Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5\times10^{15}$ cm^{-3}. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. (N_c = [unclear: 4.7×10^{17}] cm^{-3}, [unclear: ϵ_r] 13.1, $\phi_m = 4.55$, $\chi = 4.07$) 10

3. (a) Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors. 3+3+3

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(b) Discuss the significance of critical thickness in epitaxial growth. Draw enegy [*sic*: should be energy] band structure of Si and GaAs and show optical transitions in such semiconductors. 3+3+3

(c) Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact? 7

Part B

4. (a) From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates. 8

(b) What is bandgap narrowing? 3

(c) Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage." 9

(d) Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from 10^{18} to 10^{19} cm^{-3}. Determine the new value of p'_{E0} and determine the ratio p'_{E0}/p_{E0}. 5

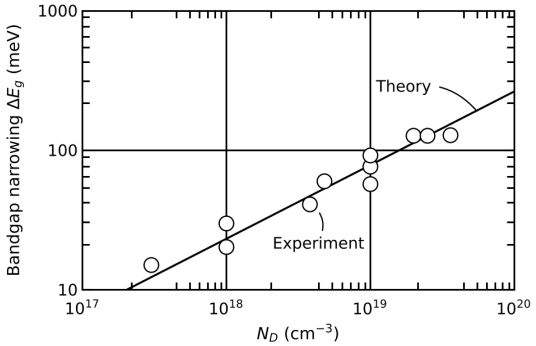


Figure 1: Fig. for Q. no. 4(d).

5. (a) With neat sketch, describe basic operation mechanism of TEBT. 4

(b) What is 2DEG? With quantum well structure describe the operation of HEMT. 5

(c) With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters. 4+3+4

(d) Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N - Al_{0.3}Ga_{0.7}As$ layer doped to 10^{18} cm^{-3} and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2. 5

6. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltages respectively)

(i) $V_g > V_p$ and $V_d = 0V$
(ii) $V_g > V_p$ and $V_d < V_{d,sat}$
(iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$
(iv) $V_g > V_p$ and $V_d > V_{d,sat}$ 8

(b) In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \dfrac{ea^2N_d}{2\epsilon_s}$ 6

(c) The n^+p junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}$ cm^{-3} and $N_a = 10^{16}$ cm^{-3}. The metallurgical channel thickness is $a = 0.40$ μm. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET. 5

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(d) Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET. 6

Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

[Course: EEE459, Term Test 1, Time: 30 Minutes, Full Marks: 20]

7. (a) If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively. 8

(b) Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field. 4+2+6

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1. (a) Define mono crystalline, polycrystalline and amorphous materials. Marks: 3

   (b) What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram. Marks: 4

   (c) How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap. Marks: 9

   (d) For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system. [Marks: 7+2]

Semiconductor	Lattice Constant (Å)	Bandgap (eV)
InAs	6.06	0.38
GaAs	5.65	1.43
InSb	6.48	0.18
GaSb	6.09	0.7

1.(a) Define mono crystalline, polycrystalline and amorphous materials. Marks: 3

1. Theory

1. Order in a solid (Core Idea)

The atoms of a solid can be arranged with more or less regularity. Two kinds of order are distinguished:

- **Short-range order:** each atom has about the same number of neighbours at about the same distances (set by its bonds).
- **Long-range order:** the pattern repeats exactly over very many atoms, so the position of a far-away atom can be predicted.

How far the order extends sorts solids into three types (Neamen, Section 1.2).

2. Answer

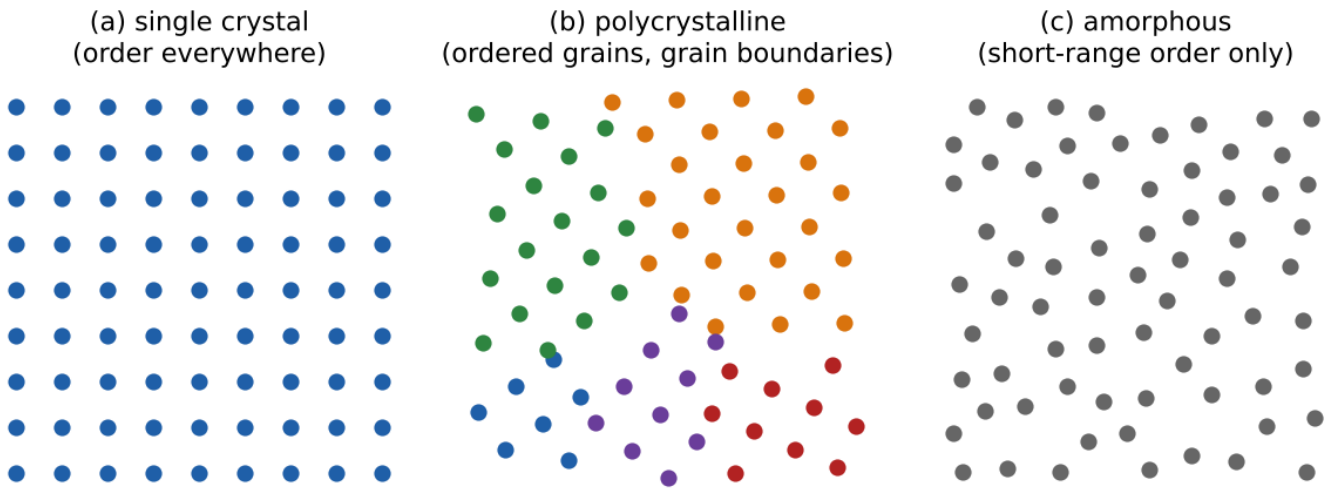


Figure 1: Arrangement of atoms in (a) a single crystal, (b) a polycrystalline solid with grains of different orientation (colours), (c) an amorphous solid

Reading the figure. (a) One regular lattice fills the whole region. (b) Several small regions (grains,

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each in its own colour) are each perfectly ordered, but their lattices point in different directions; they meet at grain boundaries. (c) The atoms keep roughly equal spacing to their neighbours, but there is no repeating pattern.

- **Single-crystal (monocrystalline) material:** a high degree of order (a regular, repeating arrangement of atoms) extends through the **entire** volume of the material. Example: the Si and GaAs wafers used for devices. Electrons move with the highest mobility because nothing breaks the lattice.
- **Polycrystalline material:** made of many small single-crystal regions (**grains**), each ordered, but with **different orientations**; the regions meet at **grain boundaries**, where the order is broken. Example: polysilicon gates and resistors, thin-film solar cells. The grain boundaries scatter carriers and trap charge.
- **Amorphous material:** order only over a few atomic distances (**short-range order**), **no long-range order** at all. Example: amorphous silicon (a-Si:H) in thin-film transistors of displays, glass, SiO_2 films. It has band tails and a low mobility.

Type	Order	Example
Single crystal	long-range, throughout	Si wafer
Polycrystalline	long-range within grains only	polysilicon
Amorphous	short-range only	a-Si:H, SiO_2

Table 1: Summary

3. Keyword Explanation

1. Grain boundary

↩ back

The surface where two crystal grains of different orientation meet; a planar defect.

2. Long-range order

↩ back

A periodic arrangement that repeats over distances very large compared with the atomic spacing.

4. Notes and Remarks

- Devices with the best performance (transistors, lasers) need single crystals, grown by the Czochralski method and by epitaxy (Neamen, Section 1.7).

1.(b) What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram. Marks: 4

1. Theory

1. Why a mass needs "effective" (Core Idea)

An electron in a crystal moves through the periodic electric potential of the atoms. Its response to an external force F is not that of a free electron of mass m_0. Quantum mechanics shows that it still obeys Newton's law, $F = m^*a$, if we use an **effective mass** m^* that includes the effect of the crystal. The effective mass is set by the shape of the **E–k diagram** (energy against wave number k).

2. Free electron as the reference

A free electron has momentum $p = \hbar k$ and energy

$$E = \frac{p^2}{2m_0} = \frac{\hbar^2 k^2}{2m_0}$$

a parabola in k. Its second derivative is $d^2E/dk^2 = \hbar^2/m_0$: the mass is hidden in the **curvature** of the E–k curve.

2. Answer

Effective mass: the parameter m^* that relates the acceleration of an electron (or hole) in a crystal to an external force, $F = m^*a$, taking into account the internal forces of the lattice; it is defined from the curvature of the E–k diagram:

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$$

Mathematical verification (Neamen, Section 3.2.2):

Step 1: Band edge as a parabola

Near the bottom of the conduction band (at $k = 0$) any smooth $E(k)$ can be expanded in a Taylor series:

$$E(k) = E_c + \left(\frac{dE}{dk}\right)_{k=0} k + \frac{1}{2}\left(\frac{d^2E}{dk^2}\right)_{k=0} k^2 + \ldots$$

At a minimum $dE/dk = 0$, so near the edge $E - E_c \approx \frac{1}{2}(d^2E/dk^2)k^2$.

Step 2: Compare with the free particle

A free particle has $E = \hbar^2k^2/2m$. Matching the two parabolas gives

$$\frac{1}{2}\frac{d^2E}{dk^2} = \frac{\hbar^2}{2m^*} \quad \Rightarrow \quad \boxed{\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}}$$

Step 3: Check with Newton's law

The electron's group velocity is $v_g = \frac{1}{\hbar}\frac{dE}{dk}$. A force F changes the crystal momentum as $\hbar\, dk/dt = F$. Then

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d^2E}{dk^2}\frac{dk}{dt} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}F = \frac{F}{m^*}$$

which is Newton's law with the same m^*: the definition is consistent.

Step 4: Method from a measured or computed E–k diagram

1. Read points (k_i, E_i) near the band extremum.
2. Fit a parabola $E = E_c + Ck^2$ (or take the second difference $d^2E/dk^2 \approx (E_{i+1} - 2E_i + E_{i-1})/\Delta k^2$ at the extremum).
3. Then $m^* = \hbar^2/(2C) = \hbar^2/(d^2E/dk^2)$. A **sharply curved** band means a **small** m^*; a flat band a large one. At a band maximum (valence band) $d^2E/dk^2 < 0$, a negative electron mass, which is described as a hole of positive mass.

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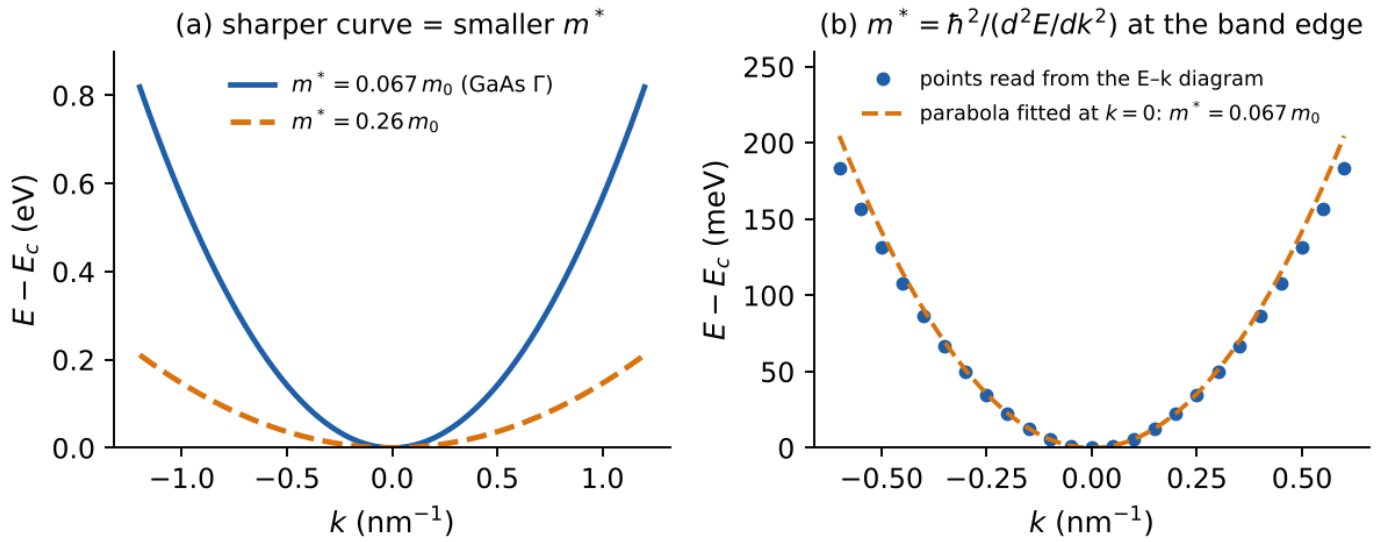


Figure 2: (a) Two parabolic bands with different effective masses; (b) the effective mass found from points of a GaAs-like E-k diagram by fitting a parabola at the band edge

Reading the figure. Panel (a): the solid blue parabola ($m^* = 0.067\,m_0$, like GaAs) rises much more steeply than the dashed orange one ($m^* = 0.26\,m_0$): at the same k the light electron has more energy. Panel (b): blue dots are points of a (slightly non-parabolic) GaAs conduction band; the second difference at $k = 0$ gives d^2E/dk^2 and the dashed orange parabola with $m^* = 0.067\,m_0$, which fits well near the edge and less well further out.

3. Keyword Explanation

1. Curvature

↩ back

The second derivative d^2E/dk^2; large curvature means a light carrier.

2. Group velocity

↩ back

$v_g = (1/\hbar)\,dE/dk$: the speed of the electron wave packet.

4. Notes and Remarks

- In anisotropic bands (Si X valleys) the curvature depends on direction, giving m_l and m_t; then $1/m^*$ is a tensor.
- Experimentally m^* is measured by cyclotron resonance: $\omega_c = eB/m^*$.

1.(c) How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap. Marks: 9

1. Theory

1. The alloy $GaAs_{1-x}P_x$ (Core Idea)

$GaAs_{1-x}P_x$ is a ternary alloy: on the group-V sites a fraction x of the atoms are P and $1 - x$ are As. At $x = 0$ it is GaAs (1.42 eV, direct gap); at $x = 1$ GaP (2.26 eV, indirect gap). Changing x tunes the band gap and the colour of an LED, $\lambda(\mu\text{m}) = 1.24/E_g(\text{eV})$.

2. Two valleys, two gaps

The conduction band has a Γ valley at $k = 0$ and X valleys at the zone edge. The **direct gap** E_Γ is from the valence-band top (at $k = 0$) to Γ; the **indirect gap** E_X is to X. The band gap is the smaller of the two. A direct gap gives efficient light emission (vertical transition, no phonon needed); an indirect gap needs a phonon and gives very weak emission.

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3. Isoelectronic trap

An **isoelectronic** impurity belongs to the same column of the periodic table as the atom it replaces. Nitrogen replacing P in GaAsP is neither donor nor acceptor, but because N is small and strongly electronegative it creates a short-range potential that can capture an electron: an **isoelectronic trap**, with a level just below E_c (about 10 meV in GaP). (The paper writes "iso-electric trap".)

2. Answer

Variation of the energy gap with mole fraction (Sze & Ng; Neamen gives the crossover at $x \approx 0.45$):

$$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2 \text{ eV}, \qquad E_X(x) = 1.900 + 0.144x + 0.211x^2 \text{ eV}$$

- Both gaps grow as P replaces As (smaller lattice constant, stronger bonds).
- E_Γ rises fast (1.33 eV from GaAs to GaP), E_X slowly (0.36 eV).
- They cross at $x_c \approx 0.45$–0.48, $E_g \approx 2.0$ eV. For $x < x_c$ the alloy is **direct** ($E_g = E_\Gamma$); for $x > x_c$ it is **indirect** ($E_g = E_X$).

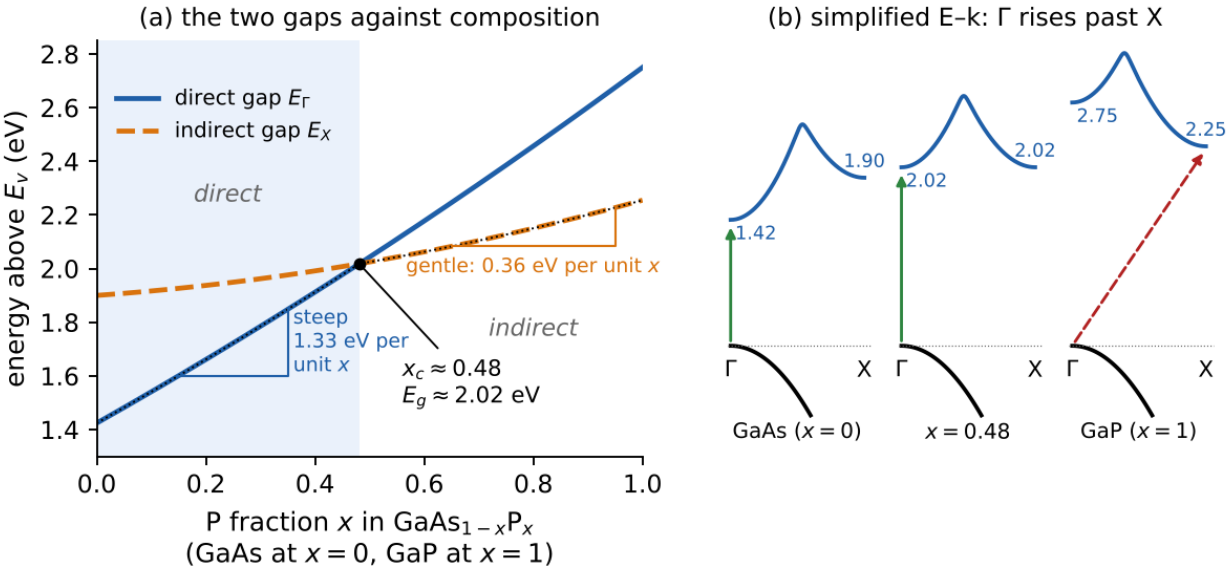


Figure 3: Direct and indirect gaps of $GaAs_{1-x}P_x$ against x and simplified E-k diagrams for GaAs, the crossover and GaP (reused from the 2021–22 paper)

Reading the figure. Panel (a): the solid blue line is the direct gap, the dashed orange curve the indirect gap; the lower one (dotted black) is the band gap; they cross at $x_c \approx 0.48$ and the shaded part is direct. Panel (b): simplified E–k diagrams (conduction band solid blue, valence band solid black). In GaAs the lowest conduction state is at Γ and the transition is vertical (solid green arrow); at the crossover both valleys are at the same height; in GaP the X valley is lowest and the transition is slanted (dashed red arrow), needing a phonon.

Light-emission efficiency without an isoelectronic trap:

- $x < 0.45$ (direct): electrons at the Γ minimum recombine with holes at $k = 0$ directly; radiative recombination is fast and the efficiency is high. Emission from infrared (0.87 µm at $x = 0$) to red (about 0.65 µm at $x = 0.4$): $GaAs_{0.6}P_{0.4}$ red LEDs.
- $x > 0.45$ (indirect): electrons collect in the X valleys at large k; recombination with holes at $k = 0$ needs a phonon to conserve momentum, so it is slow and non-radiative paths win; the efficiency drops by orders of magnitude. Bright orange, yellow and green LEDs are impossible.

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With an isoelectronic trap (nitrogen):

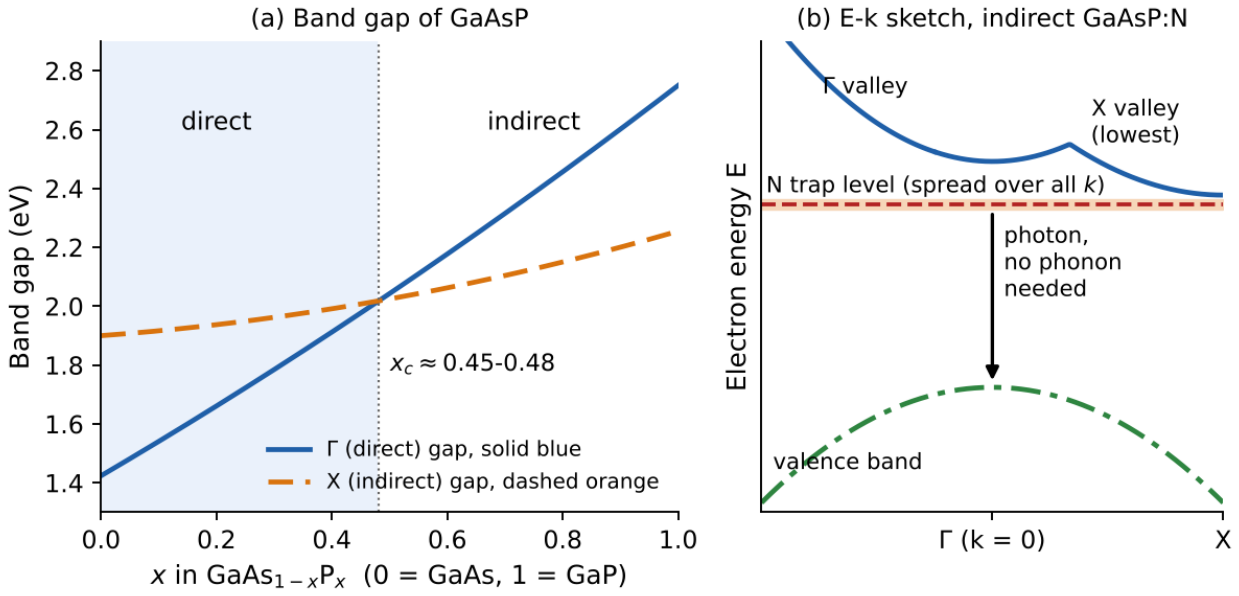


Figure 4: E-k sketch of indirect GaAsP with the nitrogen trap level just below the conduction band (reused from the 2020–21 paper)

Reading the figure. Panel (a) repeats the two gaps against x. Panel (b) is an indirect alloy: the conduction band (solid blue) has its minimum at X, the valence band (dash-dot green) its maximum at $k = 0$, and the dashed red line is the nitrogen level just below E_c, drawn across all k: the trapped electron can recombine vertically.

- The N trap captures an electron and holds it within a very small distance Δx. By the uncertainty principle, $\Delta x\,\Delta k \gtrsim 1$, the trapped electron's wavefunction spreads over a wide range of k, including $k = 0$.
- The trapped (negative) electron then attracts a hole: a **bound exciton**, which recombines **without a phonon**, vertically in k.
- So indirect GaAsP:N (and GaP:N) emits efficiently, at an energy slightly below E_g: orange-red $GaAs_{0.35}P_{0.65}$:N (about 632 nm), yellow $GaAs_{0.15}P_{0.85}$:N, green GaP:N (about 565 nm).

Summary: without N, high efficiency only for $x < 0.45$; with N, efficient emission also for $x > 0.45$, extending GaAsP LEDs from red to yellow and green.

3. Keyword Explanation

1. Direct–indirect crossover

↩ back

The composition at which the lowest conduction-band valley changes from Γ to X.

2. Bound exciton

↩ back

An electron and a hole held together by their attraction at an impurity; it recombines efficiently.

4. Notes and Remarks

- The fits cross at $x = 0.48$ ($E_g = 2.02$ eV); Neamen and many texts quote $x \approx 0.45$.
- The N level in GaAsP lies deeper below E_c as x decreases; near the crossover the trap merges with the band and its benefit fades.
- Also treated in 2(c) of the 2020–21 paper and 7(a) of the 2021–22 paper (extra reading).

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1.(d) For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system. [Marks: 7+2]

1. Theory

1. Quaternary alloys and Vegard's law (Core Idea)

$In_xGa_{1-x}As_ySb_{1-y}$ mixes two group-III atoms (In fraction x, Ga $1-x$) and two group-V atoms (As fraction y, Sb $1-y$). Two independent compositions let us choose *two* properties at once, typically the lattice constant (to suit the substrate) and the band gap (to suit the wavelength). The simplest model, **Vegard's law**, interpolates linearly between the four binary corners:

$$P(x,y) = xy\,P_{\text{InAs}} + x(1-y)P_{\text{InSb}} + (1-x)y\,P_{\text{GaAs}} + (1-x)(1-y)P_{\text{GaSb}}$$

for the lattice constant a and (approximately, ignoring bowing) the band gap E_g.

2. Absorption edge and band gap

A photon is absorbed only if its energy exceeds the band gap, $h\nu > E_g$. Light of wavelength shorter than λ_c is absorbed and longer is transmitted, so

$$E_g = \frac{hc}{\lambda_c}, \qquad E_g(\text{eV}) = \frac{1.24}{\lambda_c(\mu\text{m})}$$

3. Misfit and critical thickness

The **lattice mismatch** of a layer (lattice constant a_e) on a substrate (a_s) is

$$f = \frac{a_e - a_s}{a_s}$$

A thin mismatched layer can grow **strained**, taking the substrate's lattice spacing in the plane. The strain energy grows with thickness; above a **critical thickness** h_c it is cheaper to form misfit dislocations, and the layer relaxes with defects. Sze & Ng give the empirical estimate

$$h_c \approx \frac{a_s}{2|f|}$$

(for $f = 2$ % and $a_s = 5$ Å about 10 nm).

2. Answer

Given (paper's table): a: InAs 6.06, GaAs 5.65, InSb 6.48, GaSb 6.09 Å; E_g: InAs 0.38, GaAs 1.43, InSb 0.18, GaSb 0.7 eV. Unstrained layer $a = 5.93$ Å; cut-off $\lambda_c = 1243$ nm; substrate InP, $a_s = 5.8687$ Å (Sze & Ng).

Step 1: Required band gap

$$E_g = \frac{1.24}{1.243} = 0.998 \approx 1.0 \text{ eV}$$

Step 2: Vegard expressions

Expanding the bilinear interpolation:

$$a(x,y) = 6.09 + 0.39x - 0.44y + 0.02xy \text{ Å}$$

$$E_g(x,y) = 0.70 - 0.52x + 0.73y - 0.53xy \text{ eV}$$

(constant = GaSb; coefficient of x = InSb − GaSb; of y = GaAs − GaSb; of xy = InAs − InSb − GaAs + GaSb).

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Step 3: Lattice condition

$a = 5.93$ Å gives

$$y = \frac{6.09 + 0.39x - 5.93}{0.44 - 0.02x} = \frac{0.16 + 0.39x}{0.44 - 0.02x}$$

At $x = 0$: $y = 0.364$; y reaches 1 at $x = 0.683$ (the InAs–GaAs edge).

Step 4: Band gap along the lattice line

Substituting $y(x)$ into $E_g(x, y)$:

x	**0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.683**
y	0.364	0.454	0.546	0.638	0.731	0.826	1.000
E_g (eV)	0.965	0.956	0.937	0.908	0.871	0.824	0.713

Table 2: Summary

The band gap is largest at $x = 0$ and falls with x: **along $a = 5.93$ Å it is never more than 0.965 eV**, so no alloy of the table reaches $E_g = 1.0$ eV exactly.

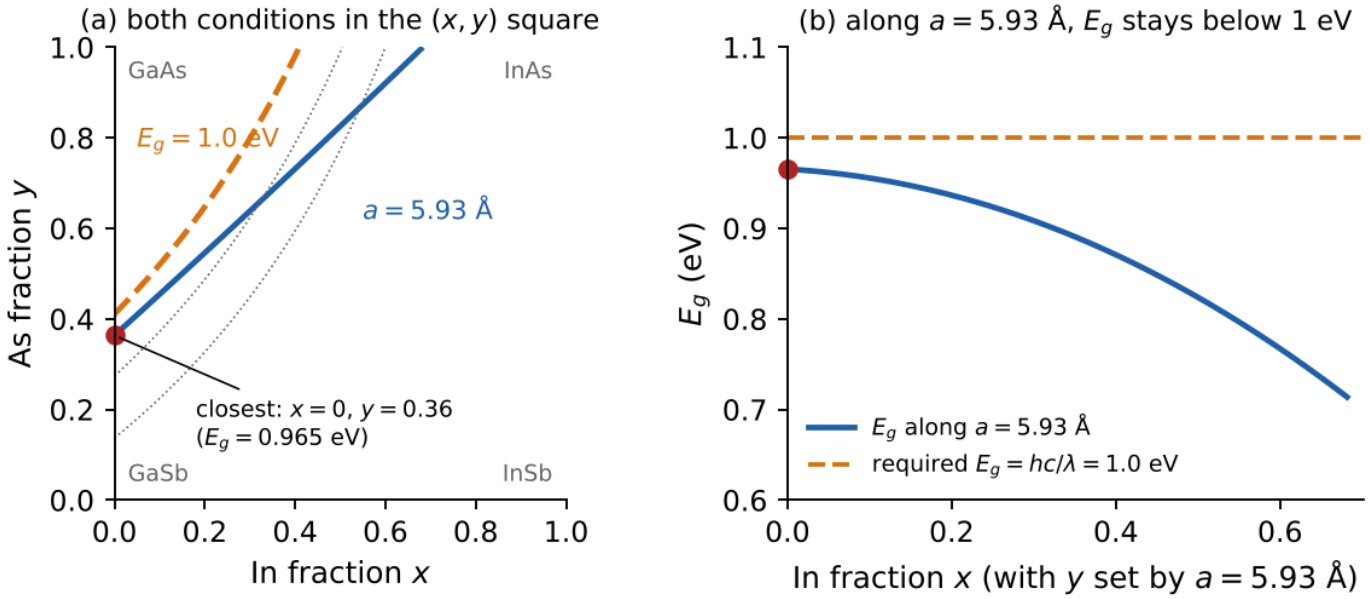


Figure 5: The $a = 5.93$ Å line and the $E_g = 1.0$ eV line in the composition square (they do not meet), and E_g along the lattice line

Reading the figure. Panel (a): the solid blue curve is every (x, y) with $a = 5.93$ Å; the dashed orange curve every (x, y) with $E_g = 1.0$ eV (dotted grey: 0.9 and 0.8 eV). The two never cross inside the square; they come closest at $x = 0$ (red dot). Panel (b): E_g along the blue curve (solid blue) stays below the required 1.0 eV (dashed orange); its maximum, 0.965 eV, is at $x = 0$.

Step 5: Closest composition

The composition nearest to both requirements is the end of the lattice line:

$$\boxed{x = 0, \quad y = 0.36}$$

that is $GaAs_{0.36}Sb_{0.64}$ (no In), with $a = 5.93$ Å and $E_g = 0.965$ eV ($\lambda_c = 1.28$ µm), 0.035 eV short of the target.

Step 6: Critical thickness on InP

$$f = \frac{5.93 - 5.8687}{5.8687} = 0.01045 = 1.04\ \%$$

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$$h_c \approx \frac{a_s}{2f} = \frac{5.8687}{2(0.01045)} = 281 \text{ Å}$$

$$\boxed{h_c \approx 28 \text{ nm}}$$

Below about 28 nm the layer grows pseudomorphically (strained, defect-free); thicker layers relax with misfit dislocations.

Final Answer: with the table's values, $a = 5.93$ Å and $E_g = 1.0$ eV cannot both be met; the closest alloy is $x = 0$, $y = 0.36$ ($GaAs_{0.36}Sb_{0.64}$, $E_g = 0.965$ eV). Critical thickness on InP: $f = 1.04$ %, $h_c \approx 28$ nm.

3. Keyword Explanation

1. Vegard's law

↩ back

Linear interpolation of a property (here the lattice constant) between the end compounds of an alloy.

2. Pseudomorphic layer

↩ back

A strained layer that takes the substrate's in-plane lattice constant; possible only below h_c.

4. Notes and Remarks

- **Note on the data:** with the corner values given in the table, the equations have no solution with $0 \le x, y \le 1$ (and none at all); the paper's 1.0 eV is just above the largest gap on the 5.93 Å line. Real InGaAsSb band gaps also bow downwards (lower than the straight line), which makes 1 eV even harder; the closest answer is given as the user chose (2026-11-13).
- **Matthews–Blakeslee** (force balance on a 60° dislocation, $b = a/\sqrt{2}$, $\nu = 1/3$) gives a larger $h_c \approx 50$ nm for $f = 1.04$ %; the empirical $a_s/2f$ is an order-of-magnitude estimate.
- A similar problem, with the band gaps read from a chart, is 2(d) of the 2020–21 paper (extra reading).

5. Final Concise Answer

(a) Single crystal: long-range order throughout (Si wafer). Polycrystalline: ordered grains of different orientation separated by grain boundaries (polysilicon). Amorphous: short-range order only (a-Si:H, SiO_2).

(b) Effective mass: $F = m^*a$ for a carrier in a crystal; from the E–k diagram $1/m^* = (1/\hbar^2)d^2E/dk^2$. Verification: Taylor expansion near the band edge matched to $\hbar^2k^2/2m^*$, and $a = (1/\hbar)(d^2E/dk^2)(dk/dt)$ with $\hbar\, dk/dt = F$ gives $a = F/m^*$. Method: fit a parabola to E–k points at the extremum.

(c) $E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ eV; direct for $x < 0.45$, indirect above. Without N: efficient only below 0.45; above, phonon-assisted recombination makes it weak. With the N isoelectronic trap: localised electron spread in k, bound exciton, phonon-free recombination, efficient emission for $x > 0.45$ (orange, yellow, green).

(d) $E_g = 1.24/1.243 = 1.0$ eV; Vegard: $a = 6.09 + 0.39x - 0.44y + 0.02xy$, $E_g = 0.70 - 0.52x + 0.73y - 0.53xy$; on $a = 5.93$ Å, $E_g \le 0.965$ eV, so the closest is $x = 0$, $y = 0.36$ ($GaAs_{0.36}Sb_{0.64}$). $f = 1.04$ % on InP, $h_c \approx a_s/2f \approx 28$ nm.

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2. (a) What are the assumptions made in Anderson's model? Determine ΔE_C, ΔE_V and V_{bi} for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, n_i = [unclear: 2.5×10^{13}] cm^{-3} (for Ge), $N_d = 1.2 \times 10^{16}$ cm^{-3}, and $N_a = 1 \times 10^{16}$ cm^{-3}. [Marks: 2+5]

(b) Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for:

(i) [unclear: N^+]-AlGaAs, intrinsic GaAs.
(ii) P^+-AlGaAs, n-GaAs.

Assume E_g of AlAs = 2.1 eV and $\Delta E_C = \Delta E_V$. [Marks: 4+4]

(c) Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5 \times 10^{15}$ cm^{-3}. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. (N_c = [unclear: 4.7×10^{17}] cm^{-3}, [unclear: ϵ_r] 13.1, $\phi_m = 4.55$, $\chi = 4.07$) Marks: 10

2.(a) What are the assumptions made in Anderson's model? Determine ΔE_C, ΔE_V and V_{bi} for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, n_i = [unclear: 2.5×10^{13}] cm^{-3} (for Ge), $N_d = 1.2 \times 10^{16}$ cm^{-3}, and $N_a = 1 \times 10^{16}$ cm^{-3}. [Marks: 2+5]

1. Theory

1. What Anderson's model does (Core Idea)

To draw the band diagram of a heterojunction we must know how the bands of the two different materials line up. **Anderson's model** (1962), also called the **electron-affinity rule**, does this using only the bulk properties of each material: its electron affinity χ, band gap E_g, permittivity and doping (Neamen, Section 9.3.2).

2. Vacuum level and electron affinity

The vacuum level E_0 is the energy of an electron at rest just outside the material. The **electron affinity** $e\chi = E_0 - E_c$ is a material constant. Before contact both materials are drawn on the same vacuum level, so their conduction-band edges differ by the difference of their affinities.

2. Answer

Assumptions of Anderson's model:

1. The two semiconductors are **single crystals with an abrupt, perfect interface**: no interface states, no dipoles, no defects (the lattices are matched).
2. The **vacuum level is continuous** across the junction, and the bands of each material keep their bulk electron affinity and band gap right up to the interface.
3. Hence the band offsets are set by the electron affinities alone: $\Delta E_c = e(\chi_1 - \chi_2)$ and $\Delta E_v = \Delta E_g - \Delta E_c$.
4. **Thermal equilibrium**, the depletion approximation, and the bands bend in each material according to Poisson's equation with its own permittivity, as in a pn homojunction; the built-in potential is the difference of the work functions.

Data: $\chi_{Ge} = 4.13$ eV, $\chi_{GaAs} = 4.07$ eV; E_g: Ge 0.67 eV, GaAs 1.43 eV (Neamen's values for this example); n_i(Ge) $= 2.5 \times 10^{13}$ cm^{-3}; $N_d = 1.2 \times 10^{16}$ cm^{-3} (n-Ge), $N_a = 10^{16}$ cm^{-3} (p-GaAs); N_v: Ge 6.0×10^{18}, GaAs 7.0×10^{18} cm^{-3} (Neamen, not given in the paper); $kT = 0.0259$ eV.

Step 1: Conduction-band offset

$$\Delta E_c = e(\chi_{Ge} - \chi_{GaAs}) = 4.13 - 4.07 = 0.06 \text{ eV}$$

Step 2: Valence-band offset

$$\Delta E_v = \Delta E_g - \Delta E_c = (1.43 - 0.67) - 0.06 = 0.70 \text{ eV}$$

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Step 3: Built-in potential

For an nP heterojunction (Neamen, Eq. 9.39):

$$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{p0}}{p_{n0}} \cdot \frac{N_{vn}}{N_{vP}}\right)$$

with $p_{p0} = N_a = 10^{16}$ cm^{-3} and $p_{n0} = n_i^2/N_d = (2.5 \times 10^{13})^2/(1.2 \times 10^{16}) = 5.21 \times 10^{10}$ cm^{-3}:

$$eV_{bi} = 0.70 + 0.0259 \ln\left[\frac{(10^{16})(6 \times 10^{18})}{(5.21 \times 10^{10})(7 \times 10^{18})}\right] = 0.70 + 0.0259 \times 12.01 = 1.011 \text{ eV}$$

$$\boxed{\Delta E_c = 0.06 \text{ eV}, \quad \Delta E_v = 0.70 \text{ eV}, \quad V_{bi} = 1.01 \text{ V}}$$

Check with the work functions: $\phi(\text{n-Ge}) = 4.13 + 0.0259 \ln(1.04 \times 10^{19}/1.2 \times 10^{16}) = 4.305$ V and $\phi(\text{p-GaAs}) = 4.07 + 1.42 - 0.0259 \ln(7 \times 10^{18}/10^{16}) = 5.320$ V give $V_{bi} = 1.015$ V, the same to within the rounding of the band gaps.

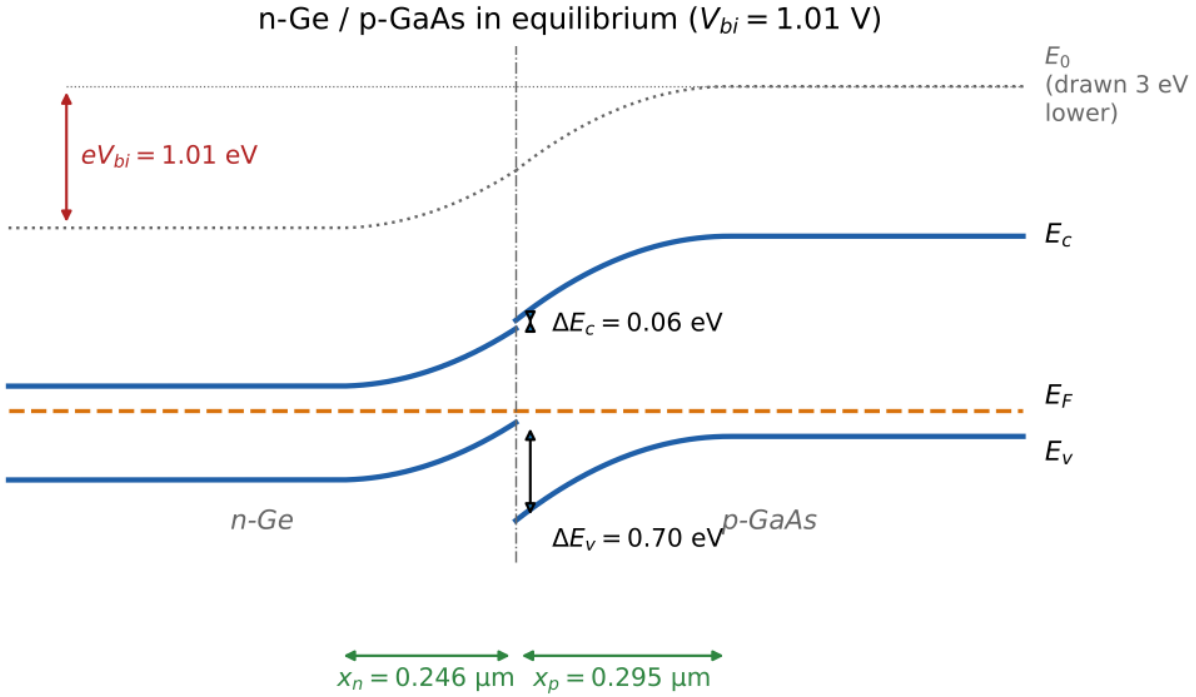


Figure 6: n-Ge / p-GaAs heterojunction in equilibrium with the computed offsets and depletion widths (vacuum level drawn 3 eV lower)

Reading the figure. The dashed orange E_F is flat; the bands (solid blue) bend up from the n-Ge (left) to the p-GaAs (right). At the interface E_c steps by only 0.06 eV but E_v drops by 0.70 eV. The dotted vacuum level (drawn lower to fit) rises by $eV_{bi} = 1.01$ eV (red arrow). The depletion widths are $x_n = 0.246$ µm and $x_p = 0.295$ µm (green double-headed arrows), from Neamen's Eq. 9.47 with $\epsilon_r = 16$ and 13.1.

3. Keyword Explanation

1. Electron-affinity rule

↩ back

$\Delta E_c = e(\chi_1 - \chi_2)$: the conduction-band offset of Anderson's model.

2. Band offsets

↩ back

The steps ΔE_c and ΔE_v in the band edges at a heterojunction.

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4. Notes and Remarks

- **Data choices:** the paper gives $n_i = 2.5 \times 10^{13}$ cm^{-3} for Ge (faint on the scan; Neamen 2.4); N_v values are Neamen's. With $n_i = 2.4 \times 10^{13}$, $V_{bi} = 1.013$ V.
- Measured offsets often differ from Anderson's rule because of interface dipoles and states.
- The same junction is 2(d) of the 2018–19 paper (extra reading).

2.(b) Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for: (i) [unclear: N^+]-AlGaAs, intrinsic GaAs. (ii) P^+-AlGaAs, n-GaAs. Assume E_g of AlAs = 2.1 eV and $\Delta E_C = \Delta E_V$. [Marks: 4+4]

1. Theory

1. Rules for an equilibrium heterojunction diagram (Core Idea)

1. E_F is flat everywhere.
2. Far from the junction each material has its bulk band positions relative to E_F (N^+: E_F at or above E_c; P^+: at or below E_v; intrinsic: near mid-gap; n: just below E_c).
3. The vacuum level is continuous; E_c and E_v follow it on each side.
4. At the interface E_c and E_v jump by ΔE_c and ΔE_v.

2. The AlGaAs band gap

With Vegard's law between GaAs (1.42 eV) and AlAs (2.1 eV, as given):

$$E_g(\mathrm{Al_{0.3}Ga_{0.7}As}) = 0.7(1.42) + 0.3(2.1) = 1.624 \text{ eV}$$

so $\Delta E_g = 0.204$ eV and, with $\Delta E_c = \Delta E_v$ (given), $\Delta E_c = \Delta E_v = 0.102$ eV.

2. Answer

Offsets: $\Delta E_g = 1.624 - 1.42 = 0.204$ eV; $\Delta E_c = \Delta E_v = 0.102$ eV; the AlGaAs gap straddles the GaAs gap (E_c higher, E_v lower in AlGaAs).

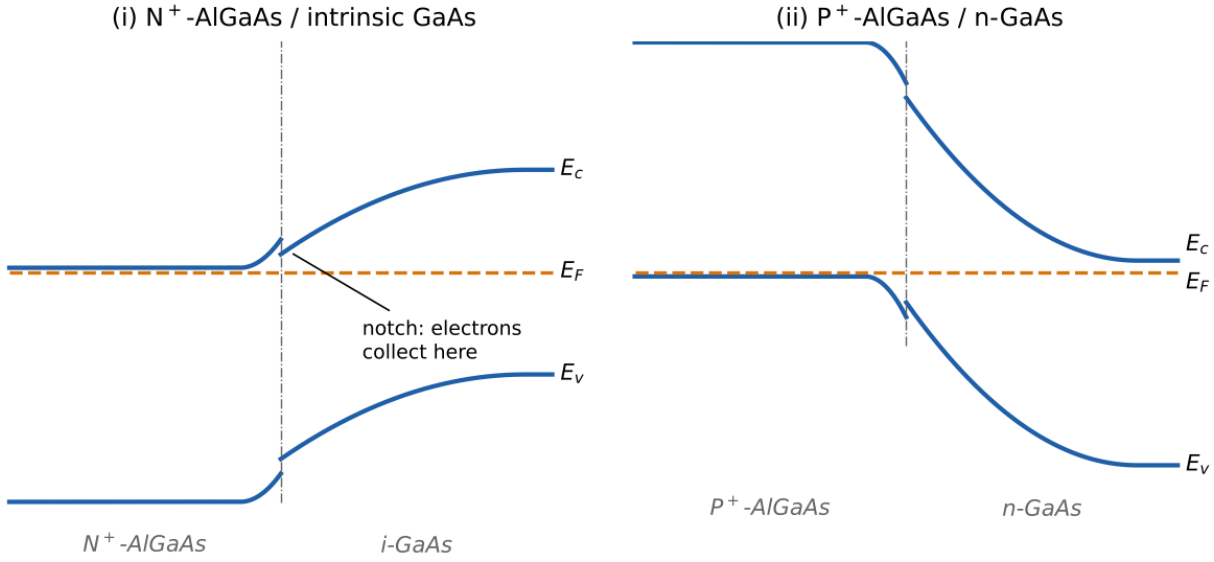


Figure 7: Equilibrium band diagrams: (i) N^+-AlGaAs / intrinsic GaAs, (ii) P^+-AlGaAs / n-GaAs (solid blue E_c, E_v; dashed orange E_F)

Reading the figure. In each panel the AlGaAs is on the left and the GaAs on the right; the dash-dot line is the interface.
(i) N^+-AlGaAs / intrinsic GaAs:

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- Before contact the N^+ AlGaAs has E_F at E_c; the intrinsic GaAs has E_F near mid-gap, much lower. Electrons flow from the AlGaAs into the GaAs.
- The AlGaAs next to the interface is depleted (thin, because heavily doped); the GaAs bands bend **down towards the interface** over a long distance (undoped, little charge).
- At the interface E_c drops by $\Delta E_c = 0.102$ eV into the GaAs, forming a **notch** (triangular well) where the transferred electrons collect: a two-dimensional electron gas, the basis of the HEMT.
- E_v steps up by 0.102 eV going into the GaAs.

(ii) P^+-AlGaAs / n-GaAs:

- The P^+ AlGaAs has E_F near E_v; the n-GaAs has E_F near E_c. Holes flow into the GaAs and electrons into the AlGaAs until E_F is flat.
- Almost all the band bending (about E_g, the built-in potential) lies in the **lightly doped n-GaAs**, which is depleted over a wide region; the P^+ side bends only a little.
- At the interface E_c jumps down by ΔE_c going into the GaAs (no spike blocks electrons in the GaAs) and E_v steps *up* by ΔE_v, which forms a small notch for holes at the GaAs side and a barrier for electrons from GaAs into AlGaAs: a Pn heterojunction diode (as used in the emitter of a pnp HBT).

3. Keyword Explanation

1. Two-dimensional electron gas

↩ back

Electrons confined in the notch at the interface, free to move only in the plane.

2. Straddling alignment

↩ back

The narrow gap lies inside the wide gap: E_c higher and E_v lower in the wide-gap material.

4. Notes and Remarks

- **Unclear on the scan:** the first item is faint; it is read as "N^+-AlGaAs, intrinsic GaAs", the pair to item (ii), and the classic modulation-doped structure.
- The paper's $\Delta E_c = \Delta E_v$ is a simplification; measured GaAs/AlGaAs offsets are about $0.6\,\Delta E_g$ and $0.4\,\Delta E_g$ (Sze & Ng). AlAs is indirect; its direct gap is about 3.0 eV, but 2.1 eV is used as given.

2.(c) Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5 \times 10^{15}$ cm^{-3}. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. (N_c = [unclear: 4.7×10^{17}] cm^{-3}, [unclear: ϵ_r] 13.1, $\phi_m = 4.55$, $\chi = 4.07$) Marks: 10

1. Theory

1. Ideal Schottky barrier (Core Idea)

An ideal metal–n-semiconductor contact with $\phi_m > \phi_s$ forms a barrier $\phi_{B0} = \phi_m - \chi$ for electrons in the metal. In the semiconductor the bands bend up by the built-in potential $V_{bi} = \phi_{B0} - \phi_n$, where $\phi_n = (kT/e)\ln(N_c/N_d)$ is the depth of E_F below E_c in the bulk (Neamen, Section 9.1).

2. Field and width

With the depletion approximation, Poisson's equation gives a field that falls linearly from the metal to the edge x_n:

$$E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x), \qquad x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}, \qquad |E_{max}| = \frac{eN_d x_n}{\epsilon_s}$$

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2. Answer

Given: $\phi_m = 4.55$ V (tungsten), $\chi = 4.07$ V (GaAs), $N_d = 5 \times 10^{15}$ cm^{-3}, $N_c = 4.7 \times 10^{17}$ cm^{-3}, $\epsilon_r = 13.1$ ($\epsilon_s = 1.159 \times 10^{-12}$ F/cm), $T = 300$ K.

Step 1: Barrier height

$$\phi_{B0} = \phi_m - \chi = 4.55 - 4.07 = 0.48 \text{ V}$$

Step 2: Fermi level below E_c

$$\phi_n = 0.0259 \ln\left(\frac{4.7 \times 10^{17}}{5 \times 10^{15}}\right) = 0.0259 \times 4.543 = 0.1177 \text{ V}$$

Step 3: Built-in potential

$$V_{bi} = 0.48 - 0.1177 = 0.3623 \text{ V}$$

Step 4: Depletion width

$$x_n = \left[\frac{2(1.159 \times 10^{-12})(0.3623)}{(1.6 \times 10^{-19})(5 \times 10^{15})}\right]^{1/2} = 3.24 \times 10^{-5} \text{ cm} = 0.324\ \mu\text{m}$$

Step 5: Maximum field

$$|E_{max}| = \frac{(1.6 \times 10^{-19})(5 \times 10^{15})(3.24 \times 10^{-5})}{1.159 \times 10^{-12}} = 2.24 \times 10^{4} \text{ V/cm}$$

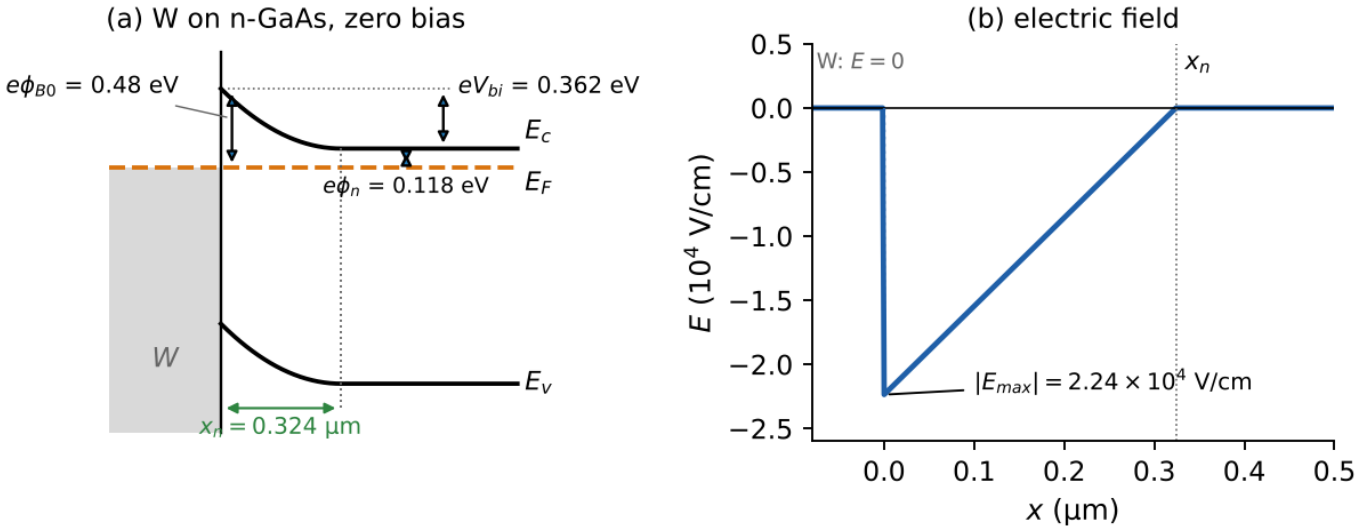


Figure 8: W on n-GaAs at zero bias: band diagram with the values and the electric field

Reading the figure. Panel (a): the GaAs bands (solid black) bend up over $x_n = 0.324$ µm next to the tungsten; the barrier from the metal is 0.48 eV, the built-in barrier 0.362 eV, and E_c is 0.118 eV above E_F (dashed orange) in the bulk. Panel (b): the field (solid blue) is -2.24×10^4 V/cm at the surface and rises linearly to zero at x_n.

Final Answer: $\phi_{B0} = 0.48$ V, $V_{bi} = 0.362$ V, $|E_{max}| = 2.24 \times 10^4$ V/cm (Neamen's Exercise 9.1 answers).

3. Keyword Explanation

1. Schottky barrier height

↩ back

$\phi_{B0} = \phi_m - \chi$: the barrier seen by electrons in the metal.

2. Built-in potential barrier

↩ back

V_{bi}: the band bending in the semiconductor at zero bias.

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4. Notes and Remarks

- **Unclear on the scan:** N_c is overwritten by hand ("47×10^{17}"); it is read as 4.7×10^{17} cm^{-3}, the value for GaAs (Neamen), and the permittivity symbol before 13.1 is faint.
- Real W/GaAs barriers are near 0.8 V because of Fermi-level pinning by interface states.

5. Final Concise Answer

(a) Anderson: perfect abrupt interface without states, continuous vacuum level, bulk χ and E_g up to the interface, so $\Delta E_c = e(\chi_1 - \chi_2)$, $\Delta E_v = \Delta E_g - \Delta E_c$. n-Ge/p-GaAs: $\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV, $V_{bi} = 0.70 + 0.0259 \ln[(10^{16})(6 \times 10^{18})/((5.21 \times 10^{10})(7 \times 10^{18}))] = 1.01$ V.

(b) $E_g(\mathrm{Al_{0.3}Ga_{0.7}As}) = 1.624$ eV, $\Delta E_c = \Delta E_v = 0.102$ eV. (i) $\mathrm{N^+/i}$: electrons transfer to the GaAs, bands bend down towards the interface, a notch (2DEG) forms in the GaAs. (ii) $\mathrm{P^+/n}$: bending mostly in the n-GaAs (depleted), E_c steps down and E_v up into the GaAs.

(c) $\phi_{B0} = 0.48$ V, $\phi_n = 0.118$ V, $V_{bi} = 0.362$ V, $x_n = 0.324$ µm, $|E_{max}| = 2.24 \times 10^4$ V/cm.

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3. (a) Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors. [Marks: 3+3+3]

(b) Discuss the significance of critical thickness in epitaxial growth. Draw enegy [*sic*: should be energy] band structure of Si and GaAs and show optical transitions in such semiconductors. [Marks: 3+3+3]

(c) Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact? Marks: 7

3.(a) Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors. [Marks: 3+3+3]

1. Theory

1. Work function and electron affinity (Core Idea)

The **work function** $e\phi$ is the energy needed to lift an electron from the Fermi level to the vacuum level E_0; the **electron affinity** $e\chi$ is the energy from E_c to E_0. For a semiconductor, $\phi_s = \chi + (E_c - E_F)/e$, which depends on doping.

2. What happens on contact

When a metal and a semiconductor touch, electrons flow from the side with the higher Fermi level (smaller work function) to the other until the Fermi levels are equal. The charge that moves leaves a **depletion layer** (or an accumulation layer) in the semiconductor, and the bands bend. A **rectifying** (Schottky) contact is one where this leaves a **barrier for the majority carriers**: current flows easily in one direction only.

2. Answer

Conditions (ideal contact, no interface states; Neamen, Section 9.1):

1. **n-type semiconductor:** $\phi_m > \phi_s$. Electrons flow from the semiconductor into the metal, leaving positive donors; the bands bend *up* and electrons see a barrier $\phi_{B0} = \phi_m - \chi$ from the metal and $V_{bi} = \phi_m - \phi_s$ from the semiconductor.
2. **p-type semiconductor:** $\phi_m < \phi_s$. Electrons flow from the metal into the semiconductor and fill acceptor states (holes are repelled), leaving negative acceptors; the bands bend *down* and holes see a barrier $\phi_{Bp} = \chi + E_g/e - \phi_m$.
3. In both cases the semiconductor must be **moderately doped** (below about 10^{17}–10^{18} cm^{-3}), so the depletion layer is wide enough that carriers cannot tunnel through it (heavy doping gives an ohmic tunnelling contact), and the interface should be clean (interface states can pin the barrier).

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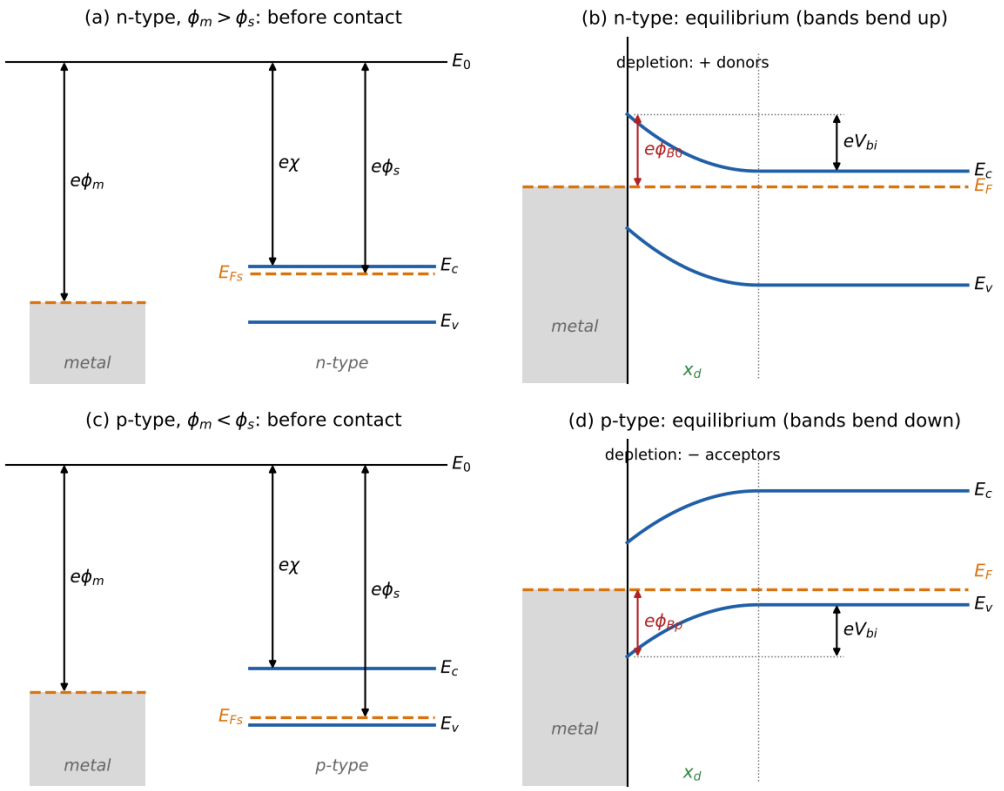


Figure 9: Rectifying contacts: (a) metal and n-type semiconductor with $\phi_m > \phi_s$ before contact, (b) in equilibrium; (c) metal and p-type with $\phi_m < \phi_s$ before contact, (d) in equilibrium

Reading the figure. In (a) and (c) both materials share the vacuum level E_0; the metal's Fermi level (dashed orange, top of the grey block) is $e\phi_m$ below E_0, the semiconductor's E_{Fs} is $e\phi_s$ below it. (a) n-type: E_{Fs} is higher than the metal's. (c) p-type: E_{Fs} is lower. In the equilibrium diagrams (b) and (d) E_F is flat. (b) The n-type bands bend *up* over the depletion width x_d: electrons in the metal face the barrier $e\phi_{B0}$ (red arrow), electrons in the semiconductor face eV_{bi}. (d) The p-type bands bend *down*: holes face the barrier $e\phi_{Bp}$ (red arrow) from the metal and eV_{bi} from the semiconductor.

Key relations:

	n-type	p-type
Condition	$\phi_m > \phi_s$	$\phi_m < \phi_s$
Barrier from the metal	$\phi_{B0} = \phi_m - \chi$	$\phi_{Bp} = \chi + E_g/e - \phi_m$
Built-in potential	$V_{bi} = \phi_m - \phi_s$	$V_{bi} = \phi_s - \phi_m$
Depletion charge	$+eN_d$	$-eN_a$

Table 3: Summary

3. Keyword Explanation

1. Schottky barrier

↩ back

The energy barrier at a rectifying metal–semiconductor contact.

2. Rectifying contact

↩ back

A contact that conducts much more easily in one bias direction than the other.

4. Notes and Remarks

- The opposite conditions ($\phi_m < \phi_s$ on n-type, $\phi_m > \phi_s$ on p-type) give ohmic contacts by accumulation.

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- For a given metal on n- and p-type material, $\phi_{B0} + \phi_{Bp} = E_g/e$.

3.(b) Discuss the significance of critical thickness in epitaxial growth. Draw enegy [*sic*: should be energy] band structure of Si and GaAs and show optical transitions in such semiconductors. [Marks: 3+3+3]

1. Theory

1. Epitaxy and lattice mismatch (Core Idea)

In **epitaxy** a thin single-crystal layer grows on a single-crystal substrate and copies its arrangement. If the layer's natural lattice constant a_e differs from the substrate's a_s, the **mismatch** is $f = (a_e - a_s)/a_s$. A thin layer can still grow perfectly by **straining**: it compresses (or stretches) in the plane to match the substrate and distorts in the perpendicular direction (a **pseudomorphic** layer). The stored elastic energy grows with thickness.

2. Critical thickness

Above a **critical thickness** h_c it costs less energy to break the strain with **misfit dislocations** at the interface; the layer then relaxes to its own lattice constant but contains dislocations, some of which thread up through the layer. Sze & Ng's empirical estimate is $h_c \approx a_s/(2|f|)$, so h_c falls roughly as $1/f$.

3. Direct and indirect gaps

The band gap is **direct** when the conduction-band minimum and the valence-band maximum are at the same k (GaAs, both at $k = 0$), and **indirect** when they are at different k (Si: maximum at $k = 0$, minimum near X). A photon carries almost no momentum, so an optical transition is vertical in k unless a phonon supplies the difference.

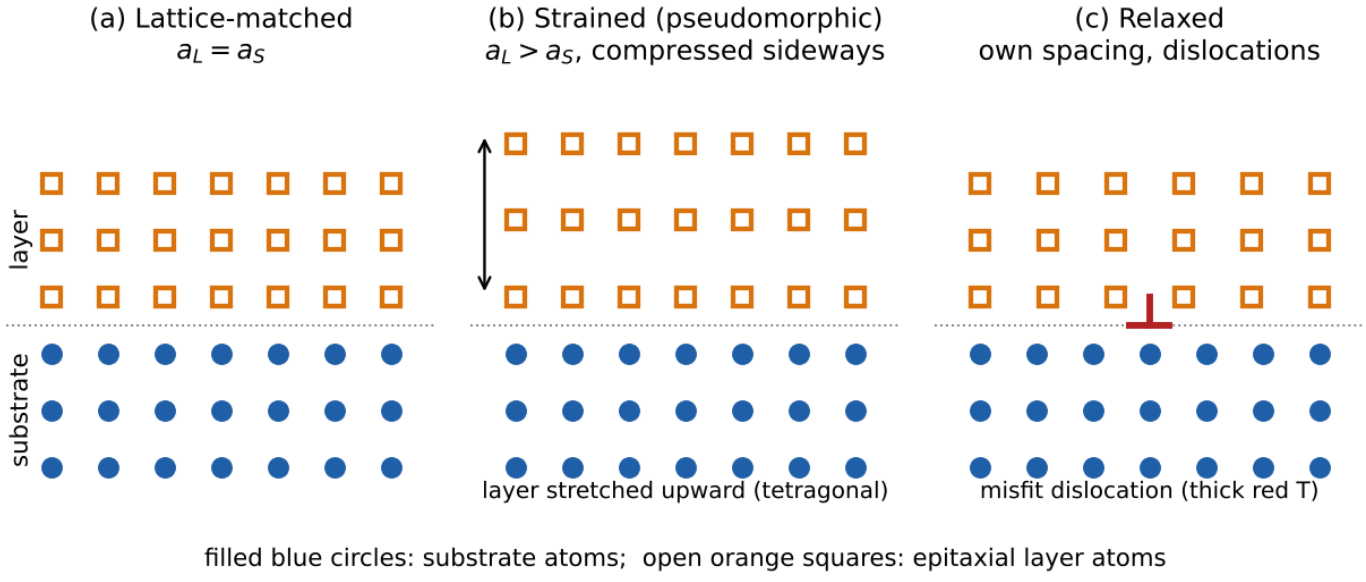


Figure 10: Matched, strained (pseudomorphic) and relaxed layers on a substrate (reused from the 2020–21 paper)

Reading the figure. Left: a lattice-matched layer continues the substrate perfectly. Middle: a thin mismatched layer is squeezed in the plane to fit the substrate and stretched vertically. Right: above h_c the layer relaxes to its own spacing, and a misfit dislocation (the thick red T, an extra half-plane) forms at the interface. Filled blue circles are substrate atoms, open orange squares layer atoms.

2. Answer

Significance of the critical thickness:

1. **It sets the largest defect-free thickness of a mismatched layer.** Below h_c the layer is coherent and strained with no dislocations; above it misfit and threading dislocations form.

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2. **Dislocations harm devices:** they are non-radiative recombination centres (they kill LED and laser efficiency and lifetime), scatter carriers (lower mobility), cause leakage and grow with operation (degradation).
3. **It allows strained-layer devices:** keeping each layer below h_c lets designers use mismatched materials (InGaAs on GaAs in pHEMTs and 980 nm lasers, SiGe on Si in HBTs). The strain itself shifts and splits the bands (lower threshold in strained quantum-well lasers, higher hole mobility).
4. **Design rule:** thick layers must be lattice-matched ($f \approx 0$); mismatched layers must be thinner than $h_c \approx a_s/(2|f|)$ (for example $f = 1$ % on GaAs: $h_c \approx 28$ nm).

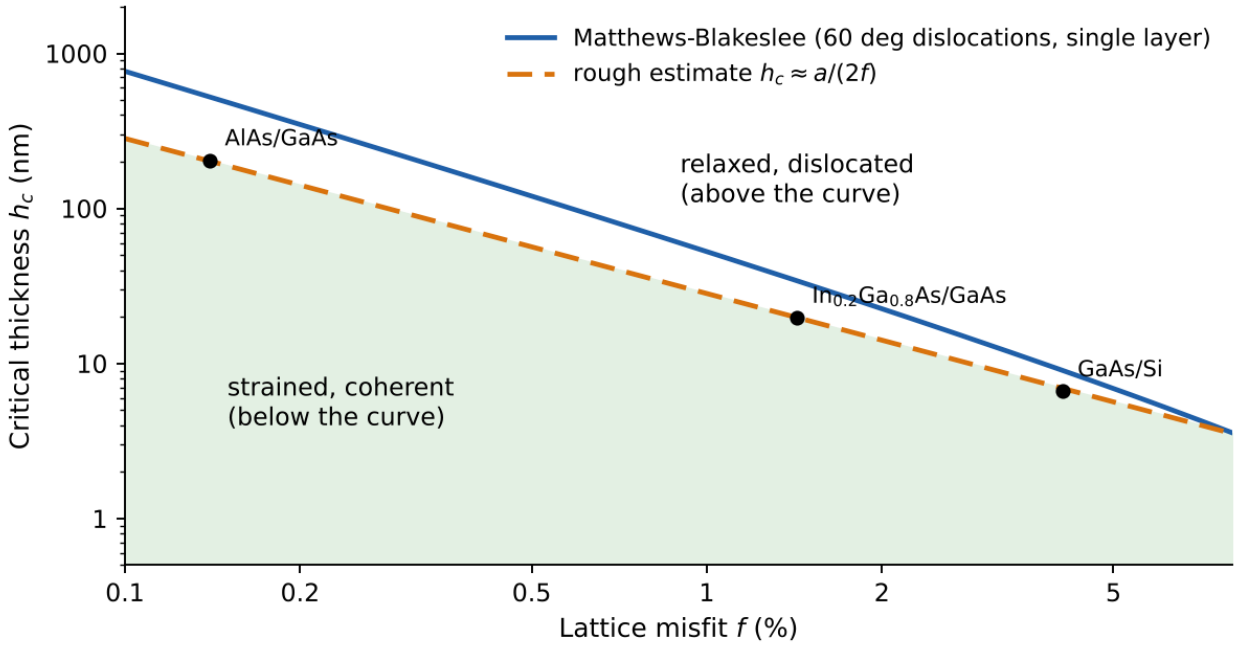


Figure 11: Critical thickness against mismatch (reused from the 2020–21 paper)

Reading the figure. Critical thickness against misfit on log scales: the solid blue curve is the Matthews–Blakeslee model and the dashed orange line the estimate $h_c \approx a/(2f)$. Both fall steeply as the misfit grows: a few per cent allows only a few nanometres of coherent growth. Layers in the shaded region below the curve stay strained and coherent; layers above it relax with dislocations (examples: AlAs/GaAs, $In_{0.2}Ga_{0.8}As$/GaAs, GaAs/Si).

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Energy-band structures and optical transitions:

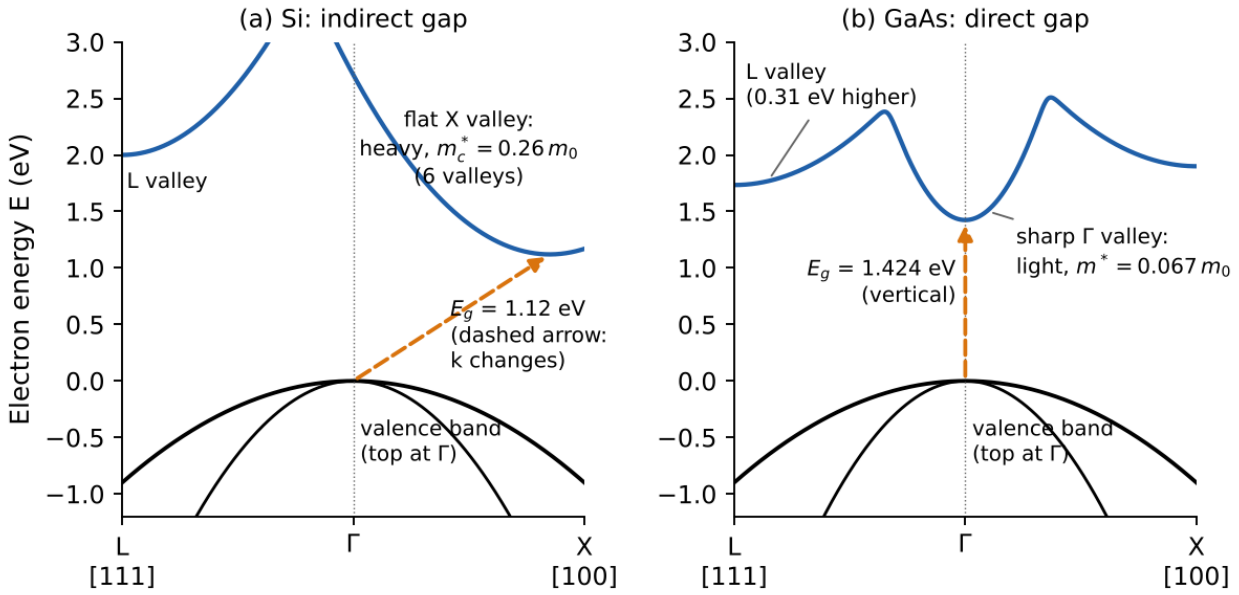


Figure 12: Simplified E-k diagrams of Si (indirect) and GaAs (direct) with the lowest-energy transitions (reused from the 2020–21 paper)

Reading the figure. The conduction band (solid blue) and valence bands (solid black) are drawn from L through Γ to X. In Si (left) the conduction-band minimum is near X and the dashed orange arrow from the valence-band top is slanted: an **indirect** transition. In GaAs (right) the minimum is at Γ and the arrow is vertical: a **direct** transition.

- **GaAs (direct gap, $E_g = 1.42$ eV):** absorption and emission are **vertical** transitions at $k = 0$; only the photon is needed. Absorption rises sharply above E_g ($\alpha \sim 10^4$ cm^{-1}), and radiative recombination is fast (ns): efficient LEDs and lasers at 0.87 µm.
- **Si (indirect gap, $E_g = 1.12$ eV):** the lowest transition goes from Γ to near X, so a **phonon** must be absorbed or emitted to supply the momentum: $h\nu = E_g \pm E_{phonon}$. This second-order process is slow, so absorption near the edge is weak (thick absorbers needed in solar cells) and light emission is very inefficient; direct (vertical) transitions in Si need about 3.4 eV (at Γ).

3. Keyword Explanation

1. Misfit dislocation

↩ back

A line defect at the interface that relieves the strain of a mismatched layer.

2. Phonon-assisted transition

↩ back

An optical transition in an indirect semiconductor in which a lattice vibration supplies the crystal momentum.

4. Notes and Remarks

- The paper writes “enegy” for energy ([sic]).
- Matthews–Blakeslee gives a more exact h_c (force balance on a dislocation); the simple $a_s/2f$ is Sze & Ng’s estimate.

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3.(c) Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact? Marks: 7

1. Theory

1. Width of a Schottky barrier (Core Idea)

At a metal–n-semiconductor contact the depletion layer has width

$$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$$

so it **shrinks as the doping rises**: for Si with $V_{bi} = 0.67$ V it is 0.3 µm at 10^{16} cm^{-3} but only about 11 nm at 7×10^{18} cm^{-3} (Neamen, Example 9.7).

2. Quantum tunnelling

An electron is a wave; it can pass through a potential barrier that is higher than its energy if the barrier is thin (a few nanometres). The tunnelling probability falls exponentially with the barrier width times the square root of its height.

2. Answer

Tunnelling junction: a metal on a **heavily doped** semiconductor (n$^+$, $N_d \gtrsim 10^{19}$ cm^{-3}). A barrier ϕ_{Bn} still forms, but the depletion layer is only a few nanometres thick, so electrons tunnel through it.

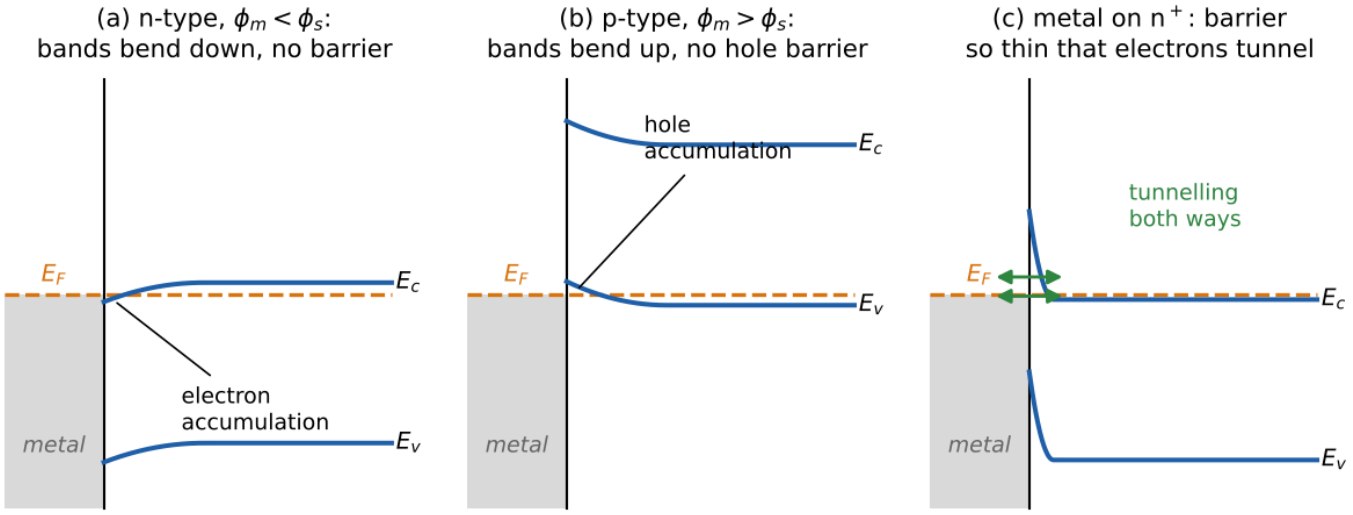


Figure 13: Ohmic contacts; panel (c) is the tunnelling junction: metal on n$^+$ with a very thin barrier (reused from the 2018–19 paper)

Reading the figure. Panel (c): the grey metal on the left, the flat dashed orange E_F and the n$^+$ semiconductor's bands (solid blue). E_F lies at or above E_c in the degenerate semiconductor; at the interface E_c rises into a tall but very thin spike; the green double arrows show electrons tunnelling through it in both directions. (Panels (a) and (b) are the barrier-free ohmic contacts.)

Why it is ohmic:

1. The barrier is thin ($x_n \approx$ 1–10 nm), so **tunnelling (field emission)** dominates over thermionic emission; the current $J_t \propto \exp(-e\phi_{Bn}/E_{00})$ with $E_{00} = (e\hbar/2)\sqrt{N_d/(\epsilon_s m^*)}$ (Neamen, Eqs. 9.29–9.30) is large.
2. Tunnelling works **in both directions**: with a small positive or negative bias, electrons on the higher-energy side tunnel through the same thin barrier to empty states on the other side. The current is nearly proportional to the voltage and symmetric: a **linear I–V**.
3. The specific contact resistance $R_c \propto \exp\left[\frac{2\sqrt{\epsilon_s m^*}}{\hbar} \cdot \frac{\phi_{Bn}}{\sqrt{N_d}}\right]$ (Eq. 9.33) falls exponentially with doping, to below 10^{-6} Ω cm^2.

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So, although a barrier exists, it does not rectify: the contact behaves as a small resistor, an **ohmic contact**. This is how practical ohmic contacts are made, especially on wide-gap materials where low barriers are impossible.

3. Keyword Explanation

1. Field emission

↩ back

Tunnelling of electrons near the Fermi level through a thin barrier.

2. Degenerate semiconductor

↩ back

A semiconductor doped so heavily that E_F lies in the conduction (or valence) band.

4. Notes and Remarks

- At intermediate doping (10^{17}–10^{19} cm^{-3}) electrons are thermally excited part of the way and tunnel through the thinner top of the barrier: **thermionic-field emission**.

5. Final Concise Answer

(a) Rectifying: $\phi_m > \phi_s$ on n-type (bands bend up, barrier $\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_m - \phi_s$) and $\phi_m < \phi_s$ on p-type (bands bend down, barrier $\phi_{Bp} = \chi + E_g/e - \phi_m$), with moderate doping and a clean interface; diagrams before contact (common vacuum level) and in equilibrium (flat E_F, depletion layer).

(b) h_c is the largest thickness a mismatched layer can grow strained without dislocations; above it misfit and threading dislocations reduce lifetime, efficiency and mobility; below it strained-layer devices are possible; $h_c \approx a_s/2|f|$. GaAs: direct gap, vertical transitions at $k = 0$, strong absorption and emission. Si: indirect gap, transitions need a phonon, weak absorption near the edge and very weak emission.

(c) Metal on n$^+$ ($\gtrsim 10^{19}$ cm^{-3}): the depletion layer is a few nm thick, electrons tunnel through the barrier in both directions, the I–V is linear and R_c is small: ohmic.

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4. (a) From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates. Marks: 8
   (b) What is bandgap narrowing? Marks: 3
   (c) Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage." Marks: 9
   (d) Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from 10^{18} to 10^{19} cm^{-3}. Determine the new value of p'_{E0} and determine the ratio p'_{E0}/p_{E0}. Marks: 5

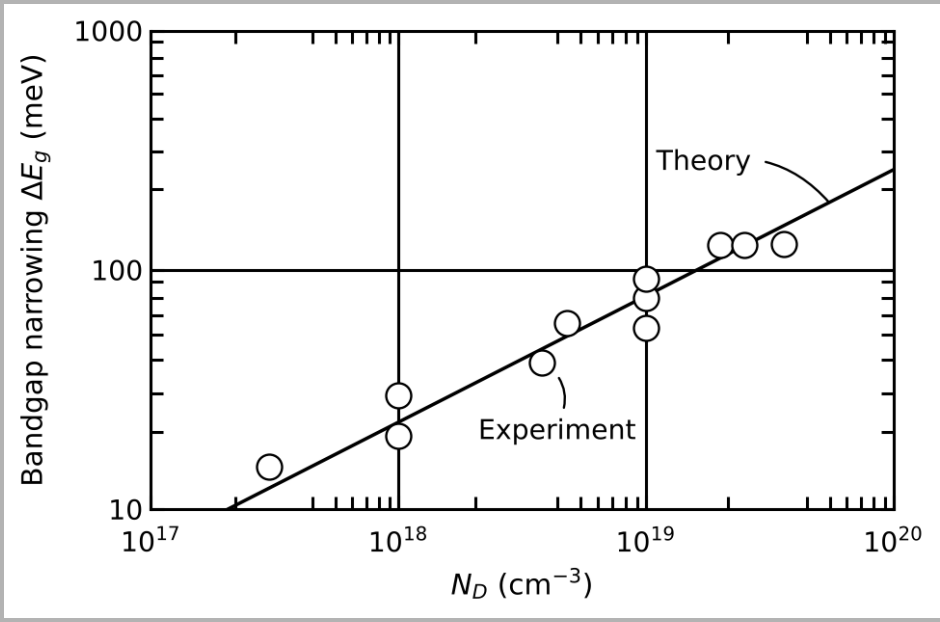


Figure 14: Figure 1: Fig. for Q. no. 4(d).

4.(a) From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates. Marks: 8

1. Theory

1. Two junctions sharing a thin base (Core Idea)

An npn transistor is two pn junctions back to back: base–emitter (B–E) and base–collector (B–C). Each alone is a diode. Because the base is much thinner than the minority-carrier diffusion length, electrons injected by one junction reach the other and are collected there: the junctions are **coupled**. This coupling is the transistor action.

2. The Ebers–Moll model

The Ebers–Moll model (Neamen, Section 12.5.1) describes any combination of junction biases with:

- two diode currents, $I_F = I_{ES}[\exp(V_{BE}/V_t) - 1]$ (B–E) and $I_R = I_{CS}[\exp(V_{BC}/V_t) - 1]$ (B–C);
- two dependent current sources, $\alpha_F I_F$ (the part of the B–E diode current that reaches the collector) and $\alpha_R I_R$ (the part of the B–C diode current that reaches the emitter);
- the reciprocity relation $\alpha_F I_{ES} = \alpha_R I_{CS}$.

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2. Answer

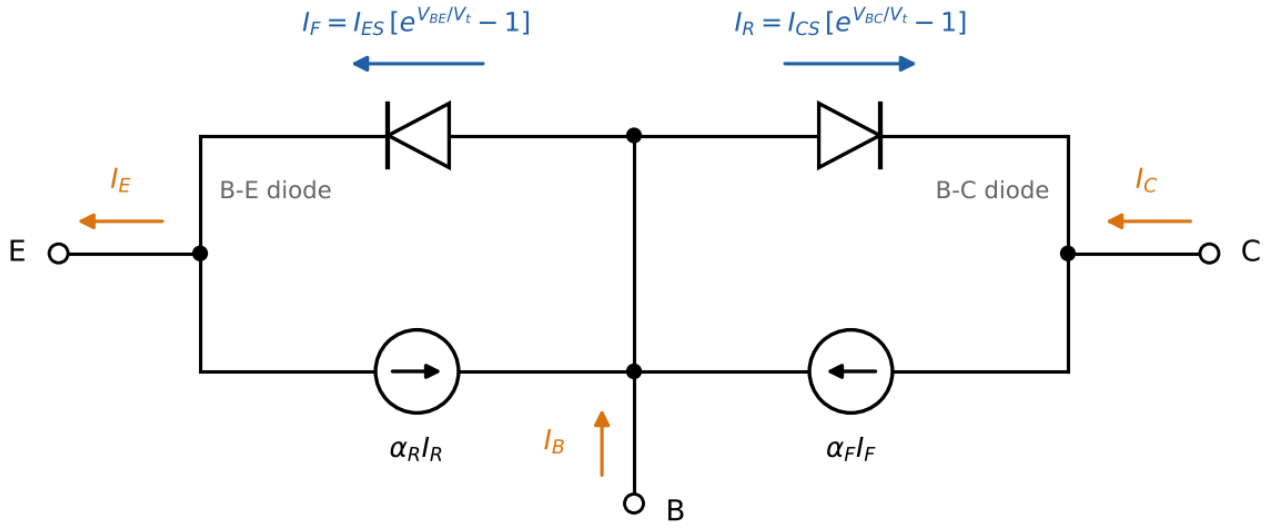


Figure 15: Ebers-Moll equivalent circuit of an npn transistor (reused from the 2020–21 paper)

Reading the figure. The B–E diode (current I_F) sits between the emitter and the internal base node, the B–C diode (current I_R) between that node and the collector. In parallel with the B–C diode is the source $\alpha_F I_F$; in parallel with the B–E diode the source $\alpha_R I_R$. The base terminal joins the two diodes.

Terminal equations (npn, currents into the collector, out of the emitter):

$$I_C = \alpha_F I_F - I_R, \qquad I_E = I_F - \alpha_R I_R, \qquad I_B = I_E - I_C = (1-\alpha_F)I_F + (1-\alpha_R)I_R$$

How the transistor operates in each mode:

1. **Forward active** ($V_{BE} > 0$, $V_{BC} < 0$): the B–E diode conducts, $I_F = I_{ES}e^{V_{BE}/V_t}$; the B–C diode is reverse biased, $I_R \approx -I_{CS}$ (tiny). So $I_E \approx I_F$ and $I_C \approx \alpha_F I_F = \alpha_F I_E$: almost all the emitter current reaches the collector (the source $\alpha_F I_F$), independent of V_{CB}. The base supplies only $(1-\alpha_F)I_F$, so $I_C = \beta_F I_B$ with $\beta_F = \alpha_F/(1-\alpha_F)$ (about 100): **amplification**.
2. **Saturation** (both junctions forward): both diodes conduct; I_R subtracts from the collector current, so $I_C < \beta_F I_B$, and $V_{CE}(\text{sat}) = V_{BE} - V_{BC}$ is only about 0.1–0.2 V: the transistor is a **closed switch**.
3. **Cut-off** (both junctions reverse): $I_F \approx -I_{ES}$, $I_R \approx -I_{CS}$; only tiny leakage flows: an **open switch**.
4. **Inverse active** ($V_{BE} < 0$, $V_{BC} > 0$): the roles swap, $I_E \approx \alpha_R I_C$; since α_R is small (0.1–0.5, the collector is a poor emitter), the gain β_R is low.

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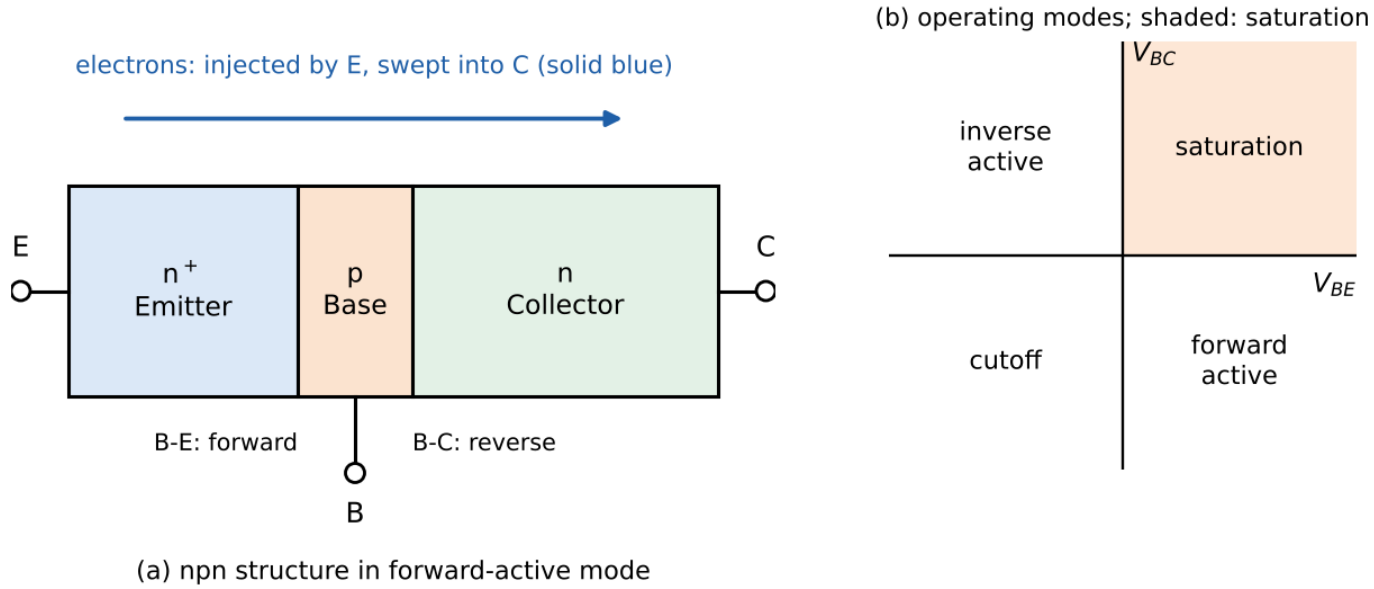


Figure 16: Operating modes of an npn transistor (reused from the 2020–21 paper)

Reading the figure. Panel (a): the npn structure in forward-active mode; electrons (solid blue arrow) are injected by the emitter and swept into the collector. Panel (b): the four modes on the V_{BE}–V_{BC} plane: forward active (B–E forward, B–C reverse), saturation (both forward, shaded), cut-off (both reverse) and inverse active.

3. Keyword Explanation

1. Forward common-base gain

↩ back

α_F: the fraction of the emitter current that reaches the collector in forward-active mode.

2. Reciprocity

↩ back

$\alpha_F I_{ES} = \alpha_R I_{CS}$, which reduces the model to three independent parameters.

4. Notes and Remarks

- The saturation voltage from the model is $V_{CE}(\text{sat}) = V_t \ln\{[I_C(1-\alpha_R) + I_B]\alpha_F/([\alpha_F I_B - (1-\alpha_F)I_C]\alpha_R)\}$ (Neamen, Eq. 12.70), derived in 6(a) of the 2020–21 paper (extra reading).

4.(b) What is bandgap narrowing? Marks: 3

1. Theory

1. Impurity levels at high doping (Core Idea)

At low doping each donor atom has its own discrete level just below E_c. When the doping is very high (above about 10^{18} cm^{-3} in Si), the donors are so close together that their electron wavefunctions overlap: the discrete levels broaden into an **impurity band**, which merges with the conduction band. The many-body interaction of the dense carriers also lowers the band edges.

2. Answer

Bandgap narrowing is the **reduction of the effective band gap** of a semiconductor when it is **heavily doped** (Neamen, Section 12.4.3):

$$E_g = E_{g0} - \Delta E_g$$

where E_{g0} is the gap at low doping and ΔE_g the bandgap narrowing factor, which rises with doping (about 20–25 meV at 10^{18}, 80 meV at 10^{19} and about 250 meV at 10^{20} cm^{-3} in Si; Fig. 1 of 4(d)).

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Causes: the donor levels broaden into an impurity band that merges with the conduction band, and band tails form; carrier–carrier and carrier–impurity interactions shift the band edges.

Effect in a BJT: the intrinsic concentration in the heavily doped emitter rises, $n_{iE}^2 = n_i^2 \exp(\Delta E_g/kT)$, so the emitter's minority holes increase, $p'_{E0} = (n_i^2/N_E)\exp(\Delta E_g/kT)$. More holes are injected back into the emitter, the **emitter injection efficiency and the current gain fall** as the emitter doping rises beyond about 10^{19} cm^{-3}, instead of improving. HBTs avoid this by using a wide-gap emitter with lighter doping.

3. Keyword Explanation

1. Impurity band

↩ back

A band of states formed when closely spaced impurity levels overlap at high doping.

4. Notes and Remarks

- In the heavily doped emitter the bandgap narrowing outweighs the benefit of more doping; that is why β falls at very high N_E (Neamen, Figure 12.24).

4.(c) Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage." Marks: 9

1. Theory

1. Avalanche multiplication (Core Idea)

In a reverse-biased junction a carrier crossing the high-field depletion region can gain enough energy to knock an electron out of a bond (impact ionisation), creating an electron–hole pair; the new carriers can do the same. The current entering the region is multiplied by a factor M, which grows without limit as the voltage approaches the **avalanche breakdown voltage**. Empirically (Neamen, Eq. 12.56):

$$M = \frac{1}{1 - (V_{CB}/BV_{CBO})^n}$$

where BV_{CBO} is the B–C breakdown voltage with the emitter open and n is an empirical constant (3 to 6).

2. Two test configurations

- **Open emitter (CBO):** only the B–C junction leakage I_{CBO} flows; it is multiplied to MI_{CBO}, and breakdown occurs when $M \to \infty$, at $V_{CB} = BV_{CBO}$.
- **Open base (CEO):** no external base current; the leakage itself acts as base current and is amplified by the transistor.

2. Answer

Step 1: Currents in the open-base configuration

With the base open, the collector current I_{CEO} also flows through the emitter. A fraction αI_{CEO} of it reaches the B–C junction from the emitter, and the junction leakage I_{CBO} adds to it. Both are multiplied in the B–C depletion region (Eq. 12.57):

$$I_{CEO} = M(\alpha I_{CEO} + I_{CBO})$$

Step 2: Solve for I_{CEO}

$$I_{CEO}(1 - \alpha M) = MI_{CBO} \quad \Rightarrow \quad I_{CEO} = \frac{MI_{CBO}}{1 - \alpha M} \qquad (12.58)$$

Step 3: Breakdown condition

I_{CEO} becomes infinite when the denominator vanishes:

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$$\alpha M = 1 \qquad (12.59)$$

Since α is just below 1, this happens when M is only slightly above 1, long before $M \to \infty$.

Step 4: Insert the empirical M

With $V_{CB} \approx V_{CE}$ at breakdown, $V_{CE} = BV_{CEO}$:

$$\frac{\alpha}{1-(BV_{CEO}/BV_{CBO})^n} = 1 \quad \Rightarrow \quad \left(\frac{BV_{CEO}}{BV_{CBO}}\right)^n = 1-\alpha$$

$$BV_{CEO} = BV_{CBO}\sqrt[n]{1-\alpha} \qquad (12.61)$$

Step 5: Express with β

$\beta = \alpha/(1-\alpha)$, and since $\alpha \approx 1$, $1-\alpha \approx 1/\beta$. Therefore

$$\boxed{BV_{CEO} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}} \qquad (12.63)$$

The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$, than the actual avalanche junction breakdown voltage (open emitter), as required. Physically, the transistor's own gain amplifies the avalanche current: a small multiplication ($M = 1/\alpha$) already gives runaway.

Example

$BV_{CBO} = 120$ V, $\beta = 100$, $n = 3$: $BV_{CEO} = 120/100^{1/3} = 120/4.64 = 25.8$ V.

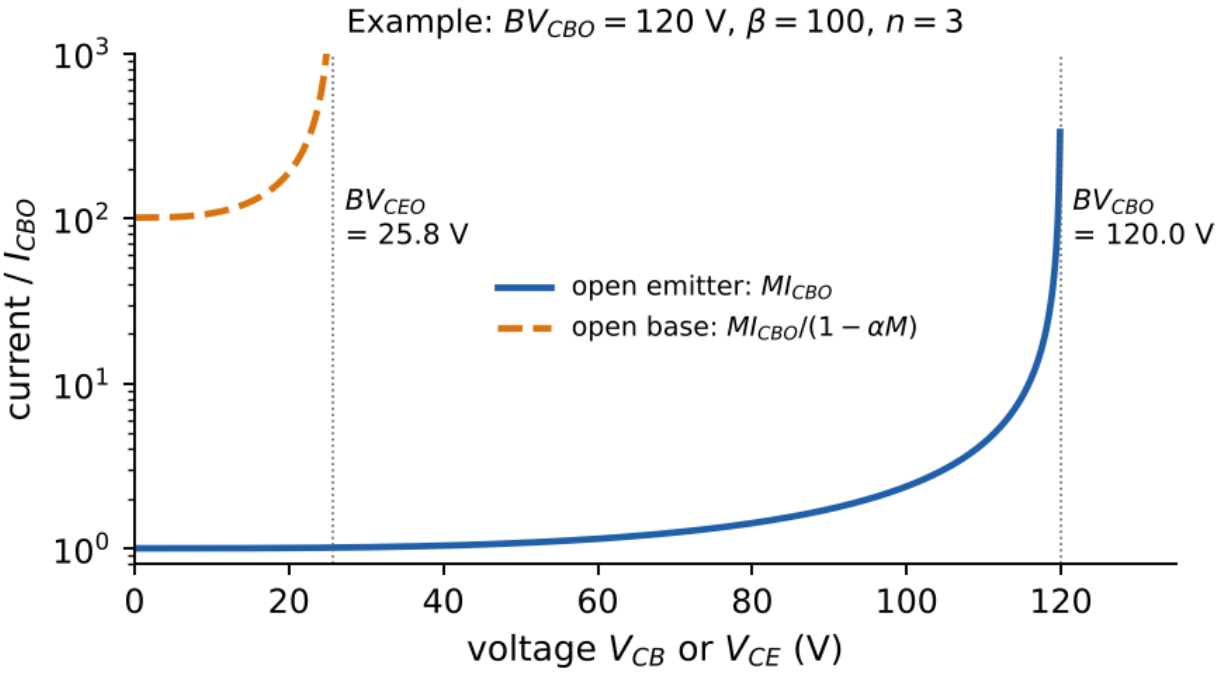


Figure 17: Open-emitter and open-base currents against voltage: the open-base current runs away near 25.8 V, the open-emitter current only near 120 V

Reading the figure. The solid blue curve is the open-emitter current MI_{CBO}, rising sharply only near $BV_{CBO} = 120$ V. The dashed orange curve is the open-base current $MI_{CBO}/(1-\alpha M)$: it starts at about βI_{CBO} and runs away near $BV_{CEO} = 25.8$ V (dotted lines).

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Steps to remember

$I_{CEO} = M(\alpha I_{CEO} + I_{CBO}) \rightarrow I_{CEO} = MI_{CBO}/(1-\alpha M) \rightarrow$ breakdown at $\alpha M = 1 \rightarrow$ put $M = 1/[1-(V/BV_{CBO})^n] \rightarrow BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} \rightarrow 1-\alpha \approx 1/\beta$.

3. Keyword Explanation

1. Multiplication factor

↩ back

M: the factor by which avalanche multiplication increases the current crossing a depletion region.

2. BV_{CEO}

↩ back

The collector–emitter breakdown voltage with the base open; the usual rated maximum V_{CE}.

4. Notes and Remarks

- The result sets the usable V_{CE} of a transistor: a high-gain device has a lower BV_{CEO} (Neamen, Figure 12.35).
- In the printed question the root index is faint; it is n, the empirical exponent of M.

4.(d) Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from 10^{18} to 10^{19} cm^{-3}. Determine the new value of p'_{E0} and determine the ratio p'_{E0}/p_{E0}. Marks: 5

1. Theory

1. Minority carriers in the emitter (Core Idea)

In the n-type emitter of an npn transistor the holes are minority carriers. In equilibrium $n_0 p_0 = n_i^2$, so with $n_0 = N_E$ (Neamen):

$$p_{E0} = \frac{n_i^2}{N_E}$$

Raising the emitter doping should lower p_{E0}, which is what makes the injection efficiency high.

2. Effect of bandgap narrowing

With bandgap narrowing the emitter's band gap is $E_{g0} - \Delta E_g$. Since $n_i^2 \propto \exp(-E_g/kT)$, the emitter's intrinsic concentration grows by $\exp(\Delta E_g/kT)$ (Eq. 12.50):

$$p'_{E0} = \frac{n_i^2}{N_E}\exp\left(\frac{\Delta E_g}{kT}\right), \qquad \frac{p'_{E0}}{p_{E0}} = \exp\left(\frac{\Delta E_g}{kT}\right)$$

ΔE_g is read from Fig. 1 at the emitter doping.

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2. Answer

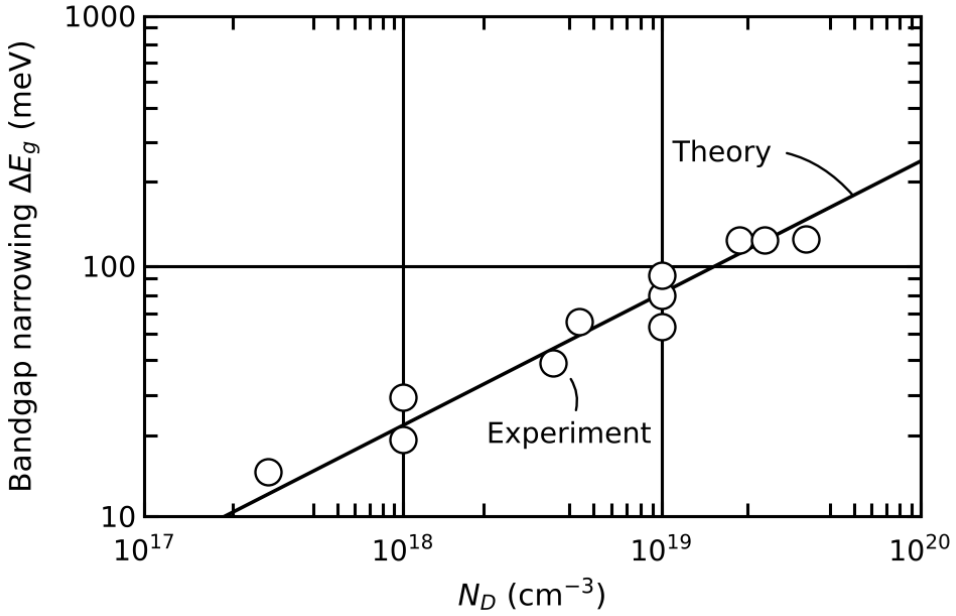


Figure 18: Repeated from the question: Figure 1 of the paper (redrawn)

Given: Si, $T = 300$ K, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $kT = 0.0259$ eV, $N_E = 10^{18}$ and 10^{19} cm^{-3}. From the "Theory" line of Fig. 1: $\Delta E_g \approx 23$ meV at 10^{18} cm^{-3} and ≈ 78 meV at 10^{19} cm^{-3}.

Step 1: Without bandgap narrowing

$$p_{E0}(10^{18}) = \frac{(1.5 \times 10^{10})^2}{10^{18}} = 225 \text{ cm}^{-3}, \qquad p_{E0}(10^{19}) = \frac{(1.5 \times 10^{10})^2}{10^{19}} = 22.5 \text{ cm}^{-3}$$

Step 2: With bandgap narrowing at 10^{18} cm^{-3}

$$\frac{p'_{E0}}{p_{E0}} = \exp\left(\frac{0.023}{0.0259}\right) = 2.44, \qquad p'_{E0} = 225 \times 2.44 = 549 \text{ cm}^{-3}$$

Step 3: With bandgap narrowing at 10^{19} cm^{-3}

$$\frac{p'_{E0}}{p_{E0}} = \exp\left(\frac{0.078}{0.0259}\right) = 20.2, \qquad p'_{E0} = 22.5 \times 20.2 = 454 \text{ cm}^{-3}$$

$$\boxed{p'_{E0}(10^{19}) \approx 4.5 \times 10^2 \text{ cm}^{-3}, \qquad \frac{p'_{E0}}{p_{E0}} \approx 20}$$

Meaning: raising the emitter doping tenfold should cut p_{E0} tenfold ($225 \to 22.5$ cm^{-3}), but bandgap narrowing raises it back to about 454 cm^{-3}, almost the same as the 549 cm^{-3} at 10^{18} cm^{-3}. The extra doping gives almost no improvement in emitter injection efficiency.

Final Answer: at $N_E = 10^{19}$ cm^{-3}, $p'_{E0} \approx 454$ cm^{-3} and $p'_{E0}/p_{E0} \approx 20$ (at 10^{18} cm^{-3}: $p'_{E0} \approx 549$ cm^{-3}, ratio 2.4).

3. Keyword Explanation

1. Bandgap narrowing factor

↩ back

ΔE_g: the reduction of the band gap caused by heavy doping.

2. Thermal-equilibrium minority carrier concentration

↩ back

$p_{E0} = n_i^2/N_E$: the hole density in the n-type emitter with no bias.

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4. Notes and Remarks

- **Reading of Fig. 1:** this is Neamen's Example 12.9; Neamen rounds the readings to 20 and 80 meV, which give $p'_{E0} = 487$ and 494 cm^{-3} and ratios 2.16 and 21.95. The values read from the printed "Theory" line are 23 ± 1 and 78 ± 3 meV (checked against the scan and the book), giving the numbers above; the conclusion is the same.
- At $N_E = 10^{20}$ cm^{-3} ($\Delta E_g \approx 0.26$ eV) p'_{E0} is even larger than at 10^{18}: heavy emitter doping eventually lowers the gain.

5. Final Concise Answer

(a) Ebers–Moll: diodes $I_F = I_{ES}(e^{V_{BE}/V_t} - 1)$, $I_R = I_{CS}(e^{V_{BC}/V_t} - 1)$ and sources $\alpha_F I_F$, $\alpha_R I_R$; $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$. Forward active: $I_C \approx \alpha_F I_E$, $I_C = \beta_F I_B$ (amplifier). Saturation: both diodes on, $V_{CE} \approx 0.1$–0.2 V (switch on). Cut-off: leakage only (switch off). Inverse: low gain β_R.

(b) Bandgap narrowing: at high doping the impurity levels form a band that merges with E_c and many-body effects lower the edges, so $E_g = E_{g0} - \Delta E_g$ (about 80 meV at 10^{19} cm^{-3} in Si); it raises p_{E0} by $e^{\Delta E_g/kT}$ and lowers the emitter injection efficiency and gain.

(c) $I_{CEO} = M(\alpha I_{CEO} + I_{CBO})$ gives $I_{CEO} = MI_{CBO}/(1 - \alpha M)$; breakdown at $\alpha M = 1$ with $M = 1/[1 - (V/BV_{CBO})^n]$ gives $BV_{CEO} = BV_{CBO}(1 - \alpha)^{1/n} = BV_{CBO}/\sqrt[n]{\beta}$.

(d) $p_{E0} = 225$ and 22.5 cm^{-3}; with $\Delta E_g \approx 23$ and 78 meV, $p'_{E0} = 549$ and 454 cm^{-3}; $p'_{E0}/p_{E0} = 2.4$ at 10^{18} and about 20 at 10^{19} cm^{-3}.

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5. (a) With neat sketch, describe basic operation mechanism of TEBT. Marks: 4
   (b) What is 2DEG? With quantum well structure describe the operation of HEMT. Marks: 5
   (c) With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters. [Marks: 4+3+4]
   (d) Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N-Al_{0.3}Ga_{0.7}As$ layer doped to $10^{18}\ cm^{-3}$ and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2. Marks: 5

5.(a) With neat sketch, describe basic operation mechanism of TEBT. Marks: 4

1. Theory

1. What limits a homojunction BJT (Core Idea)

In an npn bipolar transistor the forward-biased emitter–base junction sends two currents across: **electrons** from the n^+ emitter into the p base (the useful current, which reaches the collector) and **holes** from the base back into the emitter (wasted base current). The **emitter injection efficiency**

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}}$$

must be close to 1 for a high current gain. In a homojunction both carriers see the same barrier, so the only way to make J_{pE} small is to dope the emitter much more heavily than the base. That forces a lightly doped base, which has a high base resistance and limits the speed. (The heterojunction bipolar transistor, HBT, of 5(c) solves this with a wide-gap emitter.)

2. Quantum-mechanical tunnelling

A particle meeting a potential barrier higher than its energy is not always reflected: if the barrier is thin, its wavefunction leaks through and it appears on the other side. For a rectangular barrier of height V_0 above the particle's energy and thickness d, the transmission probability is approximately

$$T \approx \exp\left(-2d\sqrt{\frac{2m^*V_0}{\hbar^2}}\right)$$

It falls exponentially with the barrier thickness d, with the square root of the barrier height V_0 and with the square root of the effective mass m^*. A light particle meeting a low barrier passes easily; a heavy particle meeting a higher barrier is almost completely reflected.

3. The idea of the TEBT

The **tunnelling-emitter bipolar transistor** (TEBT; proposed by Xu and Shur, 1986) puts a very thin (about 2–5 nm) layer of a wide-gap semiconductor, for example AlGaAs or AlAs, between the n^+ emitter and the p^+ base of a GaAs transistor. The layer is a barrier to both carriers, but a very unequal one:

- For electrons: the conduction-band step ΔE_c is small and the electron effective mass in GaAs is small ($0.067\,m_0$), so electrons tunnel through easily.
- For holes: the valence-band step ΔE_v seen from the base is the barrier, and holes are heavy ($m_h^* \approx 0.5\,m_0$), so the transmission is many orders of magnitude lower: holes are reflected.

The barrier therefore acts as a filter that passes electrons and blocks holes. This gives the same benefit as a heterojunction emitter (high γ even with a heavily doped base), but the emitter itself can be the same material as the base.

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2. Answer

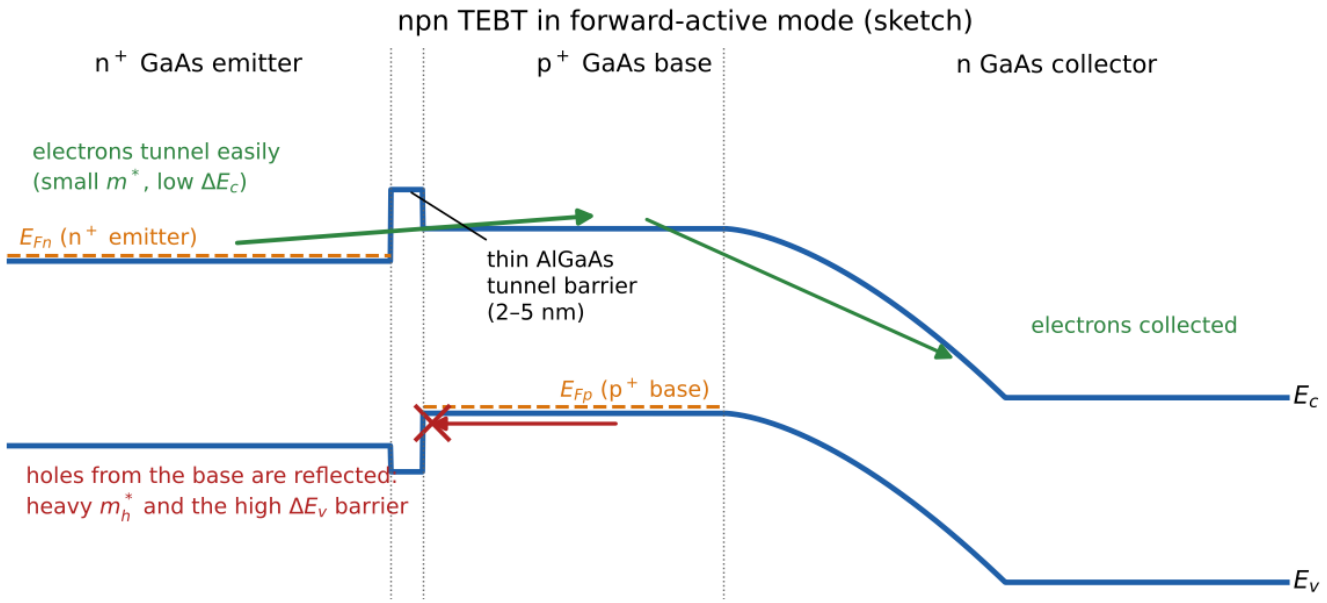


Figure 19: Band diagram of an npn GaAs TEBT in forward-active mode, with a thin AlGaAs tunnel barrier between the emitter and the base

Reading the figure. From left to right: the n^+ GaAs emitter (its Fermi level, dashed orange, near E_c), the thin AlGaAs barrier (the narrow rectangle in both bands), the p^+ GaAs base and the reverse-biased base–collector junction, where the bands fall into the n collector. The green arrows show electrons tunnelling through the thin barrier into the base and then falling into the collector. The red arrow and cross show holes from the base being reflected by the barrier in the valence band.

Structure: n^+ GaAs emitter / thin (2–5 nm) undoped AlGaAs (or AlAs) tunnel barrier / heavily doped p^+ GaAs base / n GaAs collector / n^+ subcollector.

Operation (forward-active mode, $V_{BE} > 0$, $V_{BC} < 0$):

1. **Electron injection:** the forward bias V_{BE} lowers the emitter–base barrier. Electrons of the emitter reach the thin AlGaAs layer and tunnel through it, because the conduction-band step is small and their effective mass is small. They enter the base as minority carriers.
2. **Transport and collection:** the electrons diffuse across the thin base (little recombination) and are swept into the collector by the field of the reverse-biased base–collector junction. This is the collector current $I_C \approx \alpha_T I_{nE}$, as in an ordinary BJT.
3. **Hole blocking:** holes injected from the base towards the emitter meet the valence-band step of the barrier. With their large effective mass the tunnelling probability is tiny, so the back-injected hole current J_{pE} is strongly suppressed.
4. **Result:** the injection efficiency is close to 1 independent of the emitter/base doping ratio. The base can be doped very heavily (low base resistance, no punch-through, small Early effect), and the gain stays high.

Advantages over an HBT: no graded wide-gap emitter is needed (simpler growth), the turn-on voltage is lower (the emitter is narrow-gap material) and the idea also works in silicon (a thin oxide or wide-gap layer at the emitter, as in polysilicon-emitter transistors).

3. Keyword Explanation

1. Tunnelling

↩ back

The passage of a particle through a potential barrier higher than its energy; possible only when the barrier is very thin. Its probability falls exponentially with $d\sqrt{m^* V_0}$.

2. Emitter injection efficiency

↩ back

$\gamma = J_{nE}/(J_{nE} + J_{pE})$: the fraction of the emitter current carried by the useful (electron) carriers.

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4. Notes and Remarks

- The TEBT is not covered by Neamen or Sze and Ng; the description follows the original proposal (J. Xu and M. Shur, *IEEE Electron Device Letters*, 1986). Sze and Ng (Sec. 8.3.5) describe a different device with a tunnel emitter, the tunnelling hot-electron transfer amplifier (THETA), a majority-carrier hot-electron transistor; do not confuse the two.
- The barrier must be thin enough for electron tunnelling (a few nm); a thicker layer would block the electrons too and the device would behave like an abrupt HBT.

5.(b) What is 2DEG? With quantum well structure describe the operation of HEMT. Marks: 5

1. Theory

1. Energy quantisation in a narrow well (Core Idea)

An electron confined in a region comparable with its de Broglie wavelength (a few nm) cannot have any energy: its motion across the region is **quantised**. In an infinite square well of width w the allowed energies are

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2m^* w^2}, \qquad n = 1, 2, 3, \ldots$$

The electron can still move freely in the two directions parallel to the well. Each allowed level E_n is the bottom of a **subband**; when only the lowest subband is occupied, the electrons are free in two dimensions only and form a **two-dimensional electron gas** (2DEG). Its density is counted per unit area, n_s (cm^{-2}), and the density of states of each subband is a constant, $m^*/(\pi\hbar^2)$ per unit area per unit energy.

2. The well at an N-AlGaAs / GaAs heterojunction

When heavily doped N-type AlGaAs (wide gap) is grown on undoped GaAs (narrow gap), the conduction band drops by ΔE_c at the interface (about 0.22 eV for $Al_{0.3}Ga_{0.7}As$). Electrons from the AlGaAs donors move to the lower conduction band of the GaAs. The positive donor ions stay behind in the AlGaAs, and the electrons pile up against the interface, held there by the attraction of those ions. The band bends into a narrow **triangular well** (about 10 nm wide) on the GaAs side: this is the quantum well that holds the 2DEG. Its electrons are separated from the ionised donors that supplied them (**modulation doping**), so they suffer little impurity scattering and have a very high mobility (about 8500 $\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$ at 300 K, over 10^5 at 77 K).

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2. Answer

What is a 2DEG: a layer of electrons that is free to move only in a plane, because its motion in the third direction is confined to a quantum well narrow enough (about 10 nm) for the energy across it to be quantised into subbands, with only the lowest subband(s) occupied. In a HEMT it forms in the triangular well on the GaAs side of an N-AlGaAs / undoped GaAs interface; its sheet density is about 10^{12} cm^{-2}.

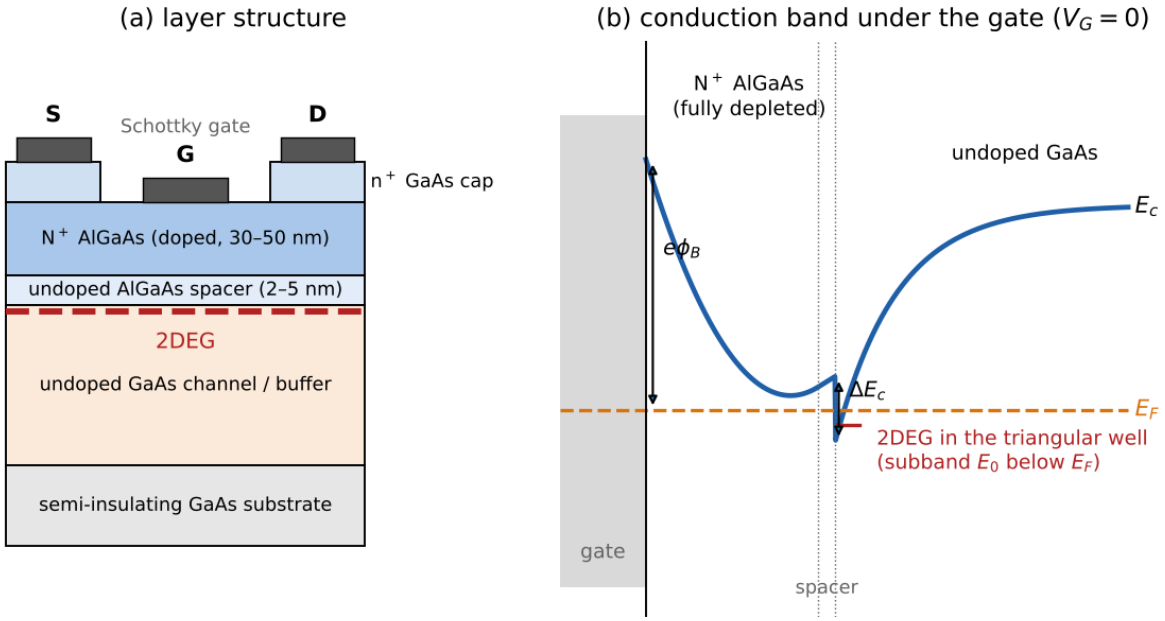


Figure 20: (a) HEMT layer structure with the 2DEG at the AlGaAs/GaAs interface; (b) the conduction band under the gate at zero gate voltage, showing the quantum well

Reading the figure. Panel (a): the layers from the top: n$^+$ GaAs cap under the source and drain, the Schottky gate on the N$^+$ AlGaAs, a thin undoped AlGaAs spacer, the undoped GaAs channel and the semi-insulating substrate; the red dashed line is the 2DEG just below the spacer. Panel (b): the conduction band (solid blue) under the gate. In the doped AlGaAs it is a parabola (fully depleted donors); at the interface it drops by ΔE_c; in the GaAs it forms a triangular quantum well that dips below the Fermi level (dashed orange). The shaded red region is the 2DEG, with its lowest subband E_0 below E_F.

Operation of the HEMT (quantum-well picture):

1. **Channel:** the triangular quantum well in the GaAs holds the 2DEG; it is the conducting channel between source and drain, with no dopants in it (high mobility, high saturation velocity).
2. **Gate control:** the Schottky gate on the AlGaAs sets the band bending under it. The 2DEG density follows a capacitor law (Neamen, Eqs. 13.68–13.71):

$$n_s = \frac{\epsilon_N}{e(d+\Delta d)}(V_g - V_{off})$$

   with d the AlGaAs thickness (doped plus spacer), $\Delta d \approx 80$ Å the 2DEG's own width and V_{off} the off (threshold) voltage.
3. **Negative gate voltage:** raises the bands under the gate, lifts the well towards E_F and lowers n_s; at $V_g = V_{off}$ the lowest subband rises above E_F, the well empties and the channel is cut off.
4. **Positive gate voltage:** deepens the well and increases n_s, until the AlGaAs conduction band reaches E_F and a parallel low-mobility channel forms in the AlGaAs; beyond that the gate loses control.
5. **Drain current:** a drain voltage drives the 2DEG along the interface. For small V_{DS}, $I_D = \frac{\epsilon_N \mu W}{2L(d+\Delta d)}[2(V_g - V_{off})V_{DS} - V_{DS}^2]$; at high fields the electrons reach the saturation velocity and

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$I_D(\text{sat}) = \dfrac{\epsilon_N W}{d + \Delta d}(V_g - V_{off} - V_0)v_{sat}$ (Eqs. 13.74–13.75).

The high mobility and velocity of the undoped channel give the HEMT a high transconductance (about 250 mS/mm), low noise and cut-off frequencies of about 100 GHz.

3. Keyword Explanation

1. Modulation doping

↩ back

Doping the wide-gap layer only, so that its electrons transfer into the undoped narrow-gap layer, away from the ionised donors.

2. Subband

↩ back

The set of states built on one quantised level E_n of a quantum well: quantised across the well, free along it.

4. Notes and Remarks

- The same structure and operation are solved in 6(b) of the 2018–19 paper (extra reading).
- The undoped spacer (2–5 nm) moves the donors further from the 2DEG: the mobility rises, the sheet density falls a little.

5.(c) With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters. [Marks: 4+3+4]

1. Theory

1. The base–collector junction at low current (Core Idea)

In a modern npn transistor the collector is a lightly doped n^- epitaxial layer (doping N_C, thickness W_C) on an n^+ subcollector. The base–collector junction is reverse biased. At low current its space-charge region holds the ionised donors $+eN_C$ on the collector side and acceptors on the base side, and the electric field peaks at the metallurgical junction (Poisson's equation, $dE/dx = \rho/\epsilon_s$).

2. Electrons in the collector add charge

The collector current is carried by electrons crossing the collector depletion region. The field there is high, so they move at the saturation velocity v_{sat} (about 10^7 cm/s), and their density is

$$n_c = \frac{J_C}{ev_{sat}}$$

These electrons are negative charge, so the net charge in the collector depletion region becomes $e(N_C - n_c)$. This is the start of the Kirk effect.

3. The HBT

A heterojunction bipolar transistor (HBT) has a wide-gap emitter (for example N-AlGaAs) on a narrow-gap base (p-GaAs). Holes from the base see an extra barrier ΔE_g, so the injection efficiency is high even with a base doped more heavily than the emitter (Neamen, Sec. 12.8.3): the hole-to-electron injection ratio falls by $\exp(-\Delta E_g/kT)$.

2. Answer

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(i) Kirk effect (base push-out).

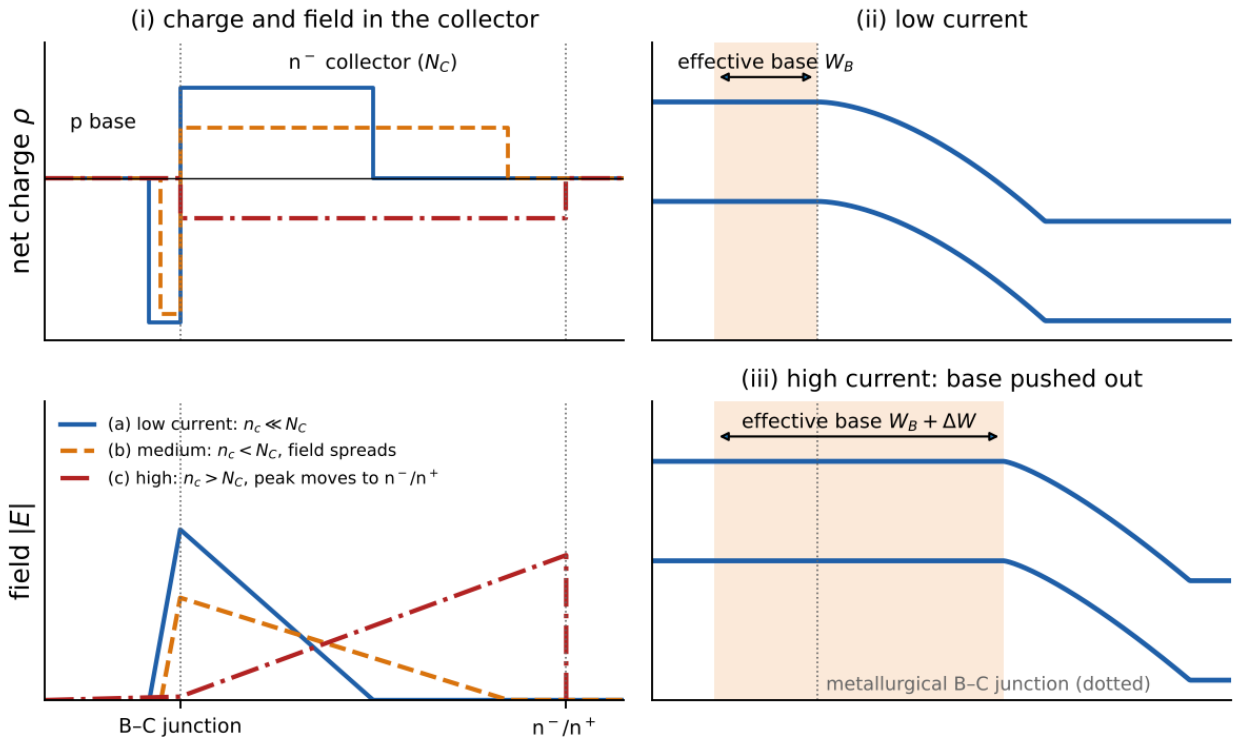


Figure 21: Kirk effect: (i) net charge and field magnitude in the collector at low, medium and high current; (ii) and (iii) the bands at low and high current, with the effective base widened

Reading the figure. Left column: the net charge (top) and the field magnitude (bottom) against position, from the base (left of the first dotted line, the B–C junction) to the n^-/n^+ interface (second dotted line). Solid blue = low current, dashed orange = medium current, dash-dot red = high current. Right column: the conduction and valence bands at low current (ii) and high current (iii); the shaded region is the effective (neutral) base.

Step 1: Low current ($n_c \ll N_C$). The collector depletion region holds the full donor charge $+eN_C$; the field is triangular, largest at the B–C junction and falling to zero inside the collector (solid blue).

Step 2: Rising current ($n_c < N_C$). The electrons in transit cancel part of the donor charge: the net charge $e(N_C - n_c)$ falls. At fixed V_{CB} the area under the field (the voltage) is fixed, so a smaller slope means a lower peak and a wider depletion region: the field spreads towards the n^+ layer (dashed orange).

Step 3: High current ($n_c > N_C$). The net charge in the collector becomes negative. The field now has its peak at the n^-/n^+ interface and falls to nearly zero at the metallurgical B–C junction (dash-dot red). With no field at the junction, holes are no longer kept out of the collector: the neutral p-type region (the base) extends into the n^- collector, up to the point where the field begins.

Step 4: Consequence. The effective base width grows from W_B to as much as $W_B + W_C$ (Sze and Ng, Sec. 5.2). The base Gummel number and the base transit time rise, so the current gain β and the cut-off frequency f_T fall sharply at high current. The effect sets in at the critical (Kirk) current density (Sze and Ng, Eq. 55)

$$J_K = ev_{sat}\left[N_C + \frac{2\epsilon_s(V_{CB} + V_{bi})}{eW_C^2}\right]$$

which is about $ev_{sat}N_C$ for a thin, lightly doped collector.

(ii) How the Kirk effect is reduced in an HBT.

- In an HBT the wide-gap emitter, not the doping ratio, gives the high injection efficiency. The base can be doped very heavily (10^{19}–10^{20} cm^{-3}) and made very thin.
- So the collector doping N_C can be raised without punch-through of the thin base or loss of gain:

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$J_K \propto N_C$ rises, and the Kirk effect starts only at much higher current.

- A **double-heterojunction** HBT (DHBT) also uses a wide-gap collector; it allows a higher collector doping and breakdown voltage and blocks hole injection into the collector, which further suppresses base push-out and quasi-saturation (Sze and Ng, Sec. 5.5.1).
- The thin base and high electron velocity keep the stored charge and transit time small even at high current densities.

(iii) Abrupt and graded emitter–base junction HBTs.

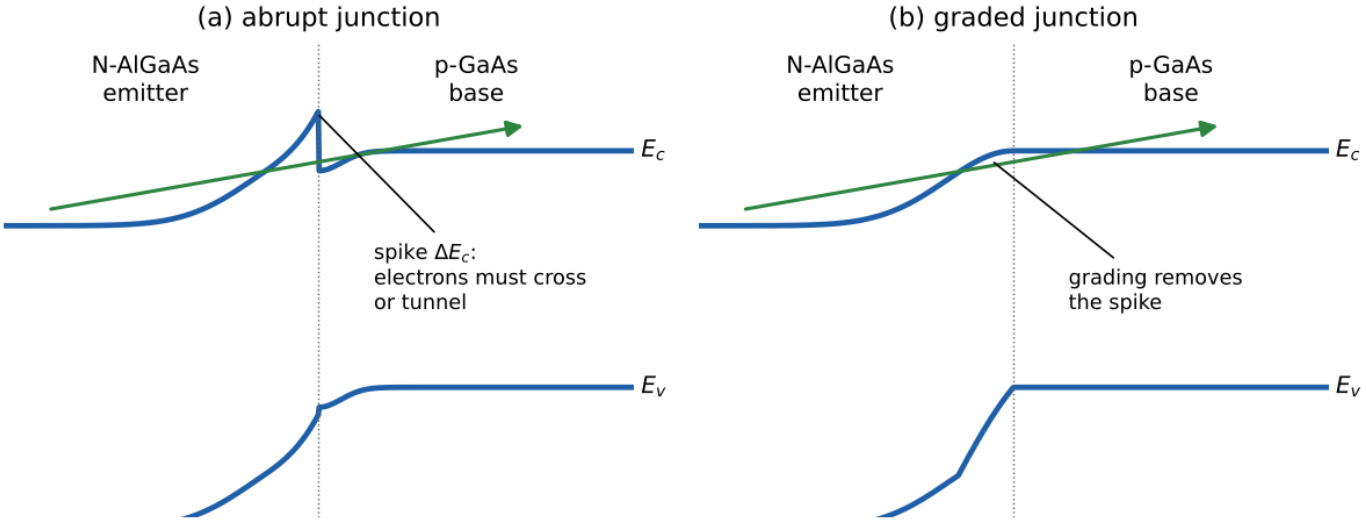


Figure 22: Emitter–base junction of an npn AlGaAs/GaAs HBT under forward bias: (a) abrupt, with the conduction-band spike; (b) compositionally graded, without the spike

Reading the figure. Each panel shows E_c and E_v across the emitter–base junction (dotted line), N-AlGaAs emitter on the left, p-GaAs base on the right; the green arrow is the injected electron flow. In (a) the conduction band has a sharp spike of height ΔE_c at the junction; in (b) the composition is graded over a few tens of nm and the spike is gone.

Parameter	Abrupt junction	Graded junction
Conduction band at the junction	spike ΔE_c and notch	smooth, no spike
Electron injection	over or through the spike (thermionic emission and tunnelling)	drift–diffusion, no extra barrier
Injection-efficiency gain over a BJT	about $\exp(\Delta E_v/kT)$ (the spike takes ΔE_c back)	the full $\exp(\Delta E_g/kT)$
Turn-on V_{BE} and V_{CE} offset	higher	lower
Injected electrons	"hot": launched into the base with energy about ΔE_c, near-ballistic, shorter base transit time	thermal (no launching energy)
Current gain β	lower	higher
Ideality factor	larger (thermionic and tunnelling)	close to 1
Fabrication	simpler	needs controlled grading (MBE, MOCVD)

Table 4: Summary

Conclusion: grading gives higher gain, lower turn-on voltage and a smaller offset; the abrupt junction's spike can be used as an electron launcher for faster base transit, at the cost of a larger V_{BE}.

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3. Keyword Explanation

1. Kirk effect

↩ back

The widening of the effective base into the collector at high current, when the mobile electron charge in the collector cancels its donor charge; also called base push-out.

2. Critical current density J_K

↩ back

The collector current density at which the Kirk effect begins; about $ev_{sat}N_C$.

3. Graded junction

↩ back

A heterojunction whose alloy composition changes gradually over a short distance, so that the band offset becomes a smooth ramp.

4. Notes and Remarks

- The Kirk effect is described in Sze and Ng (Sec. 5.2, Figs. 11–12); the HBT comparison follows Sze and Ng (Sec. 5.5) and Neamen (Sec. 12.8.3).
- A high-current drop of β also comes from high-level injection in the base and from current crowding; the Kirk effect is usually the first in a transistor with a lightly doped epitaxial collector.

5.(d) Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N - Al_{0.3}Ga_{0.7}As$ layer doped to 10^{18} cm^{-3} and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2. Marks: 5

1. Theory

1. The charge-control model (Core Idea)

In the HEMT structure of 5(b) the gate, the AlGaAs layer and the 2DEG behave like a parallel-plate capacitor. The gate is one plate and the 2DEG the other; between them is the AlGaAs, of permittivity ϵ_N and thickness $d = d_d + d_i$ (doped layer plus undoped spacer), plus a correction Δd for the thickness of the 2DEG itself (its electrons sit a few nm inside the GaAs). A capacitor holds the charge $Q = CV$ per unit area, $C = \epsilon_N/(d + \Delta d)$, so the 2DEG sheet density is (Neamen, Eq. 13.68)

$$n_s = \frac{\epsilon_N}{e(d + \Delta d)}(V_g - V_{off}), \qquad \Delta d \approx 80 \text{ Å}$$

2. The off voltage

V_{off} is the gate voltage at which $n_s = 0$. It is found by following the potential from the gate down to the 2DEG (Neamen, Eqs. 13.70–13.71):

- the Schottky barrier ϕ_B raises the AlGaAs conduction band at the gate;
- the depleted donors of the doped layer pull it back down by V_{p2}, the voltage across a fully depleted layer of thickness d_d (the same Poisson integration as for a JFET channel):

$$V_{p2} = \frac{eN_d d_d^2}{2\epsilon_N}$$

- the step ΔE_c at the interface lowers the GaAs band further.

$$V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2}$$

If $V_{off} < 0$, a 2DEG exists at $V_g = 0$: a depletion-mode (normally-on) HEMT.

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2. Answer

Given (this is Neamen's Example 13.10): N-$Al_{0.3}Ga_{0.7}As$ doped to $N_d = 10^{18}$ cm^{-3}, doped thickness $d_d = 500$ Å $= 5 \times 10^{-6}$ cm, undoped spacer $d_i = 20$ Å, $\phi_B = 0.85$ V, $\Delta E_c/e = 0.22$ V, $\epsilon_r = 12.2$, $\Delta d = 80$ Å. The 2DEG density is asked at zero gate voltage.

Permittivity: $\epsilon_N = 12.2 \times 8.85 \times 10^{-14} = 1.080 \times 10^{-12}$ F/cm.

Step 1: Voltage across the depleted AlGaAs

$$V_{p2} = \frac{eN_d d_d^2}{2\epsilon_N} = \frac{(1.6 \times 10^{-19})(10^{18})(5 \times 10^{-6})^2}{2(1.080 \times 10^{-12})} = 1.852 \text{ V}$$

Step 2: Off voltage

$$V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2} = 0.85 - 0.22 - 1.852 = -1.222 \text{ V}$$

Negative, so the device is a depletion-mode HEMT: a 2DEG exists with no gate bias.

Step 3: Effective plate distance

$$d + \Delta d = d_d + d_i + \Delta d = (500 + 20 + 80) \text{ Å} = 600 \text{ Å} = 6 \times 10^{-6} \text{ cm}$$

Step 4: 2DEG density at $V_g = 0$

$$n_s = \frac{\epsilon_N}{e(d + \Delta d)}(0 - V_{off}) = \frac{1.080 \times 10^{-12}}{(1.6 \times 10^{-19})(6 \times 10^{-6})}(1.222) = 1.375 \times 10^{12} \text{ cm}^{-2}$$

$$\boxed{n_s \approx 1.37 \times 10^{12} \text{ cm}^{-2}}$$

Meaning: about 10^{12} electrons per cm^2 is the typical sheet density of an AlGaAs/GaAs 2DEG. A gate voltage of -1.22 V would empty the channel.

3. Keyword Explanation

1. Charge-control model

↩ back

The description of the 2DEG as the second plate of a capacitor whose first plate is the gate: $en_s = C(V_g - V_{off})$.

2. Off voltage V_{off}

↩ back

The threshold voltage of the HEMT, the gate voltage at which the 2DEG disappears.

4. Notes and Remarks

- The numbers are those of Neamen's Example 13.10 (book: $V_{p2} = 1.85$ V, $V_{off} = -1.22$ V, $n_s = 1.37 \times 10^{12}$ cm^{-2}); the question does not give Δd, so Neamen's value of 80 Å (Eq. 13.69) is used.
- Without the Δd correction (plate distance 520 Å) the result would be 1.59×10^{12} cm^{-2}, about 16 % higher.

5. Final Concise Answer

(a) TEBT: n$^+$ GaAs emitter / 2–5 nm AlGaAs tunnel barrier / p$^+$ GaAs base / n collector. Electrons (small m^*, small ΔE_c) tunnel through the barrier into the base and are collected; holes (heavy, larger ΔE_v) are reflected. So $\gamma \to 1$ even with a heavily doped base: high gain, low base resistance, lower turn-on voltage than an HBT.

(b) A 2DEG is a sheet of electrons quantised across a narrow (about 10 nm) well and free in the plane. In a HEMT, N-AlGaAs donors give electrons to the triangular well in undoped GaAs (modulation doping, high mobility); the Schottky gate sets $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$; negative V_g empties the well (off at V_{off}), positive V_g fills it until the AlGaAs conducts.

(c) Kirk effect: at high current the electrons $n_c = J_C/(ev_{sat})$ cancel N_C; when $n_c > N_C$ the field peak

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moves to the n^-/n^+ interface and the base widens up to W_B+W_C, so β and f_T fall; onset $J_K \approx ev_{sat}N_C$. HBT: heavy base doping allowed, so N_C can be raised (DHBT: wide-gap collector), raising J_K. Abrupt HBT: spike ΔE_c, higher turn-on and offset, gain only $\exp(\Delta E_v/kT)$, hot-electron launch; graded HBT: no spike, full $\exp(\Delta E_g/kT)$, lower turn-on, higher β.

(d) $V_{p2} = 1.852$ V, $V_{off} = -1.222$ V, $n_s = 1.37 \times 10^{12}$ cm^{-2} at $V_g = 0$ (depletion mode).

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6. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltages respectively)

   (i) $V_g > V_p$ and $V_d = 0V$
   (ii) $V_g > V_p$ and $V_d < V_{d,sat}$
   (iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$
   (iv) $V_g > V_p$ and $V_d > V_{d,sat}$ Marks: 8

   (b) In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \frac{ea^2 N_d}{2\epsilon_s}$ Marks: 6

   (c) The n^+p junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}\ cm^{-3}$ and $N_a = 10^{16}\ cm^{-3}$. The metallurgical channel thickness is $a = 0.40\ \mu m$. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET. Marks: 5

   (d) Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET. Marks: 6

6.(a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltages respectively) (i) $V_g > V_p$ and $V_d = 0V$ (ii) $V_g > V_p$ and $V_d < V_{d,sat}$ (iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$ (iv) $V_g > V_p$ and $V_d > V_{d,sat}$ Marks: 8

1. Theory

1. What is a JFET? (Core Idea)

A **junction field-effect transistor** (JFET) is a resistor whose thickness can be squeezed by a voltage.

- A thin slab of n-type semiconductor, the **channel** (thickness a, doping N_d), carries the current. Its electrons are the moving charges.
- Its ends have contacts: the **source** (S), where electrons enter, and the **drain** (D), where they leave. A positive drain voltage V_d drives a drain current I_D from drain to source.
- In a **single-gate** JFET one heavily doped p^+ region, the **gate** (G), sits on top of the channel; the channel lies between it and the substrate.

2. How the gate controls the current

A pn junction has a **space-charge (depletion) region**: a layer with no free carriers, only fixed ionised dopants, through which no current flows.

- The gate is p^+ and the channel lightly doped n, so the depletion region lies almost entirely in the channel.
- The gate junction is reverse biased. The thickness h of the depletion layer grows with the total reverse voltage across the junction (Poisson's equation, derived in 6(b)):

$$h = \left[\frac{2\epsilon_s\left(V_{bi} - V_g + V(x)\right)}{eN_d}\right]^{1/2}$$

  where V_{bi} is the built-in potential and $V(x)$ the channel voltage at the point x.
- Only the undepleted part, $a - h$, conducts. When $h = a$ the channel is **pinched off**. The gate voltage at which this happens at the source end (with $V_d = 0$) is the **pinch-off voltage** V_p (negative for an n-channel JFET). The paper's condition $V_g > V_p$ means the gate is less negative than V_p: the channel is open at the source in all four cases.

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3. Why the depletion layer is thicker at the drain end

The gate is at V_g along its whole length, but the channel voltage $V(x)$ rises from 0 at the source to V_d at the drain because the current flows through the channel's resistance. The reverse bias $V(x) - V_g$ is therefore smallest at the source and largest at the drain: the depletion layer is a **wedge**, thicker towards the drain.

2. Answer

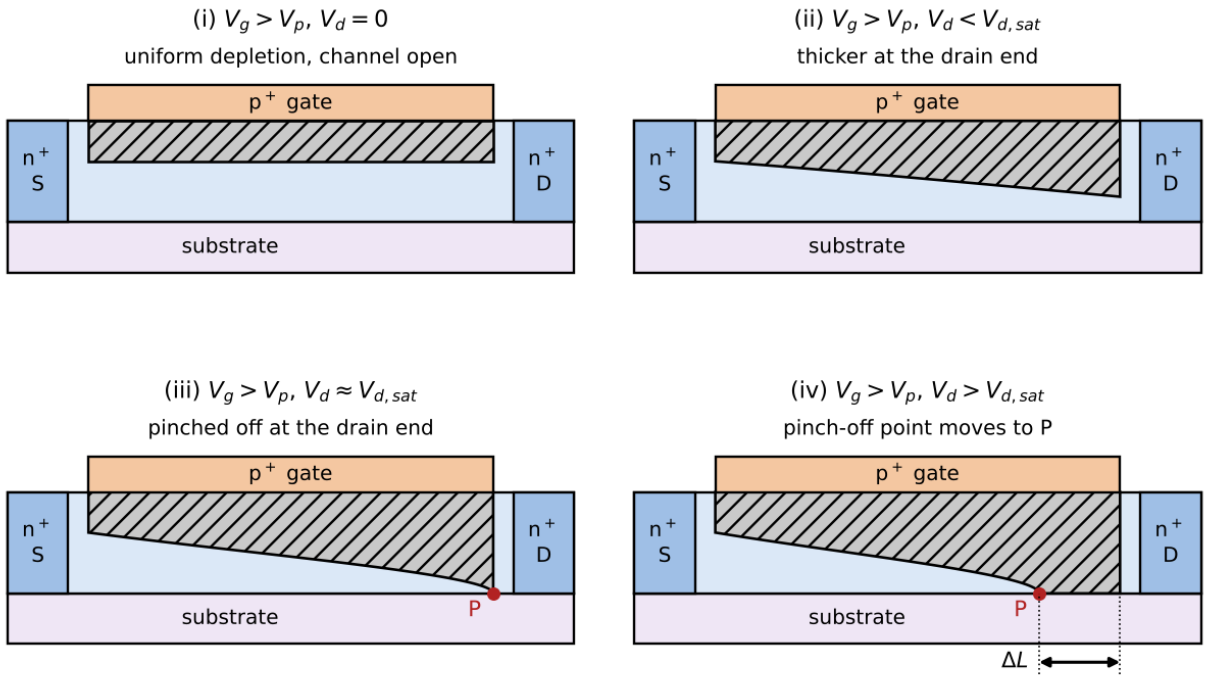


Figure 23: Channel space-charge profiles of a single-gate n-channel JFET at a fixed gate voltage above pinch-off: (i) $V_d = 0$, (ii) V_d below, (iii) at and (iv) above $V_{d,sat}$

Reading the figure. Each panel is a cross-section: p^+ gate on top, the n channel (unhatched) between the n^+ source S and drain D contacts, substrate below. The hatched grey area is the depletion region under the gate. Electrons flow from S to D through the open (unhatched) channel. P marks the pinch-off point.

(i) $V_g > V_p$, $V_d = 0$: with no drain voltage the channel is at 0 V everywhere, so the reverse bias $-V_g$ is the same along the gate. The depletion layer has a **uniform** thickness $h = [2\epsilon_s(V_{bi} - V_g)/(eN_d)]^{1/2} < a$, and the open channel $a - h$ is uniform. No current flows; the channel behaves as a resistor whose value is set by V_g.

(ii) $V_g > V_p$, $V_d < V_{d,sat}$ (linear region): the channel voltage rises from source to drain, so the depletion layer is thinnest at the source and thicker at the drain, but still $h < a$ everywhere: the channel is open from S to D. For small V_d, I_D rises linearly with V_d (a voltage-controlled resistor); as V_d grows the wedge narrows and the I_D–V_d curve bends over.

(iii) $V_g > V_p$, $V_d \approx V_{d,sat}$ (onset of saturation): the depletion layer just reaches the bottom of the channel at the drain end, $h(L) = a$: the channel is pinched off at one point P at the drain. The reverse bias there equals the pinch-off value, $V_d - V_g = -V_p$, so

$$V_{d,sat} = V_g - V_p$$

The current has reached its saturation value $I_{D,sat}$.

(iv) $V_g > V_p$, $V_d > V_{d,sat}$ (saturation): the extra drain voltage moves the pinch-off point P towards the source by a short distance ΔL. A depleted region of length ΔL separates P from the drain, and it takes the excess voltage $V_d - V_{d,sat}$. The voltage at P stays at $V_{d,sat}$, so the open channel S–P carries nearly the same current; electrons reaching P are swept across the depleted gap by its high field. I_D stays at about $I_{D,sat}$, rising only slightly because the channel is shortened to $L - \Delta L$ (channel-length

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modulation).

3. Keyword Explanation

1. Pinch-off

↩ back

The closing of the conducting channel when the depletion layer fills its whole thickness, $h = a$.

2. Saturation drain voltage $V_{d,sat}$

↩ back

The drain voltage at which the channel pinches off at the drain end, $V_{d,sat} = V_g - V_p$; above it the current is nearly constant.

3. Channel-length modulation

↩ back

The slight rise of I_D above $V_{d,sat}$, caused by the pinch-off point moving towards the source.

4. Notes and Remarks

- Pinched off does not mean zero current: in (iii) and (iv) the current is at its largest; the electrons cross the depleted gap at P like electrons crossing a reverse-biased collector junction.
- If $V_g < V_p$, the channel is closed even at the source and $I_D \approx 0$ (cut-off), the case the question excludes.
- Cases (ii) to (iv) are also solved in 5(a) of the 2020–21 paper (extra reading).

6.(b) In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \dfrac{ea^2 N_d}{2\epsilon_s}$ Marks: 6

1. Theory

1. The MESFET (Core Idea)

A **MESFET** (metal–semiconductor FET) is a JFET whose gate is a **Schottky barrier** (a rectifying metal contact) instead of a p^+ region. The n-type channel of thickness a and doping N_d lies on a semi-insulating substrate; the metal gate on top makes a depletion layer of thickness h in the channel. The channel closes (pinch-off) when $h = a$. The **internal pinch-off voltage** V_{p0} is the total potential across the gate junction (built-in potential plus reverse bias) that does this.

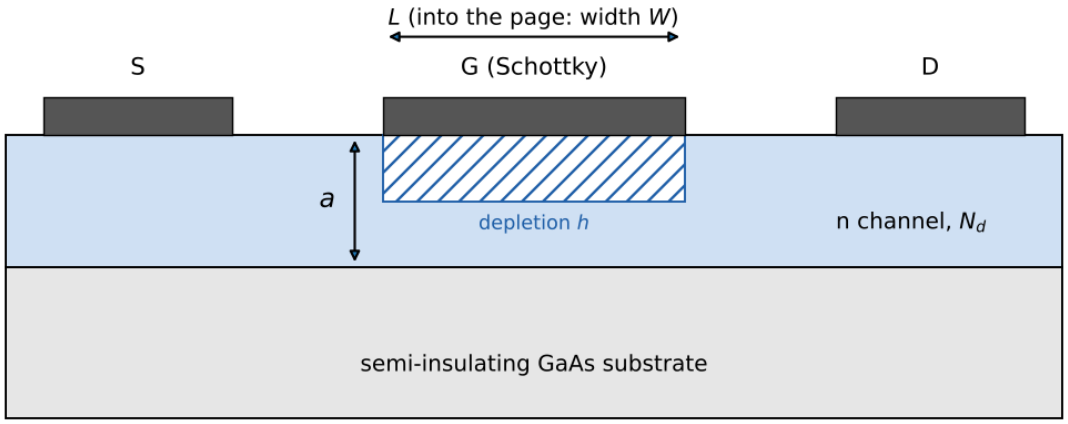


Figure 24: n-channel MESFET: channel thickness a, gate length L, the depletion region of thickness h under the Schottky gate

Reading the figure. The grey block is the semi-insulating GaAs substrate; the light blue layer is the n channel (doping N_d, thickness a). The hatched region under the Schottky gate G is the depletion layer of thickness h. S and D are ohmic contacts. Pinch-off means the hatched region reaches the substrate.

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2. The tools

- **Depletion approximation:** inside the depletion layer ($0 < y < h$, with y measured down from the metal) there are no free electrons, only ionised donors, so $\rho = +eN_d$; below h the channel is neutral. The balancing negative charge is a thin sheet on the metal surface (a metal has no depletion width), so all the voltage is dropped in the semiconductor.
- **Poisson's equation** (one dimension):

$$\frac{d^2\phi}{dy^2} = -\frac{\rho}{\epsilon_s}, \qquad E = -\frac{d\phi}{dy}$$

2. Answer

Assumptions: uniformly doped n channel (N_d), thickness a, Schottky gate with built-in potential V_{bi}, gate voltage V_{gs} (negative), source and drain at 0 ($V_{DS} = 0$, so the depletion layer is uniform); depletion approximation; the 1D field is vertical.

Step 1: Poisson's equation in the depletion layer

With $\rho = +eN_d$ for $0 < y < h$:

$$\frac{dE}{dy} = \frac{eN_d}{\epsilon_s}$$

Step 2: The electric field

Integrate once: $E(y) = \dfrac{eN_d}{\epsilon_s} y + C_1$. At the edge of the depletion layer the channel is neutral, $E(h) = 0$, so $C_1 = -\dfrac{eN_d h}{\epsilon_s}$:

$$E(y) = -\frac{eN_d}{\epsilon_s}(h - y)$$

The field is largest at the metal ($y = 0$), $|E| = eN_d h/\epsilon_s$, and falls linearly to zero at h.

Step 3: The potential

Integrate again with $\phi = -\int E\, dy$ and take $\phi(0) = 0$ at the metal–semiconductor interface:

$$\phi(y) = \frac{eN_d}{\epsilon_s}\left(hy - \frac{y^2}{2}\right)$$

The potential difference across the whole layer is

$$\phi(h) = \frac{eN_d}{\epsilon_s}\left(h^2 - \frac{h^2}{2}\right) = \frac{eN_d h^2}{2\epsilon_s}$$

Step 4: Relate it to the applied voltage

The potential across the depletion layer is the built-in potential of the Schottky junction plus the reverse bias $-V_{gs}$:

$$\frac{eN_d h^2}{2\epsilon_s} = V_{bi} - V_{gs} \quad \Longrightarrow \quad h = \left[\frac{2\epsilon_s(V_{bi} - V_{gs})}{eN_d}\right]^{1/2}$$

Step 5: Pinch-off

The open channel thickness $a - h$ becomes zero when $h = a$. Then the total junction potential is

$$V_{bi} - V_{gs} = \frac{eN_d a^2}{2\epsilon_s}$$

This total potential is the internal pinch-off voltage:

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$$\boxed{V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}}$$

as was to be shown. The gate voltage that pinches the channel off is then $V_p = V_{bi} - V_{p0}$ (Neamen, Sec. 13.2.1). Units: $\mathrm{C} \cdot \mathrm{cm}^2 \cdot \mathrm{cm}^{-3}/(\mathrm{F/cm}) = \mathrm{C/F} = \mathrm{V}$ ✔.

Steps to remember

$\rho = +eN_d \to E(h) = 0 \to E = -eN_d(h-y)/\epsilon_s \to \phi(h) = eN_d h^2/2\epsilon_s \to = V_{bi} - V_{gs} \to$ pinch-off at $h = a \to V_{p0} = ea^2 N_d/2\epsilon_s$.

3. Keyword Explanation

1. Schottky gate

↩ back

A rectifying metal–semiconductor contact used as the gate; its depletion layer lies entirely in the semiconductor.

2. Internal pinch-off voltage V_{p0}

↩ back

The total (built-in plus applied) potential across the gate junction at which the depletion layer fills the channel.

4. Notes and Remarks

- The result is the same as for a one-sided $\mathrm{p^+n}$ JFET gate, because in both cases the whole depletion layer and voltage lie in the channel. The $\mathrm{p^+n}$ version is derived in 5(b) of the 2020–21 paper (extra reading).
- $V_{p0} \propto a^2$: the threshold of a MESFET is very sensitive to the channel thickness. If $V_{p0} < V_{bi}$, then $V_p > 0$ and the MESFET is enhancement mode (normally off).

6.(c) The n^+p junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}\ cm^{-3}$ and $N_a = 10^{16}\ cm^{-3}$. The metallurgical channel thickness is $a = 0.40\ \mu m$. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET. Marks: 5

1. Theory

1. p-channel JFET (Core Idea)

In a p-channel JFET the channel is p-type (doping N_a) and the gate is $\mathrm{n^+}$ ($N_d \gg N_a$). The depletion layer lies almost entirely in the p channel. Repeating the Poisson integration of 6(b) with $\rho = -eN_a$ gives the same magnitude (Neamen, Eq. 13.6):

$$V_{p0} = \frac{ea^2 N_a}{2\epsilon_s}$$

2. Built-in potential and pinch-off voltage

The built-in potential of the $\mathrm{n^+p}$ junction is (Neamen, Eq. 7.10)

$$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

For a p-channel device the gate is made **positive** to reverse bias the junction. The total potential at pinch-off is $V_{bi} + V_p = V_{p0}$, so (Neamen, Eq. 13.8)

$$V_p = V_{p0} - V_{bi}$$

which is positive.

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2. Answer

Given: Si, $T = 300$ K, n$^+$ gate $N_d = 10^{18}$ cm^{-3}, p channel $N_a = 10^{16}$ cm^{-3}, $a = 0.40$ µm $= 0.40 \times 10^{-4}$ cm, $\epsilon_s = 11.7\epsilon_0 = 1.035 \times 10^{-12}$ F/cm, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $kT/e = 0.0259$ V.

Step 1: Internal pinch-off voltage

$$V_{p0} = \frac{eа^2 N_a}{2\epsilon_s} = \frac{(1.6 \times 10^{-19})(0.40 \times 10^{-4})^2 (10^{16})}{2(11.7)(8.85 \times 10^{-14})} = 1.236 \text{ V}$$

Step 2: Built-in potential

$$V_{bi} = 0.0259 \ln\left[\frac{(10^{16})(10^{18})}{(1.5 \times 10^{10})^2}\right] = 0.0259 \times 31.43 = 0.814 \text{ V}$$

Step 3: Pinch-off voltage

$$V_p = V_{p0} - V_{bi} = 1.236 - 0.814 = 0.422 \text{ V}$$

$$\boxed{V_{p0} = 1.24 \text{ V}, \qquad V_p = 0.42 \text{ V}}$$

Meaning: a gate voltage of +0.42 V (reverse bias on the n$^+$p gate) closes the channel; at $V_g = 0$ the channel is open, so the device is depletion mode.

3. Keyword Explanation

1. Pinch-off voltage V_p

↩ back

The gate-to-source voltage that closes the channel at the source end; positive for a p-channel JFET.

4. Notes and Remarks

- This is Neamen's Exercise Problem Ex 13.2 word for word (book answers: $V_{p0} = 1.236$ V, $V_p = 0.422$ V); the numbers here are from the calc script and agree.

6.(d) Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET. Marks: 6

1. Theory

1. Small-signal operation (Core Idea)

A transistor used as an amplifier is biased at a dc operating point (V_{GS}, V_{DS}, I_D), and a small ac signal is added. For small signals the curved $I_D(V_{GS}, V_{DS})$ characteristics can be replaced by their tangent lines, and the device by a linear circuit: the **small-signal equivalent circuit**. Its elements are the partial derivatives at the operating point.

2. The JFET drain current in saturation

In the saturation region the drain current is well approximated by (Neamen, Eq. 13.14)

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$$

where I_{DSS} is the saturation current at $V_{GS} = 0$.

2. Answer

Transconductance. The transconductance is the change in drain current produced by a change in gate voltage at a constant drain voltage (Neamen, Eq. 13.37):

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}}$$

It is the gain of the JFET: how strongly the gate controls the drain current (units S, often mS/mm of gate width). In saturation, differentiating $I_D = I_{DSS}(1 - V_{GS}/V_p)^2$:

$$g_{ms} = \frac{dI_D}{dV_{GS}} = 2I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)\left(-\frac{1}{V_p}\right) = -\frac{2I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right)$$

(Neamen, Eq. 13.41b). It is positive for an n-channel JFET ($V_p < 0$), largest at $V_{GS} = 0$, $g_{ms}(\max) = -2I_{DSS}/V_p$, and zero at $V_{GS} = V_p$.

Small-signal equivalent circuit.

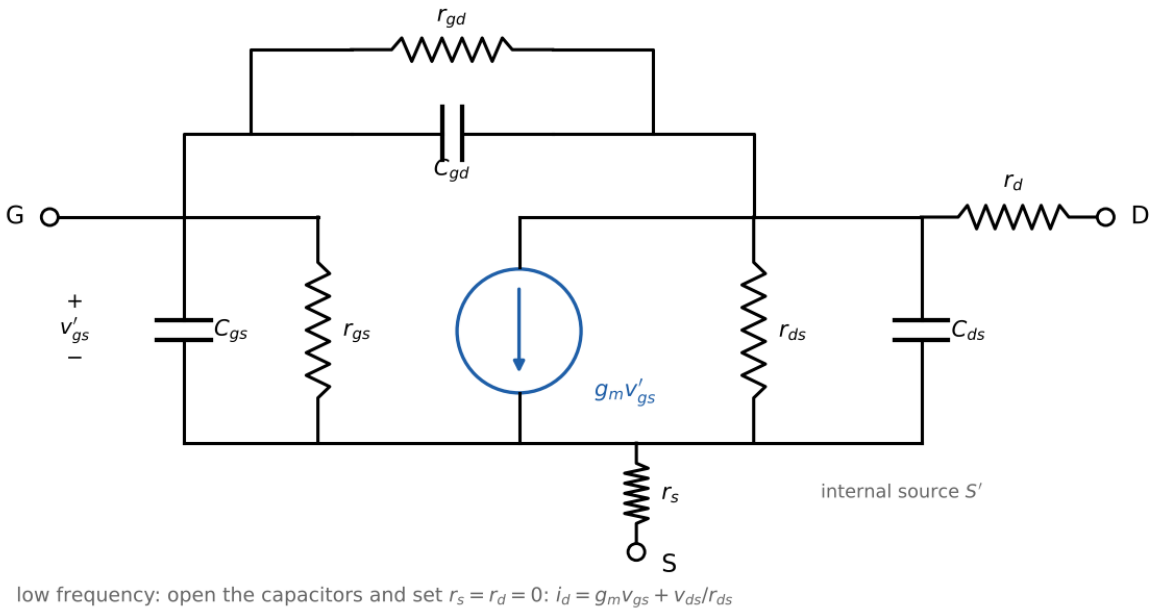


Figure 25: Small-signal equivalent circuit of a JFET with its capacitances and series resistances (after Neamen, Fig. 13.19)

Reading the figure. G, D and S are the external terminals; S' is the internal source node. Between the internal gate and source sit C_{gs} (gate–source junction capacitance) and r_{gs} (its diffusion resistance, very large for a reverse-biased gate); between gate and drain sit C_{gd} and r_{gd}. The circled current source $g_m v'_{gs}$ is controlled by the internal gate–source voltage v'_{gs}. r_{ds} is the output resistance (from channel-length modulation), C_{ds} the drain–source capacitance, and r_s, r_d are the source and drain series resistances.

- **Ideal low-frequency circuit:** the gate draws no current ($r_{gs}, r_{gd} \to \infty$), the capacitances are open, and $r_s = r_d = 0$:
$$i_d = g_m v_{gs} + \frac{v_{ds}}{r_{ds}}, \qquad i_g = 0$$
- **With the source resistance:** $v_{gs} = v'_{gs} + (g_m v'_{gs}) r_s$, so (Neamen, Eq. 13.60)
$$i_d = \frac{g_m}{1 + g_m r_s} v_{gs} = g'_m v_{gs}$$
the source resistance reduces the effective transconductance.
- **High frequency:** C_{gs} and C_{gd} load the input; the cut-off frequency is $f_T = g_m/(2\pi C_G)$ with C_G the gate capacitance.

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3. Keyword Explanation

1. Transconductance

↩ back

$g_m = \partial I_D/\partial V_{GS}$ at constant V_{DS}: the control the gate voltage has over the drain current.

2. Output resistance r_{ds}

↩ back

$r_{ds} = \partial V_{DS}/\partial I_D$ at constant V_{GS}; finite in saturation because of channel-length modulation.

4. Notes and Remarks

- Using Neamen's ideal saturation current instead of the square law gives (Eq. 13.41a)

$$g_{ms} = \frac{3I_{P1}}{V_{p0}}\left[1 - \sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right]$$

  the shape is the same.
- In the linear region g_m is proportional to V_{DS} (Eq. 13.39), so a JFET amplifier is biased in saturation.

5. Final Concise Answer

(a) (i) $V_d = 0$: uniform depletion $h < a$, channel open; (ii) $V_d < V_{d,sat}$: depletion thicker at the drain, channel open, linear I_D; (iii) $V_d \approx V_{d,sat} = V_g - V_p$: channel just pinched off at the drain end; (iv) $V_d > V_{d,sat}$: pinch-off point moves ΔL towards the source, $I_D \approx I_{D,sat}$.

(b) Poisson with $\rho = eN_d$, $E(h) = 0$ gives $\phi(h) = eN_d h^2/(2\epsilon_s) = V_{bi} - V_{gs}$; at pinch-off $h = a$: $V_{p0} = ea^2N_d/(2\epsilon_s)$.

(c) $V_{p0} = ea^2N_a/(2\epsilon_s) = 1.236$ V; $V_{bi} = 0.814$ V; $V_p = V_{p0} - V_{bi} = 0.422$ V.

(d) $g_m = \partial I_D/\partial V_{GS}|_{V_{DS}}$; in saturation $g_{ms} = -(2I_{DSS}/V_p)(1 - V_{GS}/V_p)$. Small-signal circuit: $C_{gs} \parallel r_{gs}$, $C_{gd} \parallel r_{gd}$, current source $g_m v'_{gs}$ with r_{ds} and C_{ds}, series r_s, r_d; ideal: $i_d = g_m v_{gs} + v_{ds}/r_{ds}$; with r_s: $g'_m = g_m/(1 + g_m r_s)$.

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7. (a) If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively. Marks: 8
   (b) Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field. [Marks: 4+2+6]

7.(a) If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively. Marks: 8

1. Theory

1. Lattice constant and misfit (Core Idea)

A crystal is a repeating pattern of atoms; the side of its smallest repeating cube is the **lattice constant** a. When a thin layer of one crystal is grown on a substrate of another (**epitaxy**), the layer's atoms must sit on the substrate's atoms. The mismatch between the natural lattice constant of the layer, a_L, and that of the substrate, a_s, is measured by the **misfit**

$$f = \frac{a_L - a_s}{a_s}$$

2. Vegard's law for InGaAs

$In_xGa_{1-x}As$ is an alloy of InAs and GaAs: a fraction x of the group-III sites holds In and $1-x$ holds Ga. Its lattice constant varies linearly between those of the two binaries (**Vegard's law**):

$$a_L(x) = x\, a_{\text{InAs}} + (1-x)\, a_{\text{GaAs}} = a_{\text{GaAs}} + x\,(a_{\text{InAs}} - a_{\text{GaAs}})$$

Since In atoms are larger, $a_L > a_{\text{GaAs}}$: InGaAs on GaAs is under compressive strain.

3. Critical thickness

A thin mismatched layer can grow **pseudomorphically**: it is squeezed to the substrate's in-plane spacing and stores elastic energy that grows with its thickness. Beyond the **critical thickness** h_c it is cheaper to relax the strain by forming **misfit dislocations** at the interface. The larger the misfit, the thinner the layer that can be grown without defects. A simple estimate (Sze and Ng) is

$$h_c \approx \frac{a_s}{2f}$$

which says h_c is inversely proportional to the misfit. The fuller force-balance model of Matthews and Blakeslee is in the Notes.

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2. Answer

Given: $In_xGa_{1-x}As$ on a GaAs substrate, $h_c = 50$ Å, $a_{InAs} = 6.058$ Å, $a_{GaAs} = 5.653$ Å $= a_s$.

Step 1: Misfit from the critical thickness

From $h_c = a_s/(2f)$:

$$f = \frac{a_s}{2h_c} = \frac{5.653}{2 \times 50} = 0.05653 \approx 5.65\ \%$$

Step 2: Lattice constant of the layer

From $f = (a_L - a_s)/a_s$:

$$a_L = a_s(1 + f) = 5.653 \times 1.05653 = 5.973\ \text{Å}$$

Step 3: Composition from Vegard's law

$$x = \frac{a_L - a_{GaAs}}{a_{InAs} - a_{GaAs}} = \frac{5.973 - 5.653}{6.058 - 5.653} = \frac{0.320}{0.405} = 0.789$$

$$\boxed{x \approx 0.79 : \quad In_{0.79}Ga_{0.21}As}$$

Check: the misfit of InAs itself on GaAs is $(6.058 - 5.653)/5.653 = 7.2$ %; $x = 0.79$ gives $0.79 \times 7.2 = 5.65$ % ✔.

Meaning: a layer only 50 Å thick can tolerate a large misfit, so the composition is In-rich. Thicker layers need less indium (for example $h_c \approx 200$ Å needs $f \approx 1.4$ %, $x \approx 0.2$, the usual pseudomorphic HEMT channel).

3. Keyword Explanation

1. Pseudomorphic layer

↩ back

A strained epitaxial layer that takes the in-plane lattice constant of the substrate, with no dislocations.

2. Misfit dislocation

↩ back

A line defect at the interface that relieves the strain of a layer thicker than h_c.

4. Notes and Remarks

- **Matthews–Blakeslee:** with $b = a_s/\sqrt{2} = 4.0$ Å, $\nu = 1/3$ and 60° dislocations, $h_c = \dfrac{b(1 - \nu\cos^2\alpha)}{2\pi f(1+\nu)\cos\lambda}\left[\ln\dfrac{h_c}{b} + 1\right] = 50$ Å gives $f = 0.0617$ and $x = 0.86$ (calc script). The two models agree that the layer must be In-rich ($x \approx 0.8$); the simple formula is the one the course uses.
- The same rule is used in 1(d) of this paper, and the theory of strain is in 2(b) of the 2020–21 paper (extra reading).

7.(b) Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field. [Marks: 4+2+6]

1. Theory

1. Effective mass (Core Idea)

An electron in a crystal moves through the periodic potential of the ions. Its response to an external force is not that of a free electron; it is summed up in an **effective mass** set by the curvature of the energy band $E(k)$ at the band edge:

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$$

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A sharply curved band (energy rising quickly with k) means a small effective mass; a flat band means a heavy one. The band shape, and so m^*, depends on the crystal: its atoms, its lattice and where in k-space the conduction-band minimum lies.

2. Mobility

In an electric field E an electron accelerates for an average time τ between collisions and then loses its extra momentum. Its average drift velocity is $v_d = \mu E$ with the **mobility** (Neamen, Sec. 5.1)

$$\mu = \frac{e\tau}{m^*}$$

so a lighter effective mass gives a higher mobility.

3. Velocity at high fields

At high fields the electrons gain energy faster than they lose it in collisions; they become "hot" and emit optical phonons, so the velocity stops rising: **velocity saturation**. In GaAs a second effect appears: electrons transfer to a higher, heavier valley (the **transferred-electron** or Gunn effect).

2. Answer

(i) Why the effective mass of electrons differs in GaAs and Si.

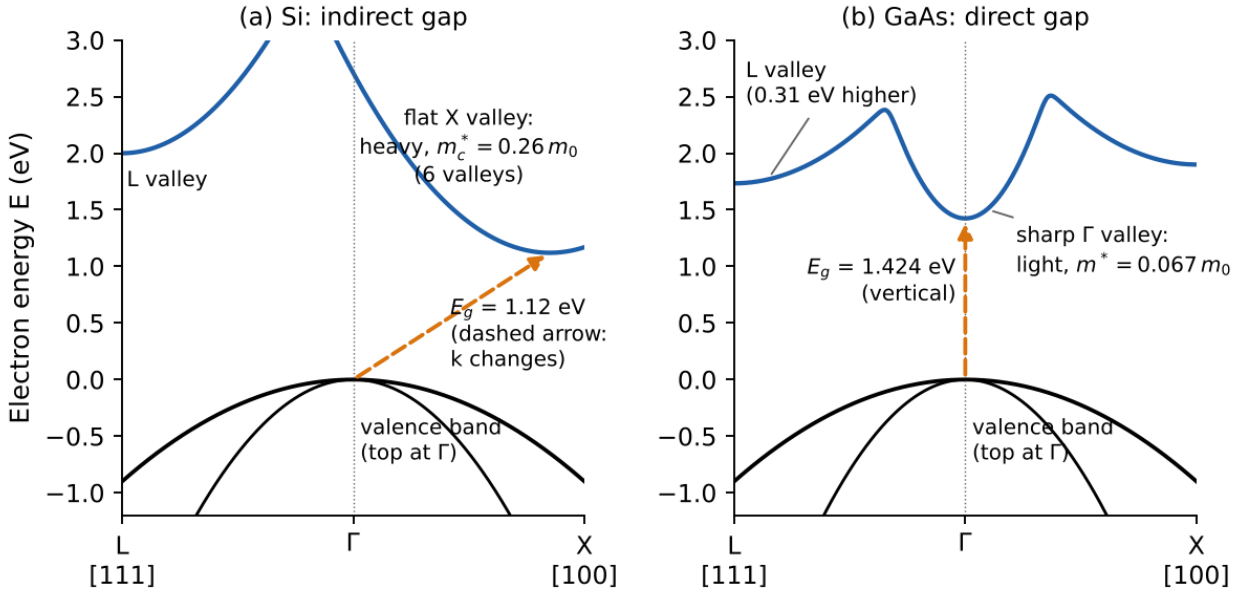


Figure 26: E–k diagrams of Si and GaAs: Si has its conduction-band minimum near X (indirect, heavy, six valleys), GaAs at Γ (direct, light)

Reading the figure. Panel (a), Si: the conduction band (solid blue) has its lowest point near X, along [100], away from Γ; it is a flat, elongated valley (six equivalent ones), and the band gap (dashed orange arrow) is indirect. Panel (b), GaAs: the lowest conduction-band point is at Γ, directly above the valence-band top; the valley is single and very sharply curved. The L valley lies 0.31 eV higher.

- The effective mass is the inverse curvature of $E(k)$ at the conduction-band minimum, and the two crystals have different band structures.
- **GaAs:** the minimum is at Γ ($k = 0$), a single spherical, sharply curved valley, strongly coupled to the valence band across the small direct gap: $m_n^* = 0.067\,m_0$, the same in every direction.
- **Si:** the minima are near X, at about 85 % of the way to the zone edge along the six ⟨100⟩ directions. Each valley is an ellipsoid, flat along its axis ($m_l = 0.98\,m_0$) and less flat across it ($m_t = 0.19\,m_0$). For conduction the average is

$$m_{cn}^* = \frac{3}{1/m_l + 2/m_t} = \frac{3}{1/0.98 + 2/0.19}\,m_0 = 0.26\,m_0$$

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- So the electron effective mass is about 4 times smaller in GaAs (0.067 against 0.26 m_0) because the crystal potential, the location of the conduction-band minimum and its curvature differ.

(ii) Which has the higher electron mobility.

GaAs: $\mu_n \approx 8500$ cm^2/V · s against $\mu_n \approx 1350$ cm^2/V · s for Si at 300 K (Neamen, Table 5.1), about 6 times higher, mainly because of its lighter effective mass ($\mu = e\tau/m^*$) and also its single valley (no intervalley scattering among equivalent valleys at low field).

(iii) Variation of electron mobility with electric field.

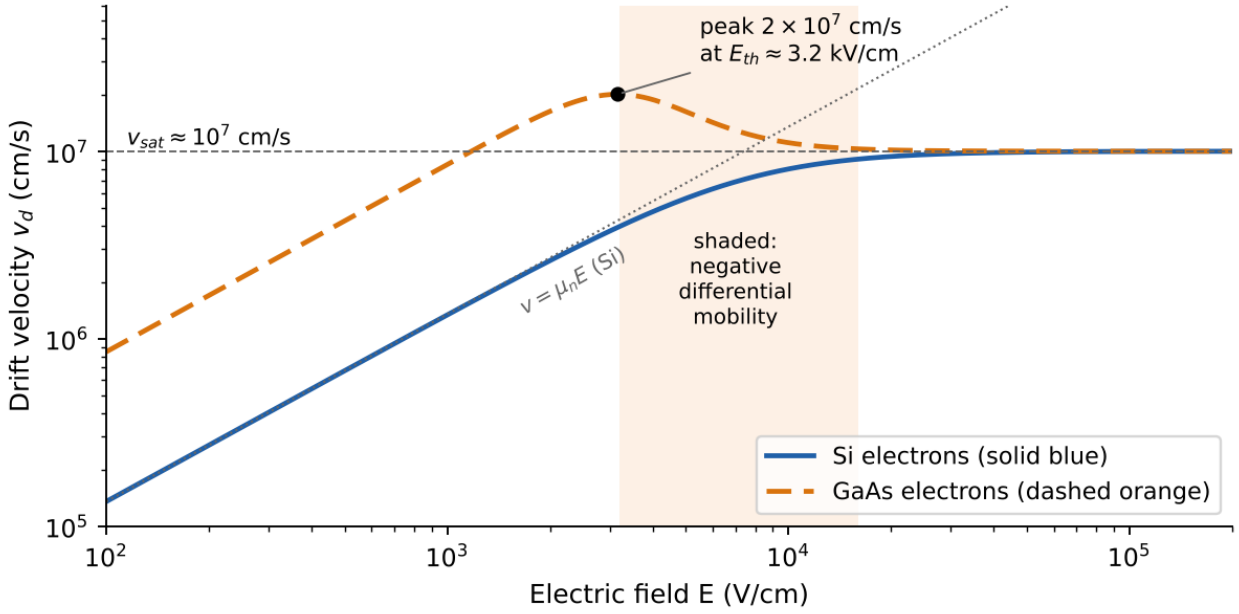


Figure 27: Electron drift velocity against electric field in Si and GaAs (log–log)

Reading the figure. The solid blue curve is Si, the dashed orange curve GaAs. At low fields both rise along straight lines of slope 1 ($v = \mu E$, constant mobility). Si bends over and saturates at $v_{sat} \approx 10^7$ cm/s. GaAs rises faster, peaks at about 2×10^7 cm/s near 3.2 kV/cm, falls in the shaded region of negative differential mobility and then saturates at about 10^7 cm/s.
The field-dependent mobility is $\mu(E) = v_d/E$.

- **Low field (both):** the electrons stay near thermal energy, τ is constant and $v_d = \mu_0 E$: the mobility is constant ($\mu_0 = 1350$ in Si, 8500 in GaAs).
- **Si at higher fields:** above about 10^3–10^4 V/cm the electrons are heated; they lose energy by emitting optical phonons (about 60 meV), so the velocity rises more and more slowly and saturates at $v_{sat} \approx 10^7$ cm/s above about 3×10^4 V/cm. The mobility falls as $\mu \approx v_{sat}/E$; an empirical form is $\mu(E) = \mu_0/[1 + (\mu_0 E/v_{sat})^\beta]^{1/\beta}$ with $\beta \approx 2$ for electrons.
- **GaAs at higher fields:** the velocity rises steeply up to a peak of about 2×10^7 cm/s at the threshold field $E_{th} \approx 3.2$ kV/cm. Above it, electrons gain enough energy (0.31 eV) to scatter into the upper L valley, where the effective mass is much larger (about $0.55\,m_0$) and the mobility low (about 100 cm^2/V · s). The average velocity therefore **falls** as the field rises: **negative differential mobility**, $dv/dE < 0$ (the basis of the Gunn diode). At still higher fields almost all electrons are in the L valley and the velocity saturates at about 10^7 cm/s, the same as in Si.
- **Result:** GaAs has the advantage at low and moderate fields (high mobility, velocity overshoot), which is why it is used for high-speed devices; at very high fields both materials reach about the same saturation velocity.

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3. Keyword Explanation

1. Conductivity effective mass

↩ back

The average $3/(1/m_l + 2/m_t)$ of the effective masses of an ellipsoidal valley, which sets the mobility in a cubic crystal.

2. Negative differential mobility

↩ back

A range of fields in which the drift velocity falls as the field rises, $dv_d/dE < 0$; in GaAs it comes from the transfer of electrons to the heavy L valley.

4. Notes and Remarks

- The same topics are solved in detail in 1(a) and 1(b) of the 2020–21 paper (extra reading); the figures are reused from there.
- Values follow Neamen (mobilities, Table 5.1; masses, Appendix B) and Sze and Ng (L-valley mass and the velocity–field curves).

5. Final Concise Answer

(a) $f = a_s/(2h_c) = 5.653/100 = 0.0565$; $a_L = a_s(1+f) = 5.973$ Å; $x = (5.973-5.653)/(6.058-5.653) = 0.79$: $In_{0.79}Ga_{0.21}As$ (Matthews–Blakeslee: $x \approx 0.86$).

(b) $m^* = \hbar^2/(d^2E/dk^2)$ depends on the band structure: GaAs has a single sharply curved Γ valley, $m^* = 0.067\,m_0$; Si has six flat X ellipsoids, $m^*_{cn} = 0.26\,m_0$. GaAs has the higher mobility (8500 against 1350 $cm^2/V \cdot s$, $\mu = e\tau/m^*$). With field: constant μ at low field; Si saturates at 10^7 cm/s, $\mu \approx v_{sat}/E$; GaAs peaks at 2×10^7 cm/s at 3.2 kV/cm, then shows negative differential mobility (transfer to the heavy L valley) and saturates at about 10^7 cm/s.

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FORMULA SHEET

EEE459

Compound Semiconductor and Hetero-Junction Devices

Formula Sheet

Session 2017–18 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2021, and Term Test 1

TOPICS

1 Bands, effective mass and transport — 7 formulas

2 Alloys and strain — 6 formulas

3 Metal–semiconductor contacts — 6 formulas

4 Heterojunctions and HEMT — 8 formulas

5 Bipolar transistors and HBTs — 10 formulas

6 JFET and MESFET — 8 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

★ **Effective mass**	$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	m^* = effective mass (kg) $\hbar$ = reduced Planck constant, $h/2\pi$ (J s) E = electron energy (J) k = wave number (m^{-1}) curvature of $E(k)$ at the band edge	1(b), 7(b)

the name you search for

the formula, exactly as derived

every new symbol with its unit

link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

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1 Bands, effective mass and transport **4**

2 Alloys and strain **5**

3 Metal–semiconductor contacts **6**

4 Heterojunctions and HEMT **7**

5 Bipolar transistors and HBTs **8**

6 JFET and MESFET **10**

7 Key formulas **11**

8 Symbol index **12**

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Conventions (Neamen): e is the magnitude of the electron charge; potentials (ϕ, V) in volts, energies in eV; $V_t = kT/e = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r \epsilon_0$, $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm; $n_i = 1.5 \times 10^{10}$ cm^{-3} (Si). References point to the solutions of the 2017–18 paper (Q7 is the term test).

Columns: Title · Equation · Notation · Ref. (reference).

1. Bands, effective mass and transport:

Title	Equation	Notation	Ref.
★ **Effective mass**	$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	m^* = effective mass (kg) $\hbar$ = reduced Planck constant, $h/2\pi$ (J s) E = electron energy (J) k = wave number (m^{-1}) curvature of $E(k)$ at the band edge	1(b), 7(b)
Band edge parabola	$E - E_c = \frac{\hbar^2 k^2}{2m^*}$	E_c = conduction-band edge (J)	1(b)
Finite difference	$\frac{d^2E}{dk^2} \approx \frac{E_{i+1} - 2E_i + E_{i-1}}{(\Delta k)^2}$	E_i = energy read from the E–k diagram at point i (J) Δk = step in k between the points (m^{-1})	1(b)
Si conductivity mass	$m^*_{cn} = \frac{3}{1/m_l + 2/m_t} = 0.26\, m_0$	m^*_{cn} = conductivity effective mass of electrons (kg) m_0 = free-electron mass (kg) m_l = longitudinal effective mass ($0.98\, m_0$) m_t = transverse effective mass ($0.19\, m_0$) GaAs: $0.067\, m_0$	7(b)
★ **Mobility**	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$	μ = mobility (cm^2/V s) e = magnitude of the electron charge (C) τ = mean time between collisions (s) v_d = drift velocity (cm/s) E = electric field (V/cm) Si 1350, GaAs 8500 cm^2/V · s	7(b)
High-field mobility	$\mu(E) = v_d/E$; $v_{sat} \approx 10^7$ cm/s	v_{sat} = saturation velocity (cm/s) GaAs peak 2×10^7 cm/s at 3.2 kV/cm	7(b)
Photon energy	$E_g = \frac{hc}{\lambda} = \frac{1.24}{\lambda(\mu\text{m})}$ eV	E_g = band gap energy (eV) h = Planck's constant (J s) c = speed of light (m/s) λ = photon wavelength (µm)	1(d)

Table 1: Bands, effective mass and transport

2. Alloys and strain:

Title	Equation	Notation	Ref.
★ **GaAsP gaps**	$E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$	E_Γ = band gap to the Γ valley (eV) E_X = band gap to the X valleys (eV) x = P fraction in $GaAs_{1-x}P_x$ direct for $x < 0.45$	1(c)
★ **Vegard (ternary)**	$a(x) = x\,a_{AC} + (1-x)\,a_{BC}$	a = lattice constant of the alloy $A_xB_{1-x}C$ (Å) x = mole fraction of AC in the alloy a_{AC}, a_{BC} = lattice constants of the binaries AC and BC (Å) linear between the binaries	7(a)
Vegard (quaternary)	$Q(x,y) = xyQ_1 + x(1-y)Q_2 + (1-x)yQ_3 + (1-x)(1-y)Q_4$	Q = a property (lattice constant, Å) of $A_xB_{1-x}C_yD_{1-y}$ y = fraction of C on its sublattice Q_1, Q_2, Q_3, Q_4 = the same property of AC, AD, BC and BD bilinear in x, y	1(d)
★ **Misfit**	$f = \dfrac{a_L - a_s}{a_s}$	f = lattice misfit a_L = lattice constant of the layer (Å) a_s = lattice constant of the substrate (Å)	1(d), 3(b), 7(a)
★ **Critical thickness**	$h_c \approx \dfrac{a_s}{2f}$	h_c = critical thickness (Å) Sze and Ng estimate	1(d), 3(b), 7(a)
Matthews–Blakeslee	$h_c = \dfrac{b(1-\nu\cos^2\alpha)}{2\pi f(1+\nu)\cos\lambda}\left[\ln\dfrac{h_c}{b}+1\right]$	b = length of the Burgers vector, $a/\sqrt{2}$ (Å) ν = Poisson ratio α = angle between the dislocation line and its Burgers vector (60°) λ = angle between the slip direction and the interface direction normal to the dislocation (60°)	7(a)

Table 2: Alloys and strain

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3. Metal–semiconductor contacts:

Title	Equation	Notation	Ref.
★ **Schottky barrier**	$\phi_{B0} = \phi_m - \chi$ (n); $\phi_{Bp} = \chi + E_g/e - \phi_m$ (p)	ϕ_{B0} = barrier height for electrons, n-type (V) ϕ_m = metal work function (V) χ = electron affinity of the semiconductor (V) ϕ_{Bp} = barrier height for holes, p-type (V) E_g = band gap energy (eV) e = magnitude of the electron charge (C) ϕ_s = work function of the semiconductor (V) rectifying if $\phi_m > \phi_s$ (n), $\phi_m < \phi_s$ (p)	2(c), 3(a)
Fermi level below E_c	$\phi_n = V_t \ln(N_c/N_d)$	$\phi_n = (E_c - E_F)/e$ in the n-type semiconductor (V) V_t = thermal voltage, kT/e (V) N_c = effective density of states in the conduction band (cm^{-3}) N_d = donor concentration (cm^{-3})	2(c)
★ **Built-in potential**	$V_{bi} = \phi_{B0} - \phi_n$	V_{bi} = built-in potential (V)	2(c)
★ **Depletion width**	$x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d}\right]^{1/2}$	x_n = depletion width in the semiconductor (cm) ϵ_s = permittivity of the semiconductor (F/cm) V_R = reverse bias (V)	2(c), 3(c)
Peak field	$\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s}$	E_{max} = peak electric field, at the metal (V/cm)	2(c)
Tunnelling contact	$R_c \propto \exp\left[\frac{2\sqrt{\epsilon_s m^*}}{\hbar}\frac{\phi_{Bn}}{\sqrt{N_d}}\right]$	R_c = specific contact resistance ($\Omega\ \mathrm{cm}^2$) m^* = effective mass (kg) $\hbar$ = reduced Planck constant (J s) ϕ_{Bn} = barrier height for electrons (V) ohmic for $N_d \gtrsim 10^{19}\ \mathrm{cm}^{-3}$	3(c)

Table 3: Metal–semiconductor contacts

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4. Heterojunctions and HEMT:

Title	Equation	Notation	Ref.
★ **Anderson offsets**	$\Delta E_c = e(\chi_n - \chi_P),\ \Delta E_v = \Delta E_g - \Delta E_c$	ΔE_c = conduction-band offset (eV) e = magnitude of the electron charge (C) χ_n = electron affinity of the narrow-gap n side (V) χ_P = electron affinity of the wide-gap P side (V) ΔE_v = valence-band offset (eV) ΔE_g = difference of the two band gaps (eV)	2(a), 2(b)
★ **Heterojunction** V_{bi}	$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0}N_{vn}}{p_{n0}N_{vP}}\right)$	V_{bi} = built-in potential (V) k = Boltzmann constant (eV/K) T = temperature (K) p_{P0} = equilibrium hole concentration in the P side (cm^{-3}) p_{n0} = equilibrium hole concentration in the n side (cm^{-3}) N_{vn}, N_{vP} = effective densities of states in the valence band of the n and P sides (cm^{-3}) $p_{n0} = n_{in}^2/N_d$ n_{in} = intrinsic carrier concentration of the n side (cm^{-3}) N_d = donor concentration of the n side (cm^{-3})	2(a)
Depletion widths	$x_n = \left[\frac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2},$ $N_{dn}x_n = N_{aP}x_P$	x_n, x_P = depletion widths in the n and P sides (cm) ϵ_n, ϵ_P = permittivities of the n and P sides (F/cm) N_{dn} = donor concentration in the n side (cm^{-3}) N_{aP} = acceptor concentration in the P side (cm^{-3})	2(a)
AlGaAs gap	$E_g(x) \approx 1.424 + (E_{g,\mathrm{AlAs}} - 1.424)x$	E_g = band gap energy (eV) x = Al fraction in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ $E_{g,\mathrm{AlAs}}$ = band gap of AlAs (eV) Vegard; $\Delta E_c = \Delta E_v$ assumed in 2(b)	2(b)
Quantum-well levels	$E_n = \frac{\hbar^2\pi^2 n^2}{2m^* w^2}$	E_n = energy of level n above the band edge in the well (J) $\hbar$ = reduced Planck constant (J s) n = level number (1, 2, ...) m^* = effective mass (kg) w = well width (m) 2D density of states $m^*/(\pi\hbar^2)$	5(b)
★ **2DEG density**	$n_s = \frac{\epsilon_N}{e(d + \Delta d)}(V_g - V_{off})$	n_s = sheet density of the 2DEG (cm^{-2}) ϵ_N = permittivity of the AlGaAs (F/cm) d = AlGaAs thickness, $d_d + d_i$ (cm) d_d = doped AlGaAs thickness (cm) d_i = undoped spacer thickness (cm) Δd = distance of the 2DEG from the interface (about 80 Å) V_g = gate voltage (V) V_{off} = off voltage (V)	5(b), 5(d)

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Title	Equation	Notation	Ref.
★ **HEMT off voltage**	$V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2},\ V_{p2} = \frac{eN_d d_d^2}{2\epsilon_N}$	ϕ_B = Schottky barrier height of the gate (V) V_{p2} = pinch-off voltage of the doped AlGaAs (V) N_d = donor concentration in the AlGaAs (cm^{-3})	5(d)
HEMT current	$I_D(\text{sat}) = \frac{\epsilon_N W}{d + \Delta d}(V_g - V_{off} - V_0)v_{sat}$	$I_D(\text{sat})$ = saturated drain current (A) W = gate width (cm) V_0 = channel voltage at velocity saturation, LE_s (V) L = gate length (cm) E_s = field at which the velocity saturates (V/cm) v_{sat} = saturation velocity (cm/s)	5(b)

Table 4: Heterojunctions and HEMT

5. Bipolar transistors and HBTs:

Title	Equation	Notation	Ref.
★ **Ebers–Moll**	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R$	I_C, I_E = collector and emitter currents (A) α_F = forward common-base current gain α_R = reverse common-base current gain I_F = forward diode current, $I_{ES}(e^{V_{BE}/V_t} - 1)$ (A) I_R = reverse diode current, $I_{CS}(e^{V_{BC}/V_t} - 1)$ (A) I_{ES}, I_{CS} = emitter and collector saturation currents (A) V_{BE}, V_{BC} = base–emitter and base–collector voltages (V) V_t = thermal voltage, kT/e (V)	4(a)
Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS}$		4(a)
Avalanche multiplication	$M = \frac{1}{1 - (V/BV_{CBO})^n}$	M = avalanche multiplication factor V = reverse collector–base voltage (V) BV_{CBO} = collector–base breakdown voltage, emitter open (V) n = empirical exponent (3 to 6)	4(c)
★ **Open-base breakdown**	$BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$	BV_{CEO} = collector–emitter breakdown voltage, base open (V) α = common-base current gain β = common-emitter current gain, $\alpha/(1-\alpha)$ breakdown at $\alpha M = 1$	4(c)
Emitter minority holes	$p_{E0} = \frac{n_i^2}{N_E}$	p_{E0} = equilibrium hole concentration in the emitter (cm^{-3}) n_i = intrinsic carrier concentration (cm^{-3}) N_E = emitter doping (cm^{-3})	4(d)

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Title	Equation	Notation	Ref.
★ **Bandgap narrowing**	$p'_{E0} = \frac{n_i^2}{N_E} \exp\left(\frac{\Delta E_g}{kT}\right)$	$p'_{E0} = p_{E0}$ with band-gap narrowing (cm^{-3}) ΔE_g = band-gap narrowing (eV), from Fig. 1 k = Boltzmann constant (eV/K) T = temperature (K)	4(b), 4(d)
Injection efficiency	$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}}$	γ = emitter injection efficiency J_{nE} = electron current density injected into the base ($\mathrm{A/cm}^2$) J_{pE} = hole current density injected into the emitter ($\mathrm{A/cm}^2$)	5(a)
Tunnelling probability	$T \approx \exp\left(-2d\sqrt{2m^* V_0}/\hbar\right)$	T = tunnelling probability d = barrier thickness (m) m^* = effective mass (kg) V_0 = barrier height above the particle's energy (J) $\hbar$ = reduced Planck constant (J s)	5(a)
★ **Kirk effect**	$n_c = \frac{J_C}{e v_{sat}}$; $J_K = e v_{sat}\left[N_C + \frac{2\epsilon_s(V_{CB} + V_{bi})}{e W_C^2}\right]$	n_c = electron concentration in the collector depletion region (cm^{-3}) J_C = collector current density ($\mathrm{A/cm}^2$) e = magnitude of the electron charge (C) v_{sat} = saturation velocity (cm/s) J_K = current density at the onset of the Kirk effect ($\mathrm{A/cm}^2$) N_C = collector doping (cm^{-3}) ϵ_s = permittivity of the semiconductor (F/cm) V_{CB} = collector–base reverse voltage (V) V_{bi} = built-in potential of the base–collector junction (V) W_C = thickness of the collector layer (cm) onset when $n_c \approx N_C$	5(c)
HBT injection ratio	$\frac{J_{pE}}{J_{nE}} \propto \exp\left(-\frac{\Delta E_g}{kT}\right)$ (graded); $\exp(-\Delta E_v/kT)$ (abrupt)	ΔE_v = valence-band offset (eV) graded junction: ΔE_g, abrupt: ΔE_v	5(c)

Table 5: Bipolar transistors and HBTs

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6. JFET and MESFET:

Title	Equation	Notation	Ref.
Depletion thickness	$h = \left[\frac{2\epsilon_s(V_{bi} - V_{gs} + V(x))}{eN_d}\right]^{1/2}$	h = depletion thickness under the gate (cm) ϵ_s = permittivity of the semiconductor (F/cm) V_{bi} = built-in potential of the gate junction (V) V_{gs} = gate–source voltage (V) $V(x)$ = channel voltage at x (V) x = distance along the channel from the source (cm) e = magnitude of the electron charge (C) N_d = donor concentration in the channel (cm^{-3}) thicker towards the drain	6(a), 6(b)
★ **Internal pinch-off**	$V_{p0} = \frac{ea^2N_d}{2\epsilon_s}$ (N_a for p channel)	V_{p0} = internal pinch-off voltage (V) a = channel thickness (cm) N_a = acceptor concentration in a p channel (cm^{-3})	6(b), 6(c)
★ **Pinch-off voltage**	$V_p = V_{bi} - V_{p0}$ (n); $V_p = V_{p0} - V_{bi}$ (p)	V_p = pinch-off voltage (V)	6(c)
pn built-in potential	$V_{bi} = V_t \ln\left(\frac{N_aN_d}{n_i^2}\right)$	V_t = thermal voltage, kT/e (V) N_a, N_d = acceptor and donor concentrations of the gate and channel (cm^{-3}) n_i = intrinsic carrier concentration (cm^{-3})	6(c)
Saturation drain voltage	$V_{d,sat} = V_g - V_p$	$V_{d,sat}$ = drain voltage at saturation (V) V_g = gate voltage (V)	6(a)
Square law	$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$	I_D = drain current (A) I_{DSS} = drain current at $V_{GS} = 0$ (A) V_{GS} = gate–source voltage (V)	6(d)
★ **Transconductance**	$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right\|_{V_{DS}}$; $g_{ms} = -\frac{2I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right)$	g_m = transconductance (S) V_{DS} = drain–source voltage (V) g_{ms} = transconductance in saturation (S)	6(d)
Source resistance	$g'_m = \frac{g_m}{1 + g_m r_s}$	g'_m = transconductance with a source resistance (S) r_s = source resistance (Ω)	6(d)

Table 6: JFET and MESFET

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7. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Effective mass	$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$
Mobility	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$
GaAsP gaps	$E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$
Vegard (ternary)	$a(x) = x\,a_{AC} + (1-x)\,a_{BC}$
Misfit	$f = \frac{a_L - a_s}{a_s}$
Critical thickness	$h_c \approx \frac{a_s}{2f}$
Schottky barrier	$\phi_{B0} = \phi_m - \chi$ (n); $\phi_{Bp} = \chi + E_g/e - \phi_m$ (p)
Built-in potential	$V_{bi} = \phi_{B0} - \phi_n$
Depletion width	$x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$
Anderson offsets	$\Delta E_c = e(\chi_n - \chi_P)$, $\Delta E_v = \Delta E_g - \Delta E_c$
Heterojunction V_{bi}	$eV_{bi} = \Delta E_v + kT\ln\left(\frac{p_{P0}N_{vn}}{p_{n0}N_{vP}}\right)$
2DEG density	$n_s = \frac{\epsilon_N}{e(d+\Delta d)}(V_g - V_{off})$
HEMT off voltage	$V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2}$, $V_{p2} = \frac{eN_d d_d^2}{2\epsilon_N}$
Ebers–Moll	$I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$
Open-base breakdown	$BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$
Bandgap narrowing	$p'_{E0} = \frac{n_i^2}{N_E}\exp\left(\frac{\Delta E_g}{kT}\right)$
Kirk effect	$n_c = \frac{J_C}{ev_{sat}}$; $J_K = ev_{sat}\left[N_C + \frac{2\epsilon_s(V_{CB}+V_{bi})}{eW_C^2}\right]$
Internal pinch-off	$V_{p0} = \frac{ea^2N_d}{2\epsilon_s}$ (N_a for p channel)
Pinch-off voltage	$V_p = V_{bi} - V_{p0}$ (n); $V_p = V_{p0} - V_{bi}$ (p)
Transconductance	$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right\|_{V_{DS}}$; $g_{ms} = -\frac{2I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right)$

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8. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
m^*	effective mass (kg)	Effective mass
$\hbar$	reduced Planck constant, $h/2\pi$ (J s)	Effective mass
E	electron energy (J)	Effective mass
k	wave number (m^{-1})	Effective mass
E_c	conduction-band edge (J)	Band edge parabola
E_i	energy read from the E–k diagram at point i (J)	Finite difference
Δk	step in k between the points (m^{-1})	Finite difference
m^*_{cn}	conductivity effective mass of electrons (kg)	Si conductivity mass
m_0	free-electron mass (kg)	Si conductivity mass
m_l	longitudinal effective mass ($0.98\,m_0$)	Si conductivity mass
m_t	transverse effective mass ($0.19\,m_0$)	Si conductivity mass
μ	mobility ($\mathrm{cm}^2/\mathrm{V\ s}$)	Mobility
e	magnitude of the electron charge (C)	Mobility
τ	mean time between collisions (s)	Mobility
v_d	drift velocity (cm/s)	Mobility
E	electric field (V/cm)	Mobility
v_{sat}	saturation velocity (cm/s)	High-field mobility
E_g	band gap energy (eV)	Photon energy
h	Planck's constant (J s)	Photon energy
c	speed of light (m/s)	Photon energy
λ	photon wavelength (µm)	Photon energy
E_Γ	band gap to the Γ valley (eV)	GaAsP gaps
E_X	band gap to the X valleys (eV)	GaAsP gaps
x	P fraction in $\mathrm{GaAs}_{1-x}\mathrm{P}_x$	GaAsP gaps
a	lattice constant of the alloy $\mathrm{A}_x\mathrm{B}_{1-x}\mathrm{C}$ (Å)	Vegard (ternary)
x	mole fraction of AC in the alloy	Vegard (ternary)
a_{AC}, a_{BC}	lattice constants of the binaries AC and BC (Å)	Vegard (ternary)
Q	a property (lattice constant, Å) of $\mathrm{A}_x\mathrm{B}_{1-x}\mathrm{C}_y\mathrm{D}_{1-y}$	Vegard (quaternary)
y	fraction of C on its sublattice	Vegard (quaternary)
Q_1, Q_2, Q_3, Q_4	the same property of AC, AD, BC and BD	Vegard (quaternary)
f	lattice misfit	Misfit
a_L	lattice constant of the layer (Å)	Misfit
a_s	lattice constant of the substrate (Å)	Misfit
h_c	critical thickness (Å)	Critical thickness
b	length of the Burgers vector, $a/\sqrt{2}$ (Å)	Matthews–Blakeslee
ν	Poisson ratio	Matthews–Blakeslee

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Symbol	Meaning	First used in
α	angle between the dislocation line and its Burgers vector (60°)	Matthews–Blakeslee
λ	angle between the slip direction and the interface direction normal to the dislocation (60°)	Matthews–Blakeslee
ϕ_{B0}	barrier height for electrons, n-type (V)	Schottky barrier
ϕ_m	metal work function (V)	Schottky barrier
χ	electron affinity of the semiconductor (V)	Schottky barrier
ϕ_{Bp}	barrier height for holes, p-type (V)	Schottky barrier
ϕ_s	work function of the semiconductor (V)	Schottky barrier
ϕ_n	$(E_c - E_F)/e$ in the n-type semiconductor (V)	Fermi level below E_c
V_t	thermal voltage, kT/e (V)	Fermi level below E_c
N_c	effective density of states in the conduction band (cm^{-3})	Fermi level below E_c
N_d	donor concentration (cm^{-3})	Fermi level below E_c
V_{bi}	built-in potential (V)	Built-in potential
x_n	depletion width in the semiconductor (cm)	Depletion width
ϵ_s	permittivity of the semiconductor (F/cm)	Depletion width
V_R	reverse bias (V)	Depletion width
E_{max}	peak electric field, at the metal (V/cm)	Peak field
R_c	specific contact resistance ($\Omega\ \mathrm{cm}^2$)	Tunnelling contact
ϕ_{Bn}	barrier height for electrons (V)	Tunnelling contact
ΔE_c	conduction-band offset (eV)	Anderson offsets
χ_n	electron affinity of the narrow-gap n side (V)	Anderson offsets
χ_P	electron affinity of the wide-gap P side (V)	Anderson offsets
ΔE_v	valence-band offset (eV)	Anderson offsets
ΔE_g	difference of the two band gaps (eV)	Anderson offsets
k	Boltzmann constant (eV/K)	Heterojunction V_{bi}
T	temperature (K)	Heterojunction V_{bi}
p_{P0}	equilibrium hole concentration in the P side (cm^{-3})	Heterojunction V_{bi}
p_{n0}	equilibrium hole concentration in the n side (cm^{-3})	Heterojunction V_{bi}
N_{vn}, N_{vP}	effective densities of states in the valence band of the n and P sides (cm^{-3})	Heterojunction V_{bi}
n_{in}	intrinsic carrier concentration of the n side (cm^{-3})	Heterojunction V_{bi}
x_n, x_P	depletion widths in the n and P sides (cm)	Depletion widths
ϵ_n, ϵ_P	permittivities of the n and P sides (F/cm)	Depletion widths
N_{dn}	donor concentration in the n side (cm^{-3})	Depletion widths
N_{aP}	acceptor concentration in the P side (cm^{-3})	Depletion widths
x	Al fraction in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$	AlGaAs gap
$E_{g,\mathrm{AlAs}}$	band gap of AlAs (eV)	AlGaAs gap
E_n	energy of level n above the band edge in the well (J)	Quantum-well levels
n	level number (1, 2, …)	Quantum-well levels

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Symbol	Meaning	First used in
w	well width (m)	Quantum-well levels
n_s	sheet density of the 2DEG (cm^{-2})	2DEG density
ϵ_N	permittivity of the AlGaAs (F/cm)	2DEG density
d	AlGaAs thickness, $d_d + d_i$ (cm)	2DEG density
d_d	doped AlGaAs thickness (cm)	2DEG density
d_i	undoped spacer thickness (cm)	2DEG density
Δd	distance of the 2DEG from the interface (about 80 Å)	2DEG density
V_g	gate voltage (V)	2DEG density
V_{off}	off voltage (V)	2DEG density
ϕ_B	Schottky barrier height of the gate (V)	HEMT off voltage
V_{p2}	pinch-off voltage of the doped AlGaAs (V)	HEMT off voltage
$I_D(\mathrm{sat})$	saturated drain current (A)	HEMT current
W	gate width (cm)	HEMT current
V_0	channel voltage at velocity saturation, LE_s (V)	HEMT current
L	gate length (cm)	HEMT current
E_s	field at which the velocity saturates (V/cm)	HEMT current
I_C, I_E	collector and emitter currents (A)	Ebers–Moll
α_F	forward common-base current gain	Ebers–Moll
α_R	reverse common-base current gain	Ebers–Moll
I_F	forward diode current, $I_{ES}(e^{V_{BE}/V_t} - 1)$ (A)	Ebers–Moll
I_R	reverse diode current, $I_{CS}(e^{V_{BC}/V_t} - 1)$ (A)	Ebers–Moll
I_{ES}, I_{CS}	emitter and collector saturation currents (A)	Ebers–Moll
V_{BE}, V_{BC}	base–emitter and base–collector voltages (V)	Ebers–Moll
M	avalanche multiplication factor	Avalanche multiplication
V	reverse collector–base voltage (V)	Avalanche multiplication
BV_{CBO}	collector–base breakdown voltage, emitter open (V)	Avalanche multiplication
n	empirical exponent (3 to 6)	Avalanche multiplication
BV_{CEO}	collector–emitter breakdown voltage, base open (V)	Open-base breakdown
α	common-base current gain	Open-base breakdown
β	common-emitter current gain, $\alpha/(1-\alpha)$	Open-base breakdown
p_{E0}	equilibrium hole concentration in the emitter (cm^{-3})	Emitter minority holes
n_i	intrinsic carrier concentration (cm^{-3})	Emitter minority holes
N_E	emitter doping (cm^{-3})	Emitter minority holes
p'_{E0}	p_{E0} with band-gap narrowing (cm^{-3})	Bandgap narrowing
ΔE_g	band-gap narrowing (eV), from Fig. 1	Bandgap narrowing
γ	emitter injection efficiency	Injection efficiency
J_{nE}	electron current density injected into the base ($\mathrm{A/cm}^2$)	Injection efficiency
J_{pE}	hole current density injected into the emitter ($\mathrm{A/cm}^2$)	Injection efficiency
T	tunnelling probability	Tunnelling probability

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Symbol	Meaning	First used in
d	barrier thickness (m)	Tunnelling probability
V_0	barrier height above the particle's energy (J)	Tunnelling probability
n_c	electron concentration in the collector depletion region (cm^{-3})	Kirk effect
J_C	collector current density ($\mathrm{A/cm}^2$)	Kirk effect
J_K	current density at the onset of the Kirk effect ($\mathrm{A/cm}^2$)	Kirk effect
N_C	collector doping (cm^{-3})	Kirk effect
V_{CB}	collector–base reverse voltage (V)	Kirk effect
W_C	thickness of the collector layer (cm)	Kirk effect
h	depletion thickness under the gate (cm)	Depletion thickness
V_{gs}	gate–source voltage (V)	Depletion thickness
$V(x)$	channel voltage at x (V)	Depletion thickness
x	distance along the channel from the source (cm)	Depletion thickness
V_{p0}	internal pinch-off voltage (V)	Internal pinch-off
a	channel thickness (cm)	Internal pinch-off
N_a	acceptor concentration in a p channel (cm^{-3})	Internal pinch-off
V_p	pinch-off voltage (V)	Pinch-off voltage
N_a, N_d	acceptor and donor concentrations of the gate and channel (cm^{-3})	pn built-in potential
$V_{d,sat}$	drain voltage at saturation (V)	Saturation drain voltage
I_D	drain current (A)	Square law
I_{DSS}	drain current at $V_{GS} = 0$ (A)	Square law
V_{GS}	gate–source voltage (V)	Square law
g_m	transconductance (S)	Transconductance
V_{DS}	drain–source voltage (V)	Transconductance
g_{ms}	transconductance in saturation (S)	Transconductance
g'_m	transconductance with a source resistance (S)	Source resistance
r_s	source resistance (Ω)	Source resistance

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